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Thermodynamic Modeling of Hydrocarbon Mixtures with Application to Polymer Phase Behavior and Asphaltene Precipitation

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ABSTRACT

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A new equation of state framework is proposed that hybridizes the classical cubic equations of state (EOS) with the chain term of the SAFT EOS. This model offers an improved description of the molecular features of long-chain molecules, such as heavy n-alkanes and polymers, whose thermodynamic properties are often poorly described by standard cubic EOS models. This hybrid EOS, called the cubic-plus-chain (CPC) equation of state, shows promising results for the n-alkanes and polymers studied and could ultimately be used for improving the predictions of liquid-liquid phase splits driven by heavy nonpolar molecules, which has potential for application to asphaltene precipitation.

Additionally, critical property correlations for nonpolar components proposed in a recent work are tested for their application to mixed solvent phase behavior modeling. Because the standard cubic EOS and CPC models require critical properties as inputs, the proposed correlations provide a means to calculate thermodynamic properties for mixed solvents given information on only molecular weight and refractive index – measurements that are simple to perform for both pure substances and mixtures – and have potential application to crude oil systems for which composition is unknown. Phase behavior modeling studies with the mixed

solvent approach and cubic EOS show exceedingly good agreement to standard modeling approaches that require a detailed compositional analysis.

Also, an asphaltic crude oil phase behavior study with a high GOR and asphaltene-rich fluid produced from a deep-water environment is presented. Thermodynamic modeling with PC-SAFT revealed interesting trends with respect to the properties of the bulk and onset phases that suggest that, though asphaltenes may be unstable in the bulk fluid, phase-splitting that is classically viewed as a precursor to deposition might sometimes produce two liquids of relatively low density whose compositions are much leaner in asphaltenes than phases commonly associated with deposition problems. As part of this study, a new algorithm for calculating saturation pressures of asphaltic crude oils is presented. This algorithm uses reliable methods to generate excellent trial phase compositions for initializing the upper and lower asphaltene onset and bubble pressure calculations.

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Nomenclature

Roman Alphabet

P	Pressure
T	Temperature
V	Volume
$\hat{V}$	Molar volume
A	Helmholtz energy
A^R	Residual Helmholtz energy, $A - A^{\mathrm{ig}}$
F	Reduced residual Helmholtz energy, A^R/RT
G	Gibbs energy
G^E	Excess Gibbs energy, $G - G^{\mathrm{is}}$
H	Enthalpy
S	Entropy
U	Internal energy
n_i	Mole number of component i
z_i	Overall molar composition of component i
y_{ik}	Molar composition of component i in phase k
$\hat{f}_{ik}$	Fugacity of component i in phase k
K_{ik}	Partition coefficient of component i in phase k
M_W	Molecular weight
N_C	Number of components
N_P	Number of phases

Greek Symbols

α	Thermal expansion
β	Reduced volume (for CPC and cubic EOS)
β_k	Phase fraction of phase k
γ	Aromaticity (for PC-SAFT parameter tuning)
δ	Solubility parameter

κ	Isothermal compressibility
$\hat{\phi}_{ik}$	Fugacity coefficient of component i in phase k
μ_{ik}	Chemical potential of component i in phase k

Abbreviations

BP	Bubble pressure
CPC	Cubic-Plus-Chain
CPA	Cubic-Plus-Association
EOS	Equation of state
LAOP	Lower asphaltene onset pressure
LLE	Liquid-liquid equilibrium
PR	Peng-Robinson
PC-SAFT	Perturbed-Chain Statistical Associating Fluid Theory
P^{VLE}	Pressure at the pseudo-VLE (for Algorithm 3.6)
RK	Redlich-Kwong
SAFT	Statistical Associating Fluid Theory
SRK	Soave-Redlich-Kwong
TPD	Tangent plane distance (for stability analysis)
UAOP	Upper asphaltene onset pressure
VDW	Van der Waals
VLE	Vapor-liquid equilibrium
VLLE	Vapor-liquid-liquid equilibrium

Chapter 1

Introduction

1.1. Research Motivation

Understanding phase behavior and making thermodynamic property predictions is important in the oil and gas, chemical products, and materials synthesis industries for helping to design strategies that optimize production and minimize downtime. In the oil and gas industry, phase behavior studies are most relevant for assessing potential flow assurance problems that can lead to costly shutdowns and cleanups of producing wells. As the thermodynamic conditions change during downhole fluid extraction or throughout the life of a project, it is important that our recovery, transportation and refining schemes change to accommodate these factors.

The phase equilibrium and rheological properties of reservoir fluids are strong functions of pressure, temperature and fluid composition. Thus, a model used to predict fluid properties must be sufficiently sophisticated to capture changes in fluid behavior as system variables change, yet simple enough to allow implementation in a programming environment and in the field. Additionally, it is difficult to design a model capable of handling complicated crude oil systems with large particle size differences and various force interactions. For this reason, much ambiguity still exists in modeling crude oils, and it is vital that industry continues to invest in the search for better models and methodologies.

1.2. Problem statement

The broad aim of this work is to propose and validate methodologies for better understanding of phase behavior and to also provide the tools capable of utilizing the research presented here for industrial application. Because of the nature of this work, it is important that every step of thermodynamic modeling is both understood and executed in-house. To this end, simulation tools were developed in the Visual Basic for Applications (VBA) environment and deployed in Microsoft Excel that automate important process calculations – including crude oil characterization, multiphase equilibrium calculations with equations of state, and simulation of common PVT experiments – without the use of third-party software. Additionally, the software tool developed as part of this dissertation is flexible enough for the implementation of new equation of state models – one of which is proposed in this work – that will facilitate rapid expansion in the capabilities of the simulation tool. Future work will include implementing new thermodynamic models, modifying existing models, and improving speed and performance for the PT-flash and saturation boundary calculations.

1.3. Thesis Outline

Chapter 2 presents the phase equilibrium thermodynamics framework on which the rest of the thesis is constructed. In this chapter, the fundamental equations of equilibrium are developed in detail and focus is placed on the thermodynamic energy functions of Gibbs and Helmholtz, which make a complete statement about the thermodynamic state of a system in terms of easily measured variables such as pressure, temperature, volume and composition. The chapter is concluded with a brief discussion on equations of state, presenting them as a means to calculate the important properties derived earlier in the chapter.

Chapter 3 presents a practical programming guide to implementing algorithms to solve the equilibrium equations presented in Chapter 2. Algorithms for stability analysis, the two-phase and multiphase PT-flash, and saturation pressure (and temperature) calculations are presented. Additionally, a new algorithm is proposed for solving saturation pressure calculations for asphaltic crude oils in which three saturation boundaries at a fixed temperature regularly exist. The proposed algorithm has shown to converge reliably to all three saturation pressures, when they exist, and to identify certain saturation pressures do not exist.

Chapter 4 presents the equation of state framework for calculating physical properties of fluid mixtures. The most commonly used equations of state for thermodynamic modeling of hydrocarbon systems, which are the classical cubic equations of state of Soave-Redlich-Kwong (SRK) and Peng-Robinson (PR) and the more recent SAFT models based in statistical thermodynamics, are presented.

Chapter 5 presents an overview of asphaltene thermodynamic modeling from crude oil characterization to simulation and prediction of equilibrium properties. Asphaltene thermodynamic modeling is usually undertaken as a key step in predicting deposition problems, and, though deposition modeling is outside the scope of this work, many of the relevant properties on which the flow models rely – such as phase densities, compositions, and solubilities – come from thermodynamic modeling. The chapter is concluded with an interesting case study showing anomalous asphaltene phase behavior when compared to many of the crude oil modeling studies presented in past publications. This anomalous behavior likely has a non-negligible effect on the deposition tendency of the asphaltenes.

Chapter 6 proposes a new equation of state framework for chain molecules that hybridizes the classical cubic equations of state with the chain equation of state from first-order perturbation theory. Both the classical cubic and chain equations of state are discussed in Chapter 4. The classical cubic equations of state treat molecules as spheres regardless of molecule size, meaning that the shape of the model molecule for propane is indistinguishable from that of n-pentadecane. For this reason, the cubic models predict liquid densities poorly for heavy molecules, one of the drawbacks which the proposed framework seeks to correct. Additionally, the new model is cubic in the physical region, making numerical algorithms for solving the classical cubic models and the proposed cubic-plus-chain (CPC) model nearly identical, and the computational burden for CPC on the order of the standard cubic models and much faster than the SAFT family of models.

Chapter 7 evaluates a recently proposed set of critical property correlations for nonpolar hydrocarbons which require only a single refractive index measurement performed at ambient conditions. The correlations are tested by performing phase equilibrium and liquid property calculations using the correlations to specify the critical properties of a mixed solvent. The ease with which the input data for the correlation can be obtained and the flexibility of the modeling approach proposed in Chapter 7 make this work attractive for crude oil modeling for which the composition of the fluid is unknown.

Chapter 8 concludes the thesis and recommends future work for improving the CPC equation of state proposed in Chapter 6, extending the modeling approach proposed in Chapter 7 to crude oil systems, and validating the anomalous asphaltene phase behavior predictions discussed in Chapter 5.

Chapter 2

Thermodynamic Equilibrium

2.1. The First and Second Laws of Thermodynamics

The first law of thermodynamics is the energy conservation principle, which postulates that the total energy of an isolated system is time-invariant and energy is transferred within the system by various mechanisms, but the total energy content of the system remains unchanged. The internal energy of a system can be changed due to heat or work that are added from the surroundings. Written in infinitesimal form, the first law states that:

$$dU = \delta Q + \delta W \tag{2.1}$$

where the infinitesimal transfer of heat (δQ) accounts for thermal effects and the infinitesimal transfer of work (δW) accounts for forces acting on a system, the most common of which are gravitational and electric forces.

The second law of thermodynamics postulates the existence of a state function that provides the direction of change for a process and the limit to which that change can occur. This state function is called entropy (S) and calculated by:

$$dS = \int_1^2 \frac{\delta Q_{\mathrm{rev}}}{T} \tag{2.2}$$

where the integral is computed along the path from equilibrium state 1 to equilibrium state 2 and δQ_{rev} is the differential change in heat added to a system for an equilibrium (reversible) process. An important consequence of the second law is that the total change in entropy for a system and its surroundings must be non-negative, which precludes transformations that decrease the total entropy from occurring:

$$\Delta S_{\mathrm{total}} = \Delta S_{\mathrm{sys}} + \Delta S_{\mathrm{surr}} \geq 0 \tag{2.3}$$

For an equilibrium system, if the only type of work allowed is that due to an external pressure, then the energy change due to work can be written:

$$\delta W = -PdV \tag{2.4}$$

This result can be combined with the entropy expression to restate the first law as:

$$dU = TdS - PdV \tag{2.5}$$

which can be written in more general terms as the total differential of the internal energy:

$$dU(S,V) = \left(\frac{\partial U}{\partial S}\right)_V dS + \left(\frac{\partial U}{\partial V}\right)_S dV \tag{2.6}$$

Revealing thermodynamic definitions for temperature and pressure:

$$T = \left(\frac{\partial U}{\partial S}\right)_V \qquad\qquad P = -\left(\frac{\partial U}{\partial V}\right)_S \tag{2.7}$$

One important contribution to internal energy that has been thus far neglected is composition. The preceding expressions for internal energy apply only to pure systems. Writing instead the total differential of internal energy with mole numbers included yields:

$$dU(S, V, n_1, \dots, n_{N_C}) = \left(\frac{\partial U}{\partial S}\right)_{V,\boldsymbol{n}} dS + \left(\frac{\partial U}{\partial V}\right)_{S,\boldsymbol{n}} dV + \sum_{i=1}^{N_C} \left(\frac{\partial U}{\partial n_i}\right)_{S,V,n_{j\neq i}} dn_i \tag{2.8}$$

where the mole number derivative of U is defined as the chemical potential, μ_i:

$$\mu_i = \left(\frac{\partial U}{\partial n_i}\right)_{S,V,n_{j\neq i}} \tag{2.9}$$

2.2. Thermodynamic Functions and the Equilibrium State

Instead of focusing on the derivation of other thermodynamic functions – which are merely the consequence of mathematical manipulations (namely, the Legendre transformation) and can be found in nearly any thermodynamics textbook – the purpose of this section is to explain the results of those derivations and connect the conceptual framework to practical applications.

Eq. (2.8) contains every term that contributes to the internal energy function (excluding those assumed negligible such as electrical and surface forces) and is thus a full statement of the thermodynamic state of a system. If changes in entropy and volume can be measured reliably, then the equation for internal energy is sufficient for the full set of thermodynamic property calculations. However, entropy and volume are rarely the most convenient properties with which to work, so alternative thermodynamic functions were derived for the sake of convenience. The full set of common thermodynamic functions are:

$$dU(S, V, n_1, \dots, n_{N_C}) = TdS - PdV + \sum_{i=1}^{N_C} \mu_i \, dn_i \tag{2.10}$$

$$dH(S, P, n_1, \dots, n_{N_C}) = TdS + VdP + \sum_{i=1}^{N_C} \mu_i \, dn_i \tag{2.11}$$

$$dG(T, P, n_1, \dots, n_{N_C}) = -SdT + VdP + \sum_{i=1}^{N_C} \mu_i \, dn_i \tag{2.12}$$

$$dA(T, V, n_1, \dots, n_{N_C}) = -SdT - PdV + \sum_{i=1}^{N_C} \mu_i \, dn_i \tag{2.13}$$

These state functions reveal a few interesting facts worth stating explicitly. First, the chemical potential is equivalent in all four formulations:

$$\mu_i = \left(\frac{\partial U}{\partial n_i}\right)_{S,V,n_{j \neq i}} = \left(\frac{\partial H}{\partial n_i}\right)_{S,P,n_{j \neq i}} = \left(\frac{\partial A}{\partial n_i}\right)_{T,V,n_{j \neq i}} = \left(\frac{\partial G}{\partial n_i}\right)_{T,P,n_{j \neq i}} \tag{2.14}$$

Second, each thermodynamic property can be calculated by at least two different state function formulations. For example, temperature, pressure, volume, and entropy can be calculated by derivatives of the energy functions:

$$\begin{aligned} T &= \left(\frac{\partial U}{\partial S}\right)_{V,\boldsymbol{n}} = \left(\frac{\partial H}{\partial S}\right)_{P,\boldsymbol{n}} \\ P &= -\left(\frac{\partial U}{\partial V}\right)_{S,\boldsymbol{n}} = -\left(\frac{\partial A}{\partial V}\right)_{T,\boldsymbol{n}} \end{aligned} \tag{2.15}$$

$$V = \left(\frac{\partial H}{\partial P}\right)_{S,\boldsymbol{n}} = \left(\frac{\partial G}{\partial P}\right)_{T,\boldsymbol{n}}$$

$$S = -\left(\frac{\partial A}{\partial T}\right)_{V,\boldsymbol{n}} = -\left(\frac{\partial G}{\partial T}\right)_{P,\boldsymbol{n}}$$

which shows that thermodynamic properties are inter-related and any formulation of the thermodynamic state of a system can reveal all thermodynamic properties of a system via mathematical transformations. Michelsen provides a comprehensive list of the relationship between the various thermodynamic functions and their derivatives[1].

Perhaps the most important thermodynamic function comes from a rearrangement of the internal energy function, written in terms of entropy:

$$dS(U, V, n_1, \ldots, n_{N_C}) = \frac{1}{T}dU + \frac{P}{T}dV - \frac{1}{T}\sum_{i=1}^{N_C} \mu_i dn_i \tag{2.16}$$

From the second law, the entropy of an isolated system is at a maximum at equilibrium and any isolated system not at equilibrium will increase its entropy to achieve equilibrium, meaning the entropy *change* at equilibrium must be zero. For a multiphase system, the equilibrium condition (entropy maximization) and entropy equation (Eq. (2.16)) require that the temperature, pressure, and chemical potentials of component i are equal in all bulk phases present at equilibrium.

$$T^1 = \cdots = T^{N_P} \qquad P^1 = \cdots = P^{N_P} \qquad \mu_i^1 = \cdots = \mu_i^{N_P} \tag{2.17}$$

Additionally, because only processes that yield non-negative values in entropy change for an isolated system are allowable, the following criteria hold for spontaneous processes:

$$\begin{aligned} &(dU)_{S,V} \leq 0 && (dA)_{T,V} \leq 0 \\ &(dH)_{S,P} \leq 0 && (dG)_{T,P} \leq 0 \end{aligned} \tag{2.18}$$

which means that each set of independent variables have a corresponding state function to minimize, where the entropy maximization principle is embedded within each state function. For equilibrium calculations, the choice of state function to minimize usually depends on the independent variable set chosen. Because temperature, pressure and volume are the most commonly specified variables for thermodynamic calculations, the Gibbs energy (G) and Helmholtz energy (A) are especially useful formulations.

2.3. Thermodynamic Stability

For a closed multi-component multiphase system at fixed temperature and pressure, the equilibrium state corresponds to the global minimum of the Gibbs energy function. The change in Gibbs energy for a system at fixed temperature and pressure is given by:

$$dG(T, P, \boldsymbol{n}) = \sum_{i=1}^{N_C} \mu_i \, dn_i \tag{2.19}$$

A phase distribution is considered stable if the Gibbs energy cannot be further decreased by transferring material between existing phases or transferring material to form a new phase. Because the Gibbs energy function can have multiple minima, searching for the state which yields the global minimum at a fixed T, P must encompass all of composition space and test the possibility of multiple phases forming. Assuming that the minimum Gibbs state has been found for a N_P-phase distribution, the test for stability requires that no new phase can be formed that lowers the Gibbs energy. The proof of this condition, called the tangent plane criterion, was originally developed by Baker et al.[2] and subsequently given practical significance by Michelsen[3].

The change in Gibbs energy for a mixture that splits into two phases is given by:

$$\Delta G = (G^I + G^{II}) - G^0 = G(N - \epsilon) + G(\epsilon) - G^0 \tag{2.20}$$

where G^I and G^{II} are the Gibbs energies of the bulk and forming phases, respectively, and G^0 is the Gibbs energy of the original single-phase mixture of amount N. A Taylor series expansion of G^I with second order terms neglected yields:

$$G^I = G(N - \epsilon) = G^0 - \epsilon \sum_{i=1}^{N_C} y_i \mu_i^0(N) \tag{2.21}$$

which gives the following change in Gibbs energy:

$$\Delta G = \epsilon \sum_{i=1}^{N_C} w_i \left(\mu_i(\boldsymbol{w}) - \mu_i^0(N) \right) \tag{2.22}$$

where w_i and μ_i are the composition and chemical potentials, respectively, of component i in the forming phase, and ϵ is the amount of material that leaves the bulk phase and enters the forming phase. Because ϵ must be a positive number, the change in Gibbs energy can only be negative if the summation yields a negative number. Stability of the N_P-phase distribution is verified when the tangent plane criterion is satisfied for any possible trial composition w_i.

$$F(\boldsymbol{w}) = \sum_{i=1}^{N_C} w_i \left(\mu_i(\boldsymbol{w}) - \mu_i^0(N)\right) \geq 0 \tag{2.23}$$

2.4. Property Calculations and Equations of State

The preceding discussion on thermodynamic functions and equilibrium revealed the existence of a particularly important property: the chemical potential, μ_i. However, instead of calculating chemical potentials, which depend on an arbitrary reference state, it is often more convenient to calculate an analogous property called the fugacity ($\hat{f}_i$) or the related fugacity coefficient ($\hat{\phi}_i$). The fugacity coefficient of component i in a mixture is calculated by:

$$RT \ln \hat{\phi}_i(T, P, \boldsymbol{n}) = \left(\frac{\partial G^R}{\partial n_i}\right)_{T,P,n_{j\neq i}} = \left(\frac{\partial A^R}{\partial n_i}\right)_{T,V,n_{j\neq i}} - RT \ln Z \tag{2.24}$$

And the fugacity is related to the fugacity coefficient by:

$$\hat{f}_i(T, P, \boldsymbol{n}) = y_i \hat{\phi}_i P \tag{2.25}$$

The fugacity and chemical potential are related by:

$$d\mu_i = RT \, d \ln \hat{f}_i \tag{2.26}$$

The most common method for calculating thermodynamic properties for a mixture is with a generating function like the excess Gibbs (G^E) or residual Helmholtz (A^R) that is differentiated with respect to its independent variables to calculate all thermodynamic properties of interest. Equations of state – which will be discussed in Chapter 4 – are often shown in textbooks in pressure-explicit form, but they are most useful in their residual Helmholtz form where any thermodynamic property of a phase can be obtained via differentiation of the Helmholtz energy.

Appendix A contains some of these important thermodynamic properties in terms of the residual Helmholtz energy, and Appendix B shows analytic residual Helmholtz derivative expressions for the cubic and cubic-plus-chain (CPC) equations of state.

Chapter 3

Phase Equilibria Calculations

3.1. Formulating the Phase Equilibrium Constraints

In the previous chapter, the phase equilibrium relations were derived from the entropy maximization principle. The conclusion was that the temperature, pressure, and chemical potential of component i must be equal in all phases present at equilibrium.

$$T^1 = \cdots = T^{N_P} \qquad P^1 = \cdots = P^{N_P} \qquad \mu_i^1 = \cdots = \mu_i^{N_P} \tag{2.17}$$

Instead of chemical potentials, which depend on a reference state, fugacities are the preferred working variable for phase equilibrium calculations, and an equivalent equilibrium condition can be derived for fugacity. Integrating the fugacity expression in Eq. (2.26) from a reference state P^0 to final state P yields:

$$\mu_i(T,P) = \mu_i^0(T,P^0) + RT\ln\frac{\hat{f}_i(T,P)}{f_i^0(T,P^0)} \tag{3.1}$$

Because the temperature and pressure are equal in all phases present at equilibrium and the reference states depend only on temperature, the reference states cancel and the equilibrium criterion in terms of fugacity is simply:

$$\hat{f}_{i1} = \hat{f}_{i2} = \cdots = \hat{f}_{iN_P} \qquad i = 1, N_C \tag{3.2}$$

In terms of fugacity coefficients, an equivalent statement is:

$$y_{i1}\hat{\phi}_{i1} = y_{i2}\hat{\phi}_{i2} = \cdots = y_{iN_P}\hat{\phi}_{iN_P} \qquad i = 1, N_C \tag{3.3}$$

The material balance for a non-reacting multi-component system that splits into multiple phases requires that the phase mole fractions sum to one and the sum of each component in the respective phases matches the overall composition:

$$1 = \sum_{k=1}^{N_P} \beta_k \tag{3.4}$$

$$z_i = \sum_{k=1}^{N_P} \beta_k y_{ik} \tag{3.5}$$

Additionally, each equilibrium phase can be related by partition coefficients (K_{ik}) that are defined with respect to an arbitrary phase. Here, N_P is the reference phase:

$$K_{ik} = \frac{y_{ik}}{y_{iN_P}} \qquad \begin{array}{l} i = 1, N_C \\ k = 1, (N_P - 1) \end{array} \tag{3.6}$$

From the equi-fugacity criterion, the partition coefficients can also be written in terms of fugacity coefficients:

$$K_{ik} = \frac{y_{ik}}{y_{iN_P}} = \frac{\hat{\phi}_{iN_P}}{\hat{\phi}_{ik}} \qquad \begin{array}{l} i = 1, N_C \\ k = 1, (N_P - 1) \end{array} \tag{3.7}$$

For a set of partition coefficients and phase fractions, the compositions of each phase can be calculated by:

$$y_{ik} = K_{ik} y_{iN_P} = \frac{z_i K_{ik}}{\sum_{k=1}^{N_P} \beta_k K_{ik}} \qquad \begin{matrix} i = 1, N_C \\ k = 1, (N_P - 1) \end{matrix} \tag{3.8}$$

Additionally, the sum of mole fractions constraint requires that the mole fractions in each phase sum to one, or, equivalently:

$$\sum_{i=1}^{N_C} \left(y_{ik} - y_{iN_P} \right) = 0 \qquad k = 1, (N_P - 1) \tag{3.9}$$

Regardless of the specifications of the problem – PT-flash, VT-flash, phase boundary, etc. – the phase equilibrium constraints (Eqs. (2.17) and (3.2)) and material balance constraints (Eqs. (3.4) - (3.9)) must hold.

3.2. Multiphase PT-Flash

The objective of the pressure and temperature specified phase equilibrium calculation (PT-Flash) is to find the phase fractions ($\boldsymbol{\beta}$) and compositions ($\boldsymbol{y}$) which minimize the Gibbs energy and satisfy the equilibrium and material balance constraints. Because the equilibrium state corresponds to the global minimum in the Gibbs energy function, the natural assumption would be that a global minimization algorithm would be most suitable. However, these algorithms rarely offer advantages over local minimization algorithms for phase equilibrium problems. The most widely-used local minimization algorithm for phase equilibrium was proposed by Michelsen. His formulation consists of stability analysis followed by a phase split calculation[3,4]. If stability analysis reveals that the formation of another phase can lower the Gibbs energy, then a phase is added, and the phase-split calculation is performed. This cycle repeats until stability analysis verifies that the current phase distribution cannot be destabilized by adding another phase. A high-level view of the PT-flash algorithm is shown in Figure 3.1.

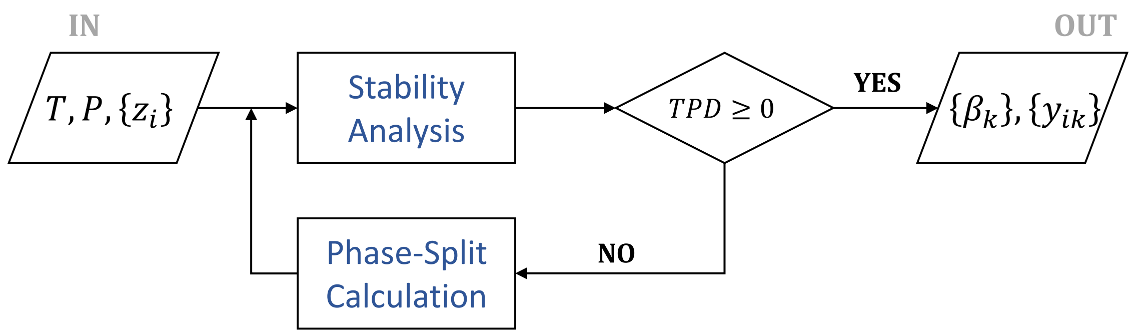

Figure 3.1. PT-flash programming structure. If stability analysis verifies thermodynamic stability of the current phase distribution (TPD ≥ 0), then the flash calculation is finished. If stability analysis identfies a phase that would destabilize the current phase distribution (TPD < 0), then the phase-split calculation commences.

3.2.1. Stability Analysis

From the criterion for phase stability derived in the previous chapter, stability is verified when no trial composition can be found that yields $F(\boldsymbol{w}) \geq 0$:

$$F(\boldsymbol{w}) = \sum_{i=1}^{N_C} w_i \left(\mu_i(\boldsymbol{w}) - \mu_i^0(N)\right) \geq 0 \tag{3.10}$$

Michelsen recast the tangent plane criterion in terms of fugacities, deriving the reduced modified tangent plane distance (TPD) function.

$$TPD(\boldsymbol{W}) = \sum_{i=1}^{N_C} W_i \left(\ln W_i + \ln \hat{\phi}_i(\boldsymbol{W}) - d_i(\boldsymbol{z}) \right)$$
$$d_i(\boldsymbol{z}) = \ln z_i + \ln \hat{\phi}_i(\boldsymbol{z}) \tag{3.11}$$

where $\boldsymbol{W}$ are the mole numbers of the trial phase (not constrained to sum to one) and $\boldsymbol{z}$ the composition of the test phase. The test phase can be any of the phases that exist at equilibrium, though Gorucu and Johns recommend the phase with the smallest molar Gibbs energy be chosen[5].

To verify stability of the test phase, the TPD function should be evaluated at every stationary point. If all stationary points of the TPD function are non-negative, then the test phase is verified as stable and no further phase-split calculations are performed. If any stationary point is negative, then a phase-split calculation is performed, taking the composition or fugacity coefficients of the trial phase that destabilizes the test phase to initialize the phase-split calculation.

The stationary points are found by minimizing the TPD function. The gradient and Hessian of the TPD function are[6]:

$$\mathrm{g}_i = \frac{\partial TPD(\boldsymbol{W})}{\partial W_i} = \ln W_i + \ln \hat{\phi}_i(\boldsymbol{W}) - d_i(\boldsymbol{z}) \tag{3.12}$$
$$H_{ij} = \frac{\partial \mathrm{g}_i}{\partial W_j} = \frac{\delta_{ij}}{W_i} + \frac{\partial \ln \hat{\phi}_i(\boldsymbol{W})}{\partial W_j}$$

A Newton step can be performed to update mole numbers of the trial phase by solving $\boldsymbol{H}\Delta\boldsymbol{W} = -\mathbf{g}$. However, the Newton method requires that the Hessian be positive definite, which is often not the case during early iterations when the current set of mole numbers are far from the solution. During these iterations, mole numbers are updated by a successive substitution iteration (SSI) procedure.

$$\ln W_i^{k+1} = d_i(\boldsymbol{z}) - \ln \hat{\phi}_i(\boldsymbol{W}) \tag{3.13}$$

Because a minimum in the TPD function occurs when the gradient vector is zero, convergence criteria based on the gradient vector is a natural choice.

$$S = \sqrt{\sum_{i=1}^{N_C} |\mathrm{g}_i|^2} < \epsilon \tag{3.14}$$

The stability analysis algorithm usually converges quickly, but Petitfrere and Nichita[6] identified cases which cause convergence issues, which are usually near phase boundaries or critical points.

Algorithm 3.1 - Stability Analysis

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1. Calculate $\ln \hat{\phi}_i\,(\boldsymbol{z})$
2. Calculate $d_i(\boldsymbol{z})$
3. Set initial trial phase composition $(\boldsymbol{W})$
4. Calculate $\ln \hat{\phi}_i\,(\boldsymbol{W})$
5. Update $\boldsymbol{W}$
6. Calculate error. If $S < \epsilon$ then converged, else return to step 4

3.2.1.1. Initial Guesses for Trial Phase Composition

The objective of stability analysis is to perform an exhaustive search of composition space to verify that the addition of another phase will not lower the Gibbs energy ($TPD \geq 0$ at all stationary points). What constitutes an "exhaustive search" is very much dependent on the types of systems under consideration and the amount of computational time dedicated to stability analysis. The initialization strategy used in this thesis is slightly modified from that proposed by Li and Firoozabadi[7]. They extended the Haugen four-sided initialization, which uses the Wilson correlation to initialize four trial partition coefficients, two vapor-like roots and two liquid-like roots:

$$\ln K_i^W = \ln\left(\frac{P_{c_i}}{P}\right) + 5.373(1+\omega_i)\left(1 - \frac{T_{c_i}}{T}\right) \tag{3.15}$$

$$K_i^S = K_i^W \qquad \text{Vapor}$$

$$K_i^S = (K_i^W)^{\frac{1}{3}} \qquad \text{Vapor}$$

$$K_i^S = (K_i^W)^{-1} \qquad \text{Liquid}$$

$$K_i^S = (K_i^W)^{-\frac{1}{3}} \qquad \text{Liquid}$$

where the initial trial compositions ($\boldsymbol{W}$) are calculated from the initial trial partition coefficients and the composition of the test phase ($\boldsymbol{z}$):

$$W_i = K_i^S z_i \tag{3.16}$$

Depending on the nature of the test phase, certain trial compositions can be eliminated as infeasible before wasting iterations performing stability analysis. For example, if a vapor already exists among the stable phases, then it is highly unlikely that initializing vapor-like roots will reveal further instability. Thus, when it is known in advance which types of phases can exist at the specified conditions ($T, P, \boldsymbol{z}$), safe assumptions can be made to bypass wasteful calculations without sacrificing accuracy.

In addition to the Haugen four-sided initialization, Li and Firoozabadi proposed near-pure trial phases, with partition coefficients of each phase calculated by:

$$K_i^S = 0.9/z_i$$
$$K_j^S = \left[\frac{0.1}{(N_C - 1)}\right]/z_j \qquad (j \neq i) \tag{3.17}$$

The full Li and Firoozabadi stability initialization scheme offers $N_C + 4$ trial phases for stability testing, which can be relatively time-consuming and unnecessary depending on the types of systems being studied. In this thesis, the near-pure trial phases used were only for the lightest component present and any components with molecular weights larger than 150 g/mol.

3.2.2. Phase-Split Calculation

The objective of the phase-split calculation is to find the minimum Gibbs state that satisfies the equilibrium relations and material balance. The Gibbs energy is calculated by:

$$G = RT \sum_{k=1}^{N_P} \sum_{i=1}^{N_C} n_{ik} \ln \hat{f}_{ik} \tag{3.18}$$

In terms of phase fractions and fugacity coefficients, the Gibbs energy can be restated in molar form as:

$$\hat{G} = RT \sum_{k=1}^{N_P} \beta_k \sum_{i=1}^{N_C} y_{ik} \ln\left(y_{ik} \hat{\phi}_{ik} P\right) \tag{3.19}$$

Additionally, the following equilibrium and mass balance constraints must be satisfied:

$$y_{i1}\hat{\phi}_{i1} = \cdots = y_{iN_P}\hat{\phi}_{iN_P} \qquad i = 1, N_C \qquad (3.3)$$

$$1 = \sum_{k=1}^{N_P} \beta_k \qquad (3.4)$$

$$z_i = \sum_{k=1}^{N_P} \beta_k y_{ik} \qquad i = 1, N_C \qquad (3.5)$$

$$\sum_{i=1}^{N_C} \left(y_{ik} - y_{iN_P}\right) = 0 \qquad k = 1, (N_P - 1) \qquad (3.9)$$

The equality of temperature and pressure in all phases present at equilibrium is automatically satisfied by the choice of specification variables.

As in stability analysis, the natural choice for the gradient equation for the phase-split calculation comes from the equi-fugacity criterion:

$$\mathrm{g}_{ik} = \ln \hat{f}_{ik} - \ln \hat{f}_{iN_P} \qquad k = 1, (N_P - 1) \qquad (3.20)$$

Also, like stability analysis, the safest and most robust method for converging to a minimum of the Gibbs energy function is by successive substitution iterations (SSI) when far from the solution and by a second order method (like Newton-Raphson) when near to the solution. Michelsen and Heidemann[8] showed that, except in extremely rare cases, the SSI algorithm converges reliably to local minima of the objective function, but progress can be very slow near critical points or, more generally, when the partition coefficients are a strong function of composition.

To leverage the reliability of SSI with the speed of a second order method, SSI is used during early iterations to provide reasonable initial guesses for the fast-converging methods. However, even when SSI provides reasonable guesses, Newton iterations could still over-shoot the solution. Thus, to protect against over-shooting, iterations that increase the Gibbs energy are rejected and a safer update procedure (like SSI or trust region method) is applied to continue progressing to the solution.

The SSI algorithm consists of an outer loop for adjusting the partition coefficients (or fugacity coefficients) of each phase and an inner loop for converging the material balance equations for the current set of partition coefficients. Resolution of the Rachford-Rice (material balance) equation for a fixed set of partition coefficients (or fugacity coefficients) is commonly called a constant-K flash. The two-phase and multiphase constant-K flashes are covered in the next two sections.

The SSI algorithm for the phase-split calculation is:

Algorithm 3.2 – Phase-Split Calculation (SSI)

1. Set initial partition (K_{ik}) or fugacity coefficients $\left(\hat{\phi}_{ik}\right)$
2. Perform constant-K flash to calculate β_k, y_{ik}
3. Recalculate fugacity coefficients for each phase from EOS
   - $\ln \hat{\phi}_{ik}$ = EOS(T, P, y_{ik})
4. Update partition coefficients for each phase
   - $\ln K_{ik} = \ln \hat{\phi}_{iN_P} - \ln \hat{\phi}_{ik}$
5. Calculate error. If $S < \epsilon$ then converged, else return to step 2

As discussed in previous sections, the phase-split calculation is performed when stability analysis detects a phase that would lower the Gibbs energy of the current phase distribution. The initial guesses for the phase-split calculation are taken from the results of stability analysis, where either the composition or fugacity coefficients of the destabilizing phase are used as the initial guess along with the compositions or fugacity coefficients of the original phase distribution. However, there is no requirement that the phase-split calculation must be preceded by stability analysis. The constant-K flash procedure should be robust enough that an initial guess for partition coefficients in the single-phase region will converge to a result in the single-phase region. Stability analysis is preferred as a precondition because it is faster than the phase-split calculation and yields initial guesses that promote convergence of the phase-split calculation to the desired solution.

3.2.2.1. Two-Phase Constant-K Flash

The Rachford-Rice formula comes from the sum of mole fractions and composition formulas (Eqs. (3.9) and (3.8)):

$$R_k(\boldsymbol{\beta}) = \sum_{i=1}^{N_C}\left(y_{ik} - y_{iN_P}\right) = \sum_{i=1}^{N_C}\frac{z_i(K_{ik}-1)}{\sum_{k=1}^{N_P}\beta_k K_{ik}} = 0 \qquad k = 1, (N_P - 1) \tag{3.21}$$

where R_k is the residual of the Rachford-Rice equation for phase k.

For the two-phase case, only one residual needs to be written and the fraction of phase 2 can be eliminated because of its dependence on phase 1. The Rachford-Rice formula then simplifies to:

$$R(\beta_1) = \sum_{i=1}^{N_C} \frac{z_i(K_i - 1)}{1 + \beta_1(K_i - 1)} = 0 \tag{3.22}$$

Michelsen[1] suggests solving the two-phase Rachford-Rice by a Newton method with interval control that does not allow β_1 to leave the feasible range ($\beta_1 \in (0,1)$). Newton's method requires that the objective be differentiated with respect to the independent variable, yielding:

$$R'(\beta_1) = \sum_{i=1}^{N_C} \frac{z_i(K_i - 1)^2}{[1 + \beta_1(K_i - 1)]^2} = 0 \tag{3.23}$$

Nichita and Leibovici proposed an alternative method for resolving the two-phase Rachford-Rice by a series of convex transformations[9,10]. Their formulation effectively rescales the Rachford-Rice formula to eliminate the need for interval control and guarantees fast convergence. We forgo the mathematical proof showing the advantageous properties of their new function and instead present the key equations and algorithm of what they call the FGH approach to resolving the two-phase Rachford-Rice[9]. Their approach requires that the components are ordered by decreasing K-values.

The reformulated Rachford-Rice equation and its derivative are:

$$F(a) = z_1 + \sum_{i=2}^{N_C - 1} \frac{z_i a}{d_i + a(1 + d_i)} - z_{N_C} a = 0 \tag{3.24}$$

$$F'(a) = \sum_{i=2}^{N_C - 1} \frac{z_i d_i}{[d_i + a(1 + d_i)]^2} - z_{N_C}$$

where:

$$d_i = \frac{c_1 - c_i}{c_{N_C} - c_1} \qquad\qquad c_i = \frac{1}{1 - K_i} \tag{3.25}$$

The solution window for the independent variable (a) is found by:

$$a > a_L = \frac{z_1}{1 - z_1} \qquad\qquad a < a_R = \frac{1 - z_{N_C}}{z_{N_C}} \tag{3.26}$$

and initial guess by:

$$a_0 = a_L - \frac{F(a_L)}{m} \qquad\qquad m = \frac{F(a_R) - F(a_L)}{a_R - a_L} \tag{3.27}$$

At convergence, the fraction of phase 1 can be calculated from:

$$\beta_1 = \frac{c_1 + a c_{N_C}}{1 + a} \tag{3.28}$$

Nichita and Leibovici defined two additional functions that have favorable convergence properties in certain parts of the solution window. These are:

$$G(a) = \frac{a + 1}{a} F(a) \tag{3.29}$$

$$G'(a) = -\frac{z_1}{a^2} - \sum_{i=2}^{N_C - 1} \frac{z_i}{[d_i + a(1 + d_i)]^2} - z_{N_C}$$

and:

$$H(a) = -aG(a) \tag{3.30}$$

$$H'(a) = -z_1 - \sum_{i=2}^{N_C - 1} \frac{z_i[d_i(1 + a)^2 + a^2]}{[d_i + a(1 + d_i)]^2} - z_{N_C}(1 + 2a)$$

Convergence is achieved when the relative error is less than the required tolerance, which Nichita and Leibovici recommend as $\epsilon = 10^{-8}$.

$$S = \left|\frac{a^{j+1} - a^{j}}{a^{j}}\right| < \epsilon \tag{3.31}$$

The proposed algorithm is presented below:

Algorithm 3.3 – Two-phase constant-K flash (FGH algorithm)[9]

---

1. Calculate solution window, a_L and a_R
2. Evaluate $F(a_L)$ and $F(a_R)$
3. Calculate initial guess, a_0
4. Evaluate $F(a)$ and $F'(a)$
5. If $F'(a) > 0$ then go to step 9, else go to step 6
6. Perform Newton step:
   - $a^{j+1} = a^{j} - F(a^{j})/F'(a^{j})$
7. Calculate error. If $S < \epsilon$ then converged, else go to step 8
8. If $a^{j+1} \in (a_L, a_R)$ then go to step 4, else go to step 9
9. Perform Newton step. If $F(a) > 0$ then use $G(a^{j})$, else use $H(a^{j})$
   - $a^{j+1} = a^{j} - G(a^{j})/G'(a^{j})$ OR
   - $a^{j+1} = a^{j} - H(a^{j})/H'(a^{j})$
10. Return to step 4
11. Once converged, calculate β_k, y_{ik}

3.2.2.2. Multiphase Constant-K Flash

Returning to the general form of the Rachford-Rice formula for multiphase systems, we see that there are $(N_P - 1)$ equations that must be resolved:

$$R_k(\boldsymbol{\beta}) = \sum_{i=1}^{N_C}\left(y_{ik} - y_{iN_P}\right) = \sum_{i=1}^{N_C}\frac{z_i(K_{ik}-1)}{\sum_{k=1}^{N_P}\beta_k K_{ik}} = 0 \tag{3.21}$$

This was often done in the same manner presented as for the two-phase Rachford-Rice formula, with the residual and its derivative calculated for each $(N_P - 1)$ equations[11]. However, these formulations rely on good initial guesses for phase fractions and the satisfaction of multiple objectives. Some researchers continue to use this type of formulation[7,12,13], but most prefer the approach derived by Michelsen[14]. His approach converts the resolution of the Rachford-Rice formulas from a nonlinear root-finding method – where each additional phase adds another objective – to a more reliable single-objective minimization of a convex function. Okuno later proposed a slight reformulation of the Michelsen approach that is more conducive to negative flashes (i.e. convergence to phase fractions outside of the physical range)[15]. Negative flashes are common in reservoir simulation, as they provide an estimate of the proximity of the phase boundary and the composition of the phase that will form upon reaching that boundary. The negative flash was not necessary for this thesis, and Michelsen's algorithm was preferred to that of Okuno in this work.

Michelsen defined an objective function that, at its minimum, guarantees satisfaction of the equi-fugacity criterion, material balance and sum of mole fractions constraint. The objective function (Q), its gradient (g_k) and Hessian (H_{kL}) are given by:

$$Q(\boldsymbol{\beta}) = \sum_{k=1}^{N_P}\beta_k - \sum_{i=1}^{N_C} z_i \ln E_i \tag{3.32}$$

$$E_i = \sum_{k=1}^{N_P} \frac{\beta_k}{\hat{\phi}_{ik}} \qquad i = 1, N_C$$

$$\mathrm{g}_k = \frac{\partial Q}{\partial \beta_k} = 1 - \sum_{i=1}^{N_C} \frac{z_i}{E_i \hat{\phi}_{ik}} \qquad k = 1, N_P$$

$$H_{kL} = \frac{\partial \mathrm{g}_k}{\partial \beta_L} = \sum_{i=1}^{N_C} \frac{z_i}{E_i^2 \hat{\phi}_{ik} \hat{\phi}_{iL}} \qquad k, L = 1, N_P$$

At convergence, phase compositions are calculated by:

$$y_{ik} = \frac{z_i}{E_i \hat{\phi}_{ik}} \qquad \begin{array}{l} i = 1, N_C \\ k = 1, N_P \end{array} \tag{3.33}$$

The objective function is semi-convex with a unique minimum, and the Hessian is positive semi-definite, which means that convergence is guaranteed using Newton's method with line-search. Michelsen implements this search technique with a restricted-step Newton method, given by:

$$\boldsymbol{\beta}^{\mathrm{new}} = \boldsymbol{\beta} + \alpha \Delta \boldsymbol{\beta} \tag{3.34}$$

where α is a scalar multiplier bound between zero (no step) and one (full Newton step) that is set to constrain the vector of phase fractions to the physical region and to avoid overshooting the solution. The error of the gradient vector is a natural choice for the convergence criterion:

$$S = \sum_{k=1}^{N_P} |\mathrm{g}_k|^2 < \epsilon \tag{3.35}$$

Michelsen's multiphase constant-K flash algorithm is presented in the following:

Algorithm 3.4 – Multiphase constant-K flash (Michelsen algorithm)[14]

1. Set initial guess, $\boldsymbol{\beta}$
2. Evaluate $Q(\boldsymbol{\beta})$ and corresponding $\boldsymbol{E}$
3. Calculate gradient and Hessian of $Q(\boldsymbol{\beta})$: $\mathbf{g}$, $\boldsymbol{H}$
4. Calculate $\Delta\boldsymbol{\beta}$. Set $\alpha = 1$
5. Perform restricted Newton step:
    - $\boldsymbol{\beta}^{\text{new}} = \boldsymbol{\beta} + \alpha\Delta\boldsymbol{\beta}$
6. Check for negative element in $\boldsymbol{\beta}^{\text{new}}$. If any exist, set $\alpha = -\beta_k/\Delta\beta_k$ such that the restricted Newton step yields all non-negative phase amounts with one equal to zero. Perform restricted Newton step with new α
7. Evaluate $Q^{\text{new}}(\boldsymbol{\beta}^{\text{new}})$ and corresponding $\boldsymbol{E}^{\text{new}}$
8. If $Q^{\text{new}} < Q$ then the step is accepted and set:
    - $\boldsymbol{\beta} = \boldsymbol{\beta}^{\text{new}}$
    - $\boldsymbol{E} = \boldsymbol{E}^{\text{new}}$

    Else bisect α (set $\alpha = \alpha/2$), perform restricted Newton step with new α, and continue bisecting until $Q^{\text{new}} < Q$. When a reduction is observed, set:

    - $\boldsymbol{\beta} = \boldsymbol{\beta}^{\text{new}}$
    - $\boldsymbol{E} = \boldsymbol{E}^{\text{new}}$

*NOTE: If any element of $\boldsymbol{\beta}$ is zero, then deactivate the phase (set corresponding row element of $\mathbf{g}$ and row and column elements of $\boldsymbol{H}$ to zero, set diagonal element of $\boldsymbol{H}$ to 1)

9. Calculate error. If $S < \epsilon$ then go to step 10, else go to step 3
10. Calculate the gradient. Check the gradient element of all inactive phases. If no inactive phases have a negative $\mathbf{g}$, then go to step 11, else reactivate the phase with the most negative $\mathbf{g}$ and return to step 3
11. Calculate y_{ik} for converged $\boldsymbol{\beta}$

3.3. Saturation Point Calculation

Saturation points represent the thermodynamic conditions at which phase-splitting or phase-combining occur. Predicting saturation points accurately is important for the design of chemical processes as important phenomena – like fluid transport, fluid mixing, reactions, etc. – are strongly dependent on the nature of the phases present. If a single-phase fluid is desired, for example, then it is important to know the operating conditions at which a second fluid-phase might form, so that care can be taken to either mitigate or eliminate potential problems associated with phase-splitting.

Whereas for a PT-flash, the pressure, temperature, and overall composition are fixed, saturation point calculations are generally performed with one of the two key process variables (P or T) fixed and an additional specification for the onset phase amount. For example, the bubble pressure calculation – which is arguably the most popular of the saturation point calculations – is performed at fixed temperature and vapor phase fraction. The asphaltene onset pressure calculation is performed at fixed temperature and onset liquid fraction. In this section, a general framework for saturation point calculations is presented and application to two and three-phase systems is discussed. A fast and robust strategy for calculating all saturation pressures along an isotherm (or, conversely, all saturation temperatures along an isobar) is also proposed.

Like the PT-flash, saturation point calculations leverage the equilibrium relations, material balance and stability analysis. In fact, stability analysis could be used exclusively to solve the saturation point problem because its main objective is to determine whether a phase distribution is stable and if that phase distribution is revealed to be unstable, stability analysis also provides the composition of the destabilizing phase. One could solve the bubble pressure problem by starting at a low pressure and successively increasing pressure until stability analysis fails to find a vapor phase that destabilizes the mixture. This approach is reliable (or as reliable as the extent of the search space chosen for stability analysis) but computationally inefficient, so instead focus is placed on efficient and robust methods for finding saturation points.

Recall from the PT-flash algorithm that the Gibbs minimization problem is solved by calculating partition coefficients in an outer loop and resolving the material balance in the inner loop (constant-K flash). For the saturation point calculations, both the partition coefficients and search variable (usually P or T) are adjusted in the outer loop and the material balance is resolved in the inner loop.

3.3.1. Problem Formulation

From the equilibrium relations and material balances, the following system of equations holds at saturation points:

$$\ln K_{ik} + \ln \hat{\phi}_{iN_P} - \ln \hat{\phi}_{ik} = 0 \qquad \begin{array}{l} i = 1, N_C \\ k = 1, (N_P - 1) \end{array} \tag{3.36}$$

$$R_k = \sum_{i=1}^{N_C} \left(y_{ik} - y_{iN_P} \right) = 0 \qquad k = 1, (N_P - 1)$$

Additionally, one phase fraction and process variable must be specified. This formulation resolves $N_C(N_P - 1)$ equilibrium relations and $(N_P - 1)$ material balances. Michelsen formulated the solution of this system of equations as a multivariable Newton-Raphson (MVNR) with the specification variable built in to the system of equations. His original formulation was for two-phase saturation lines[16]. This was later extended by Lindeloff and Michelsen[17,18] to hydrocarbon-water systems and by Agger and Sorensen[19] to asphaltene P-T and P-x diagrams. These more recent formulations are restricted to three phases, but the extension to any number of phases is obvious from the three-phase formulations.

The MVNR root-finding approach requires that an objective vector and independent variable vector be defined. These follow from Eq. (3.36):

$$g_i = \ln K_{ik} + \ln \hat{\phi}_{iN_P} - \ln \hat{\phi}_{ik} = 0 \qquad \begin{array}{l} i = 1, N_C \\ k = 1, (N_P - 1) \end{array} \tag{3.37}$$

$$g_{N_C(N_P-1)+k} = \sum_{i=1}^{N_C} \left(y_{ik} - y_{iN_P} \right) = 0 \qquad k = 1, (N_P - 1)$$

$$g_{N_C(N_P-1)+(N_P-1)} = u_{iS} - S = 0$$

where the last equation is the specification equation and S is the value of the specified variable. The vector of independent variables is then given by:

$$\boldsymbol{u} = [\ln K_{ik}, \beta_k, \ln T, \ln P] \qquad \begin{array}{l} i = 1, N_C \\ k = 1, (N_P - 1) \end{array} \tag{3.38}$$

where β_k is the fraction of the fixed phase. Michelsen suggested using natural logarithms for the set of independent variables to provide a more uniform scaling between unlike variables like K_i, T, P.

Clearly, this formulation does not require that the fixed phase be a true saturation point ($\beta_k = 0$), and the specification equation can be any one of the independent variables. Michelsen leveraged these factors to propose a Newton algorithm for tracing the full PT-diagram. In fact, what makes his phase-tracing algorithm so robust is its ability to trace saturation lines by specifying the most sensitive variable at each point along the curve (which is often one of the partition coefficients) and solving the system of equations. Whereas the conventional specification of pressure or temperature fails in the critical region, the specification of one of the partition coefficients faces no such issue.

The MVNR approach should only be used with very good initial guesses for the vector of independent variables. The saturation line tracing algorithm proposed by Michelsen guarantees good initial guesses by using the converged values at the previous point to initialize the independent variables at the next point. When good initial guesses are not available, which is the case for “blind” saturation point calculations, then the MVNR approach is unreliable. Instead, a somewhat hybrid approach with a successive substitution scheme to converge partition coefficients in an inner loop and a restricted step Newton update for the search variable in an outer loop provides more reliable convergence behavior.

3.3.2. Two-Phase Saturation Points

Two-phase saturation points are considerably easier to locate than multiphase saturation points, mainly due to a simplifying of the material balance equations: For two-phase equilibrium, the material balance comes from the classical Rachford-Rice formula:

$$R(\beta_1) = \sum_{i=1}^{N_C} \frac{z_i(K_i - 1)}{1 + \beta_1(K_i - 1)} = 0 \tag{3.22}$$

When the fraction of phase 1 is assumed to be zero, the material balance and composition calculations simplify to:

$$R(\beta_1 = 0) = \sum_{i=1}^{N_C} z_i(K_i - 1) = 0 \tag{3.39}$$

$$y_{i1} = K_i y_{i2} = z_i K_i \tag{3.40}$$

where y_{i1} is the composition of component i in the onset phase and y_{i2} is the composition of component i in the bulk phase. Clearly, the bulk composition (y_{i2}) must equal the overall composition (z_i) to satisfy the material balance and both phase fractions are known when one phase fraction is specified. These factors greatly simplify the two-phase saturation point calculation.

Because the phase fractions are known in advance, there is no need to resolve the Rachford-Rice formula for phase fractions as was done in the two-phase constant-K flash. Instead, the Rachford-Rice formula is differentiated with respect to the search variable. Assuming pressure is the search variable and temperature is fixed yields:

$$R'(P) = \sum_{i=1}^{N_C} z_i \frac{\partial K_i}{\partial P} = \sum_{i=1}^{N_C} z_i K_i \frac{\partial \ln K_i}{\partial P} = \sum_{i=1}^{N_C} z_i K_i \left(\frac{\partial \ln \hat{\phi}_{i2}}{\partial P} - \frac{\partial \ln \hat{\phi}_{i1}}{\partial P} \right) \tag{3.41}$$

This equation is true only for the form of the Rachford-Rice in which the fraction of phase 1 is zero. Other specifications (such as $\beta_1 = 1$) will yield different simplified forms of the Rachford-Rice and its derivative.

The pressure guess for subsequent iterations can be calculated with a Newton step by:

$$P^{j+1} = P^j - \frac{R}{R'(P^j)} \tag{3.42}$$

or:

$$\ln P^{j+1} = \ln P^j - \frac{R}{R'(\ln P^j)} \tag{3.43}$$

A natural choice for convergence criteria of the outer loop is the size of the step change in the search variable:

$$S = \left|\frac{R}{R'}\right| < \epsilon \tag{3.44}$$

The primary challenge in the two-phase saturation point calculation comes from the initialization of both the search variable and the composition of the onset phase. Bubble pressure calculations are usually initialized at low pressure and with vapor phase compositions calculated from the Wilson partition coefficient equation. However, what constitutes a low pressure is highly dependent on the components present in the mixture, and the initial guess determination can itself be an iterative procedure. For now, we will assume that good initial guesses are provided and present the algorithm for two-phase saturation points:

Algorithm 3.5 – Two-Phase Saturation Pressure Calculation

1. Set initial pressure P and partition coefficients $\{K_i\}$
2. Calculate bulk phase properties at $P, \{z_i\}$ from EOS
    - $\left[\ln \hat{\phi}_{i2}, {}^{\partial \ln \hat{\phi}_{i2}}/_{\partial P}\right]$ = EOS(T, P, z_i)
3. Calculate onset phase composition with current $\{K_i\}$
4. Calculate onset phase properties at $P, \{y_{i1}\}$ from EOS
    - $\left[\ln \hat{\phi}_{i1}, {}^{\partial \ln \hat{\phi}_{i1}}/_{\partial P}\right]$ = EOS(T, P, y_{i1})
5. Update partition coefficients and calculate partition coefficient derivatives
    - $\ln K_i = \ln \hat{\phi}_{i2} - \ln \hat{\phi}_{i1}$
6. Repeat steps 3-5 until $\sum y_{i1}$ is roughly equal in current and previous iterations.

NOTE: step 6 is often unnecessary and, if performed, convergence of $\{y_{i1}\}$ does not need to be tight

7. Calculate material balance and its derivative at current $P, \{K_i\}$
8. Calculate error. If $S < \epsilon$ then converged, else update P and return to step 2

During early iterations, it is advisable to control the magnitude of the search variable update because the Newton method is prone to overshooting when far from the solution. Thus, the Newton method is trusted to provide the correct search direction but not necessarily a reasonable magnitude when far from the solution. During the work performed for this thesis, search variable updates were constrained to be no more than 25% of the current value during the first 5 iterations. If the pressure changes are still too large after these iterations, then the calculation is aborted.

Additionally, a check on the partition coefficients $\{K_i\}$ is put in place to reveal if convergence to the trivial solution is occurring. Unfortunately, this check cannot distinguish between a saturation point in the neighborhood of the critical point and a trivial solution. Of course, there are checks that could be performed – such as a PT-flash near the solution – that could distinguish these two, but the computational effort is rarely worth the extremely small number of cases for which this algorithm would cause problems. When non-convergence occurs, a new set of initial conditions can be set and the algorithm restarted. Initial guesses provided by stability analysis almost always lead to successful convergence to non-trivial solutions, so this is the preferred method for initializing partition coefficients.

The algorithm for two-phase saturation temperature is identical to that for saturation pressure, except temperature is the search variable and pressure is the fixed variable. Also, the derivatives of the Rachford-Rice equation require temperature as the differentiation variable instead of pressure. This algorithm for two-phase saturation points – as opposed to the standard algorithms found in introductory thermodynamics texts – is applicable to both VLE and LLE. The standard methods often use update formulas that respond well for VLE problems but not LLE.

The primary difference between a two-phase bubble pressure calculation (vapor-liquid) and two-phase asphaltene onset calculation (liquid-liquid) is in the initialization of pressure and the partition coefficients. Once these have been initialized, both calculations can use the two-phase saturation pressure calculation outlined above because the pressure dependence on the partition coefficients is stated explicitly by the partial derivatives of the fugacity coefficient with respect to pressure. For some two-phase boundary search algorithms, where the update formulas do not include this dependence, convergence is intolerably slow or does not occur for liquid-liquid systems.

3.3.3. New Algorithm for Fast and Robust Determination of Saturation Pressures at Fixed Temperature

The algorithms for phase split and saturation boundary calculations presented in this chapter rely on good initial estimates for phase compositions or, equivalently, fugacity or partition coefficients. Stability analysis provides a reliable method for generating good initial estimates, but it is not as straightforward to use stability analysis for phase boundary calculations as it is for the PT-flash calculation. Standard stability analysis calculations are performed at fixed pressure and temperature, which is the specification of the PT-flash, but the phase boundary calculation generally fixes only one of these variables and searches for the value of the other that gives a saturation point.

For the two-phase bubble pressure problem, performing stability analysis at low pressure and the temperature of interest is usually sufficient to initialize compositions. Of course, the nature of the fluid mixture determines how careful one must be when choosing a good initial guess for pressure. Wide-boiling fluids like crude oils – with heavy components that strongly prefer the liquid state and light components that strongly prefer the gaseous state – do not require much care. Any starting pressure for stability analysis from about 1 atm to 50 atm is likely to be sufficient for stability analysis. With a narrow-boiling fluid, the initialization must be more robust. It may be appropriate to scan until a pressure is found at which one of the trial phases verifies phase instability. Once this phase is found, the two-phase bubble pressure search can commence.

However, there is no guarantee that the bubble pressure found from the two-phase algorithm is the true bubble pressure. We have assumed vapor-liquid equilibrium to do this calculation, but it might be the case that two (or more) liquid phases should exist at the actual bubble pressure. Though the two-phase bubble pressure might satisfy all necessary conditions for equilibrium (equi-fugacity, material balance, etc.), the minimum Gibbs energy for that pressure and temperature might correspond to three-phases (or more), in which case the two-phase bubble pressure is not the true bubble pressure.

To verify whether the two-phase bubble pressure is the true bubble pressure, stability analysis is performed at this pressure. If stability analysis verifies that no other liquids can form, then the two-phase bubble pressure is the true bubble pressure. If, however, stability analysis finds a new phase that would lower the Gibbs energy, then a three-phase bubble pressure calculation commences with the initial guesses given by the two-phase bubble pressure, the converged compositions for the two-phase bubble pressure calculation, and the composition for the destabilizing phase found from stability analysis. Additionally, the existence of two (or more) liquids at the bubble pressure implies that there will also be saturation pressures for liquid-liquid phase splitting (or combining) at pressures above and below the two-phase bubble pressure. For this reason, the two-phase bubble pressure (called pseudo-VLE or denoted P^{VLE} for brevity) performs a very important role.

First, the pseudo-VLE pressure is relatively easy to find for many crude oil mixtures. Temperature specifications in the dew region or near the critical point (which are rarely known a priori) cause problems for any saturation pressure algorithm, but heavy crude oils containing asphaltenes rarely have critical points in the temperature range of interest, so problems rarely appear in practice. Second, the pseudo-VLE pressure is the true bubble pressure if stability analysis verifies the two-phase solution is stable. Even if stability analysis verifies that the pseudo-VLE is NOT the solution, it has still provided a good starting pressure and compositions for subsequent calculations.

For asphaltic crude oils in the bubble region, for which this algorithm has been thoroughly tested, there are often one or three saturation pressures of interest at a fixed temperature. If no liquid phases can be initialized at the pseudo-VLE to destabilize the mixture, then the pseudo-VLE is verified as the true bubble pressure and no other phase splits are likely to be found. If a second liquid is found at the pseudo-VLE, then there are three saturation pressures that can be found and the initial guesses for pressure and phase compositions will have already been calculated from the pseudo-VLE and stability analysis calculations. The proposed algorithm for calculating these three saturation points, which are, in order of ascending pressure, the lower asphaltene onset pressure (LAOP), bubble pressure (BP) and upper asphaltene onset pressure (UAOP), is given below. The LAOP and BP calculation require the three-phase saturation pressure algorithm, while the two-phase algorithm is sufficient for the UAOP.

Algorithm 3.6 – New Saturation Pressure Algorithm for LAOP, BP, UAOP

1. Set initial pressure P and partition coefficients $\{K_i\}$
   - Choose P at which vapor and liquid are guaranteed to exist. Atmospheric pressure is usually sufficient for crude oils
2. Solve two-phase saturation pressure algorithm to obtain P^{VLE} and converged phase compositions, $\{y_{iV}, y_{iL}\}$
3. Run stability analysis at P^{VLE}. If stability analysis reveals a trial phase $\{W_i\}$ that destabilizes $\{y_{iV}, y_{iL}\}$, then go to step 4, else P^{VLE} is the true bubble pressure (BP), report no LAOP and UAOP then STOP
4. Solve three-phase saturation pressure algorithm to find LAOP
   - Initialize with P^{VLE} and $\{y_{iV}, y_{iL}, W_i\}$
   - The phase amount of $\{W_i\}$ is zero at LAOP
5. Solve three-phase saturation pressure algorithm to find BP
   - Initialize with P^{VLE} and $\{y_{iV}, y_{iL}, W_i\}$
   - The phase amount of $\{y_{iV}\}$ is zero at BP
6. Solve two-phase saturation pressure algorithm to find UAOP
   - Initialize with P^{VLE} and $\{y_{iL}, W_i\}$
   - The phase amount of $\{W_i\}$ is zero at UAOP

For asphaltic crude oils, such as those that are studied in Chapter 5, no cases have been found that this algorithm fails to detect. If two liquids exist at the bubble pressure, then the pseudo-VLE pressure has shown to ALWAYS lie above the true bubble pressure. In **Figure 3.2**, a plot for saturation pressures as a function of gas injection amount at fixed temperature for crude oil C2 shows the behavior that has been observed for all crude oils containing asphaltenes. When stability analysis fails to find the presence of a second liquid phase at P^{VLE}, then P^{VLE} is equal to the true bubble pressure. When stability analysis does find a second phase, the proximity of the true bubble pressure to P^{VLE} is a function of the distance between the UAOP and LAOP curves. Additionally, the difference between the pseudo-VLE pressure and true bubble pressure is a function of the amount of the second phase present. When the amount of the onset phase is small, which is the case for crude oils U8, J1, and S14, the true bubble pressure curve and pseudo-VLE curve show very small differences. When the amount of the onset phase is large, as is the case for C2, the pseudo-VLE curve deviates more strongly.

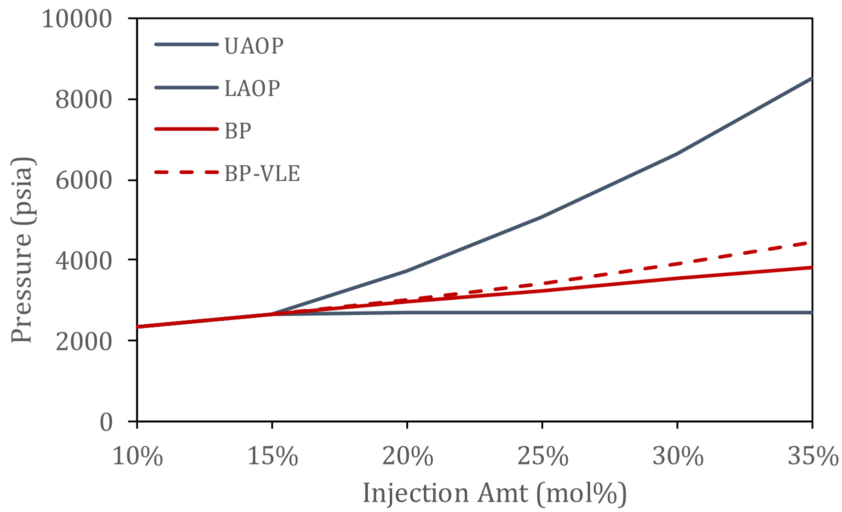


Figure 3.2. Saturation pressures for crude oil C2 as a function of gas injection at 200 °F. The pseudo-VLE pressure line (BP-VLE) lies on the true bubble pressure line when asphaltenes are stable and deviates from the line as the separation of the UAOP and LAOP lines grows.

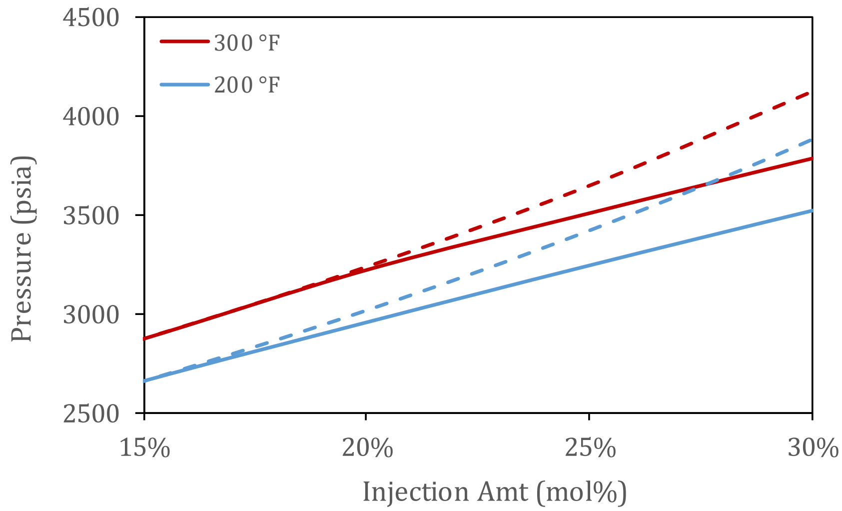


Figure 3.3. True bubble pressure (solid) and pseudo-VLE pressure (dashed) as a function of gas injection at low and high temperature (200 °F, 300 °F) for crude oil C2. The pseudo-VLE line deviates further from the BP line as onset phase amount increases.

Chapter 4

Equations of State

4.1. Introduction

One of the primary focuses of this work is a new equation of state (EOS) designed to calculate the equilibrium properties of nonpolar chain fluids by hybridizing the classical cubic equation of state[20–22] with the chain equation of state proposed by Chapman[23]. A similar hybrid approach originally proposed by Kontogeorgis[24] and modified by Li and Firoozabadi[25] combines the classical cubics with the association equation of state proposed by Chapman. Whereas the Kontogeorgis and Li and Firoozabadi approaches are designed to handle associating fluids like water and alcohols, the chain equation of state is designed for nonpolar chain-like molecules, which fits the description of polymers and is the class of components the cubic-plus-chain (CPC) equation of state is designed to handle.

In this chapter, we develop a framework for a generic equation of state, recognizing them primarily as generating functions for thermodynamic property calculations, and then address specifically the cubic equations of state, the perturbed chain version of the Statistical Associating Fluid Theory (PC-SAFT) equation of state, and the cubic-plus-association (CPA) model. These discussions will also complement the introduction of the new CPC equation of state in Chapter 6, which draws on both the cubic and SAFT EOS models. The discussion of PC-SAFT will complement the thermodynamic modeling of crude oils discussed in Chapter 5.

4.2. General Framework for Equations of State

As the preceding chapters have made clear, the calculation of the fugacity coefficients (or, equivalently, the fugacities or chemical potentials) and their derivatives play a central role in solving phase equilibrium problems. The general thermodynamics framework presented in Chapter 2 and the algorithms for solving phase equilibrium problems presented in Chapter 3 are applicable to any EOS, and the various equations of state will merely yield different results for equilibrium properties when used to solve the algorithms presented in Chapter 3.

Equations of state for fluid mixtures are commonly expressed in the residual Helmholtz form (A^R) or pressure form (P), which are related by the following equivalent expressions:

$$A^R(T,V,\boldsymbol{n}) = -\int_{\infty}^{V}\left(P - \frac{nRT}{V}\right)dV \tag{4.1}$$

$$P - \frac{nRT}{V} = -\left(\frac{\partial A^R(T,V,\boldsymbol{n})}{\partial V}\right)_{T,\boldsymbol{n}} \tag{4.2}$$

where nRT/V accounts for the ideal gas contribution to the Helmholtz and pressure equations and the other terms account for deviations from ideal gas behavior.

The standard practice for programming equations of state is to express them in the form of the residual Helmholtz, which is then differentiated with respect to the independent variables (temperature T, volume V, and mole numbers $\{n_i\}$) to obtain all thermodynamic properties. A few of the properties of interest that can be calculated from derivatives of the residual Helmholtz are:

$$Z = 1 - V\left(\frac{\partial(A^R/nRT)}{\partial V}\right)_{T,\boldsymbol{n}} \tag{4.3}$$

$$\ln\hat{\phi}_i = \left(\frac{\partial(A^R/RT)}{\partial n_i}\right)_{T,V} - \ln Z$$

$$\left(\frac{\partial P}{\partial V}\right)_{T,\boldsymbol{n}} = -\frac{nRT}{V^2} - \left(\frac{\partial^2 A^R}{\partial V^2}\right)_T$$

Though it is common to see equations of state expressed in terms of temperature T, density ρ, and composition $\{x_i\}$, it is often preferable to express the functional dependencies of the Helmholtz energy in terms of $(T, V, \mathbf{n})$ instead of $(T, \rho, \mathbf{x})$ due to the rules of differentiation. In differential calculus, a partial derivative is obtained by varying one independent variable while holding all other independent variables constant and measuring the change in the value of the dependent variable. If a function f can be calculated from the independent variables $(\mathrm{x}, \mathrm{y}, \mathrm{z})$, then the partial derivative of that function with respect to x is given by:

$$\left(\frac{\partial f(\mathrm{x},\mathrm{y},\mathrm{z})}{\partial \mathrm{x}}\right)_{\mathrm{y},\mathrm{z}} = \lim_{\Delta\mathrm{x}\to 0}\left[\frac{f(\mathrm{x}+\Delta\mathrm{x},\mathrm{y},\mathrm{z}) - f(\mathrm{x},\mathrm{y},\mathrm{z})}{\Delta\mathrm{x}}\right]$$

One such complication in specifying $(T, \rho, \mathbf{x})$ comes from composition derivatives. Because compositions are required to sum to one, varying a single composition forces all other compositions to change in order to satisfy the constraint. Though it is mathematically possible to differentiate a function with respect to a single composition, it violates the rules of differentiation and the result is not physically meaningful. The second complication is that density is a function of both $\mathbf{n}$ and V. Thus, in a partial derivative of the Helmholtz function, holding density constant amounts to holding both the mole numbers and total volume constant (or allowing them to vary proportionally). Because the set $(T, V, \mathbf{n})$ are independent of other variables and independent of each other, it is preferred to express the equation of state in terms of these variables.

4.3. Intermolecular Forces

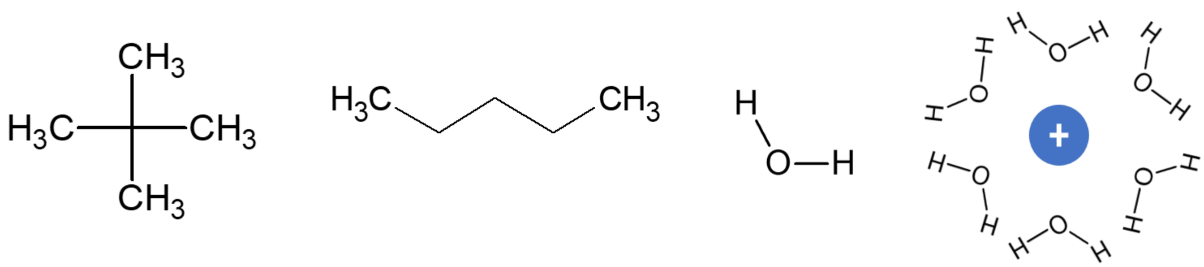
The ideal gas law was the first equation of state relating pressure P, volume V, and temperature T of dilute gases:

$$P = \frac{nRT}{V} = \frac{RT}{\hat{V}} \tag{4.4}$$

The assumptions implicit in the ideal gas law are that molecules have an infinitesimally small volume in relation to the total system volume V and that the molecules are sufficiently far apart that they do not act on each other. Consequently, the ideal gas law cannot describe condensation, liquid-liquid instability, critical points, etc. because these phenomena require that molecules interact strongly enough to form liquids. Dilute gases at low pressure adhere closely to the ideal gas law, but these assumptions break down at even moderate temperatures and pressures, where molecular size and interactions become important.

Intermolecular forces arise due to electric fields produced by charged electron clouds. The magnitude of the forces is largely dependent on the intermolecular distance, the shape of the molecules, and the types of elements that comprise them. **Figure 4.1** shows a few different components and their normal boiling points, which are used to illustrate the differences in magnitude of various intermolecular forces. The normal boiling temperature (measured at 1 atm) is a means of quantifying the thermal energy required to overcome the intermolecular attractions that keep molecules in the liquid state. The more energy (higher temperature) required to vaporize a component, the larger the magnitude of the attractive forces keeping it in the liquid state.

Neopentane | N-pentane | Water | Salt Water



$T_{\mathrm{nbp}} = 10$ °C | $T_{\mathrm{nbp}} = 36$ °C | $T_{\mathrm{nbp}} = 100$ °C | $T_{\mathrm{nbp}} > 100$ °C

Figure 4.1. Normal boiling point temperatures for selected pure fluids illustrating the effect of intermolecular forces on phase behavior.

The dominant forces driving phase behavior in nonpolar molecules like neopentane and n-pentane are dispersion. These forces occur due to weak dipoles induced by surrounding nonpolar molecules and are the weakest of the intermolecular forces. All molecules experience dispersion forces, but they are particularly important for describing the phase behavior of nonpolar substances as these substances do not interact by the stronger dipole forces that exist in water or alcohols.

The boiling point difference between neopentane and n-pentane (which have identical molecular weights) is due to charge shielding. The elements that comprise neopentane are arranged such that they shield a certain amount of force that would otherwise increase the strength of interactions with nearby molecules. This charge-shielding effect is much less pronounced in n-pentane as the electron cloud is distributed along the length of the molecule, so its normal boiling temperature is higher than that of neopentane.

Both neopentane and n-pentane show lower boiling temperatures than smaller molecules like water because dispersion forces are weaker than the polar forces that dominate phase behavior for water. Polar forces arise due to permanent dipoles, which are in general stronger than dipole-induced or ion-induced dipoles which are then stronger than standard dispersion forces. Water self-association is an example of dipole-dipole interaction, but we often differentiate it from standard dipole-dipole forces because hydrogen bonding is a particularly strong form of dipole-dipole bonding. Ion-dipole forces are even stronger than standard dipole-dipole forces, which is shown in the example of protonated water in **Figure 4.1**.

The importance in differentiating between these various types of interactions is evident from the normal boiling temperatures. If an equation of state is expected to describe the thermodynamic properties of both nonpolar substances like n-pentane and polar substances like water, then it must be able to capture the physical phenomena that manifests in a smaller molecule like water showing much stronger interactions that keep it in the liquid state at temperatures for which the larger n-pentane would have already vaporized. Clearly, this means that the expression for calculating the pressure of a fluid exhibiting hydrogen-bonding is going to be different for that of a nonpolar system which is going to be different than that of an ideal gas at the same conditions of temperature and volume. The goal of an equation of state is to describe mathematically these contributions to the residual Helmholtz such that derivatives of the residual Helmholtz yield accurate values of pressure, partial molar properties, fugacities, etc.

4.4. Equation of State Models

4.4.1. Cubic Equations of State

The cubic equations of state (EOS) are arguably the most popular class of thermodynamic models for describing the phase behavior of nonpolar hydrocarbon systems. Despite significant research efforts in the past few decades to improve the accuracy of thermodynamic models for hydrocarbon systems, cubic EOS are still widely used because of their unique combination of simplicity, sufficient accuracy for many applications, acceptable computational speed, and their broad availability across various industrial thermodynamics software packages.

The first cubic EOS was proposed by Johannes van der Waals[26] and was the first EOS capable of modeling complex VLE and LLE phenomena. The van der Waals (VDW) EOS regards liquids as compressed gases and proposes a single equation with multiple volume roots to describe both vapor and liquid phases. The VDW EOS captures deviations from ideal gas behavior due to attractive and repulsive molecular interactions, relaxing both assumptions implicit in the ideal gas law. VDW in pressure-explicit form is given by the following:

$$P = \frac{nRT}{V - B} - \frac{A}{V^2} \tag{4.5}$$

where A and B are calculated by mixing rules that average the pure-component attraction energy (a_i) and excluded volumes (b_i).

$$A = \sum_i \sum_j n_i n_j \sqrt{\mathrm{a}_i \mathrm{a}_j}\left(1 - k_{ij}\right) \qquad B = \sum_i n_i b_i \tag{4.6}$$

Rearranging VDW in terms of volume yields:

$$V^3 - \left(B + \frac{nRT}{P}\right)V^2 + \frac{A}{P}V - \frac{AB}{P} = 0 \tag{4.7}$$

The third-order polynomial in volume is the defining characteristic of all cubic equations of state.

Subsequent equations of state based on the original VDW have since been developed. These more recent cubic equations of state have added a temperature-dependence term on the molecular attraction parameter (a_i) and a new tuning parameter, known as the acentric factor (ω_i), that improves saturation pressure predictions away from the critical point. The two most popular contemporary models of the cubic class are the Soave-Redlich-Kwong (SRK)[20,21] and Peng-Robinson (PR)[22] equations of state. Instead of providing explicit expressions for SRK and PR, a generalized cubic EOS is presented that can be applied to any cubic EOS.

$$P = \frac{nRT}{V-B} - \frac{A}{(V+\delta_1 B)(V+\delta_2 B)} \tag{4.8}$$

with compressibility factor given by:

$$Z = \frac{1}{1-\beta} - \frac{A}{BRT}\frac{\beta}{(1+\delta_1\beta)(1-\delta_2\beta)} \tag{4.9}$$

where $\beta = B/V$. The coefficients of the pressure and compressibility factor equations are dependent on the EOS, and the parameters for SRK and PR in the generalized cubic EOS are given in **Table 4.1**. The pure-component attraction energy (a_i) and excluded volumes (b_i) are calculated by:

$$\mathrm{a}_i = \Omega_\mathrm{a}\frac{R^2T_c^2}{P_c}\alpha \tag{4.10}$$

$$\alpha = \left[1+\kappa\left(1-\sqrt{T/T_c}\right)\right]^2$$

$$b_i = \Omega_b\frac{RT_c}{P_c}$$

The reduced Helmholtz for the generalized cubic EOS is obtained by integrating the pressure equation (Eq. (4.1)). In terms of the repulsive (rep) and attractive (att) contributions, the residual reduced Helmholtz of the cubic EOS is written:

$$F = \frac{A^R(T,V,\boldsymbol{n})}{RT} = a^{\mathrm{rep}} + a^{\mathrm{att}} \tag{4.11}$$

$$a^{\mathrm{rep}} = -n\ln(1-\beta)$$

$$a^{\mathrm{att}} = -\frac{A}{BRT(\delta_1-\delta_2)}\ln\frac{1+\delta_1\beta}{1+\delta_2\beta}$$

The derivatives of the cubic EOS Helmholtz function are presented in Appendix B.

Table 4.1. Constants for selected cubic equations of state.

EOS	$\boldsymbol{\delta_1}$	$\boldsymbol{\delta_2}$	$\boldsymbol{\Omega_a}$	$\boldsymbol{\Omega_b}$	$\boldsymbol{\kappa(\omega)}$
SRK	1	0	0.42748	0.08664	$0.480 + 1.574\omega - 0.176\omega^2$
PR	$1+\sqrt{2}$	$1-\sqrt{2}$	0.45724	0.07780	$0.37464 + 1.54226\omega - 0.26992\omega^2$

4.4.2. PC-SAFT Equation of State

SAFT-type EOS models are based on Wertheim's first order perturbation theory[28–31] that was derived for fluids comprised of molecules that tend to associate due to directional attractions. Chapman et al. made practical use of Wertheim's theory to propose an EOS for chain molecules with strong directional forces, which they called the Statistical Associating Fluid Theory (SAFT)[32,33]. The physical basis for this model is presented in **Figure 4.2**. The reference fluid is a group of hard spheres that can form chains through covalent bonding, hydrogen bond between terminal sites of different chains, and interact through weak dispersion forces.

The residual Helmholtz energy for the SAFT EOS is written as separate contributions from the reference and perturbation terms.

$$F = \frac{A^R(T,V,\boldsymbol{n})}{RT} = a^{\text{ref}} + a^{\text{pert}} = a^{\text{hs}} + a^{\text{chain}} + a^{\text{assoc}} + a^{\text{disp}} \tag{4.12}$$

Most variants of the SAFT EOS retain this fundamental form of the EOS, though some may add terms (such as a dipole-dipole or quadrupole-quadrupole or ion-dipole etc.) or drop terms (association is neglected for nonpolar fluids) or modify the radial distribution function or dispersion term.

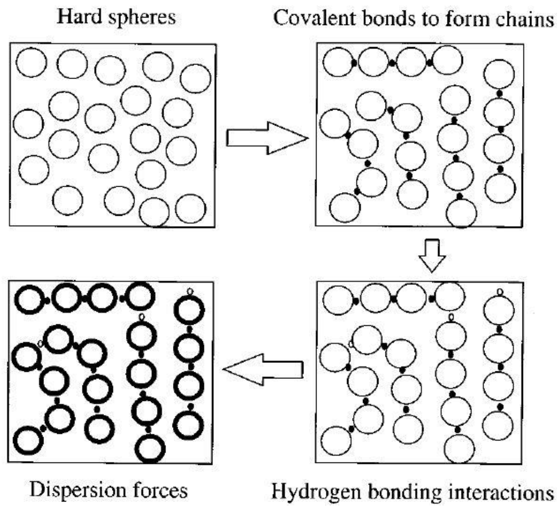


Figure 4.2. Physical basis for the SAFT model.[34]

Further development of SAFT models have led to many variations of the original EOS proposed by Chapman et al. The most popular of these models for predicting crude oil phase behavior was proposed by Gross and Sadowski and is known as Perturbed-Chain Statistical Associating Fluid Theory (PC-SAFT)[35–38]. This model has shown excellent results in predicting asphaltene phase behavior in complex crude oil mixtures and is thus an appropriate model for the work discussed in Chapter 5.

PC-SAFT retains the same fundamental form of the EOS as the original SAFT with only the dispersion term modified. SAFT applies its dispersion term to the system of hard-spheres while PC-SAFT applies its dispersion term to the system of hard-chains. The pair potential is also different from SAFT, where PC-SAFT uses the modified square-well potential of Chen and Kreglewski[39] with constant well width. The pair potential ($u(r)$) in PC-SAFT is given by:

$$u(r) = \begin{cases} \infty & r < (\sigma - s_1) \\ 3\epsilon & (\sigma - s_1) \leq r < \sigma \\ -\epsilon & \sigma \leq r < \lambda\sigma \\ 0 & r \geq \lambda\sigma \end{cases}$$

where r is the radial distance between two segments, σ is the temperature-independent segment diameter, and λ is the reduced well width. Chen and Kreglewski suggested a ratio of s_1/σ = 0.12 be used, yielding for the temperature-dependent segment diameter:

$$d(T) = \sigma\left[1 - 0.12\exp\left(-\frac{3\epsilon}{kT}\right)\right]$$

The hard-sphere (hs) term was first proposed by Carnahan and Starling[40–42] for a homogenous fluid and then extended to a mixture of hard spheres by Mansoori et al.[43]

$$a^{\text{hs}} = \frac{6V}{\pi}\left[\frac{3\zeta_1\zeta_2}{1-\zeta_3} + \frac{{\zeta_2}^3}{(1-\zeta_3)^2\zeta_3} + \left(\frac{{\zeta_2}^3}{{\zeta_3}^2} - \zeta_0\right)\ln(1-\zeta_3)\right] \tag{4.13}$$

The chain term was derived by solving Chapman's association term in the limit of infinite association strength.

$$a^{\text{chain}} = -\sum_{i=1}^{N_C} n_i(m_i - 1)\ln \mathrm{g}_{ii}^{\text{hs}}(\sigma_{ii}) \tag{4.14}$$

and the association term is written as a function of the unbonded associating sites, X_{A_i}.

$$a^{\text{assoc}} = \sum_{i}^{N_C} n_i \sum_{A_i}^{S(i)}\left[\ln X_{A_i} - \frac{1}{2}X_{A_i} + \frac{1}{2}\right] \tag{4.15}$$

where $S(i)$ is the total number of associating sites $(A, B, C, \ldots)$ on molecule i.

The dispersion term for PC-SAFT was derived using a Barker and Henderson[44–46] second-order contribution extended to chain molecules by Chiew[47]. This was then integrated with the Chen and Kreglewski pair potential to yield:

$$a_1 = -2\pi\rho m^2 \left(\frac{\epsilon}{kT}\right) \sigma^3 \int_1^{\infty} u(x) \mathrm{g}^{\mathrm{hc}} x^2 dx$$

$$a_2 = -\pi\rho m \left(1 + Z^{\mathrm{hc}} + \rho \frac{\partial Z^{\mathrm{hc}}}{\partial \rho}\right)^{-1} m^2 \left(\frac{\epsilon}{kT}\right)^2 \sigma^3 \frac{\partial}{\partial \rho}\left[\rho \int_1^{\infty} u(x)^2 \mathrm{g}^{\mathrm{hc}} x^2 dx\right]$$

$$I_1 = \int_1^{\infty} u(x) \mathrm{g}^{\mathrm{hc}} x^2 dx = \sum_{i=0}^{6} a_i \eta^i \qquad I_2 = \frac{\partial}{\partial \rho}\left[\rho \int_1^{\infty} u(x)^2 \mathrm{g}^{\mathrm{hc}} x^2 dx\right] = \sum_{i=0}^{6} b_i \eta^i$$

where reduced density (η) is bounded between 0 for an ideal gas and 0.74 for the highest allowable packing fraction. The coefficients a_i and b_i depend on chain length and constants fit to pure-component data for n-alkanes. These expressions for the first- and second-order dispersion terms were then extended to mixtures with the application of the van der Waals one-fluid mixing rule:

$$a_1 = -\frac{2\pi}{V} I_1 \sum_i \sum_j n_i n_j m_i m_j \left(\frac{\epsilon_{ij}}{kT}\right) {\sigma_{ij}}^3 \tag{4.16}$$

$$a_2 = -\frac{\pi \overline{m}}{V} C_1 I_2 \sum_i \sum_j n_i n_j m_i m_j \left(\frac{\epsilon_{ij}}{kT}\right)^2 {\sigma_{ij}}^3 \tag{4.17}$$

The total dispersion term is then a summation of the contributions:

$$a^{\mathrm{disp}} = a_1 + a_2 \tag{4.18}$$

4.4.3. CPA Equation of State

The cubic-plus-association (CPA) equation of state is a combination of the classical cubic EOS and the SAFT association term. In the form of the reduced residual Helmholtz, CPA is given as a physical contribution (cubic EOS) and chemical contribution (association). Some researchers prefer to use SRK for the physical contribution[48] while others prefer PR[25].

$$F = \frac{A^R(T, V, \boldsymbol{n})}{RT} = a^{\text{rep}} + a^{\text{att}} + a^{\text{assoc}} \tag{4.19}$$

The reduced residual Helmholtz for the cubic is given by the repulsive (rep) and attractive (att) contributions given in Eq. (4.11), and association retains the same expression as shown for PC-SAFT (Eq. (4.15)), but the radial distribution function and energy parameter are modified:

$$\mathrm{g}_{ij}^{CPA} = \frac{1}{1 - 1.9\eta} \qquad \eta = \frac{B}{4V} \tag{4.20}$$

$$\mathrm{a} = \mathrm{a}_0\left[1 + c_1\left(1 - \sqrt{T/T_c}\right)\right]^2 \tag{4.21}$$

The tuning parameters for the physical contribution are b, a_0, c_1.

The addition of the association term means that CPA is no longer cubic in volume like the classical cubic EOS models, so the numerical efficiency that makes the cubics attractive is lost. Additionally, there are five tuning parameters for CPA, which are fit to liquid density and vapor pressure data (like PC-SAFT), instead of the three parameters for cubics fit to the critical point. Thus, CPA improves dramatically upon the liquid volume predictions produced by the cubics, partly due to a retuning of the model parameters and partly due to the additional association term and more fitting parameters. Many academic works have demonstrated the improved performance of CPA over the classical cubics in predicting the phase behavior of asphaltic crude oils and polar compounds like water and alcohols[25,49–55].

4.5. Applying Equations of State to Crude Oil Systems

There are two major factors to consider when applying mathematical models to physical systems: speed and accuracy. These considerations are usually contradictory, as what is gained in accuracy is usually paid for by an increase in model complexity and subsequent increase in computational time. For applications requiring millions of calls to a physical property model routine, speed becomes the dominant consideration. For applications requiring only a few calls to a few hundred calls to these routines, accuracy is more important. Because general phase behavior calculations, which are the focus of this work, require comparatively few calls to the EOS function, speed is not a limiting factor and the choice of EOS is down to which model best represents the physical characteristics of the systems under consideration.

As discussed in Section 4.3, fluid phase behavior is driven by the strength and directionality of the intermolecular forces between the various components present in a given system, and the primary goal of EOS models is to describe these intermolecular forces as a function of easily measured variables like pressure, temperature, volume and composition. Intermolecular forces arise due to the chemical structures of the components present in a mixture; nonpolar components interact only by weak dispersion forces due to electron polarizability, while polar components have permanent dipoles with stronger interactions that require more energy to overcome the bonds that hold their molecules in the liquid state.

The intermolecular forces that characterize the phase behavior of asphaltenes are more nuanced than that of n-pentane or water because asphaltenes are a solubility class – meaning they consist of many different molecular structures with different sizes and intermolecular forces – and they are more polarizable than alkanes and aromatics but with no permanent dipoles like exist for water, alcohols, and carboxylic acids. Asphaltenes tend to self-associate even in good solvents and rarely, if ever, exist in monomeric form. Their aromatic cores interact by pi-pi bonding and stack to form nanoaggregates that are much heavier than the asphaltene monomer. This stacking phenomenon of asphaltene monomers is due to strong short-range forces that are characteristic of association, but the aggregated monomers – once formed – tend to interact with other components by dispersion forces instead of association[56–58].

This distinction offers two reasonable modeling approaches, one in which the pre-aggregated asphaltene monomer is modeled as an associating component and one in which the aggregated asphaltene is modeled as a non-associating component. Various researchers have shown that PC-SAFT is capable of modeling asphaltene precipitation for extreme ranges of pressure and temperature without the association term[59–65], while the cubic EOS are generally incapable of accurately modeling the liquid-liquid phase split of asphaltene precipitation from bulk oil without the association term. This suggests that asphaltene precipitation modeling with CPA yields results consistent with experimental data because of the additional tuning parameters and not necessarily because CPA is properly capturing the physics of asphaltene phase behavior. Given sufficient parameters tuned to enough experimental data, it is unsurprising that CPA often performs as well as PC-SAFT. However, the benefit of an EOS model is in its predictive capability. PC-SAFT has shown to predict asphaltene precipitation phenomena at different gas injection concentrations accurately with parameters tuned to only two or three experimental AOP points. Thus, for the application of asphaltene precipitation, PC-SAFT without association is our preferred model. Cubic EOS models often perform satisfactorily for vapor-liquid equilibrium (VLE) applications, but their shortcomings in describing liquid phase compressibility becomes especially pronounced in liquid-liquid equilibrium (LLE) applications.

Chapter 5

Asphaltene Thermodynamic Modeling

The work presented in this chapter is a culmination of Chapters 2-4. The thermodynamic equilibrium framework was presented in Chapter 2, algorithms designed to solve the equilibrium equations were presented in Chapter 3, and equations of state for calculating the thermodynamic properties of fluid phases were presented in Chapter 4. These works were combined to make a phase equilibrium simulator programmed in an EXCEL VBA environment capable of calculating phase distributions and equilibrium properties at any temperature and pressure for any mixture composition.

Using a slightly modified version of the characterization procedure proposed by Tavakkoli et al.[64] and the phase equilibrium simulator developed as part of this thesis, a thermodynamic modeling study of crude oil C2 is presented. This modeling study is a subset of a more comprehensive project intended to assess the thermodynamic and transport properties of crude oil C2 under varying amounts of lean gas injection and over the life of the well. This project included experimental determination of asphaltene precipitation from the indirect method[66] and precipitation kinetics[67] as well as a deposition study from packed-bed experiments[68,69]. The results of these experiments were then used to tune adjustable parameter values for both the asphaltene precipitation and deposition models.

The focus in this chapter is on asphaltene precipitation modeling as other aspects of this project were undertaken by colleagues. Asphaltene precipitation modeling in this work was accomplished using a slightly modified version of the crude oil characterization technique presented by Tavakkoli et al.[64], where experimental measurements of asphaltene precipitation were used to tune PC-SAFT parameters for the polydisperse fraction of asphaltenes. Both the general methodology of crude oil characterization and the specific characterization of crude oil C2 is discussed in Sections 5.2 and 5.3, and the results of the thermodynamic modeling for crude oil C2 is discussed in Section 5.4. To conclude Section 5.4, various crude oils published in the literature are re-investigated to highlight the anomalous thermodynamic behavior observed in crude oil C2. The focus in Section 5.1 is on the physicochemical characteristics of asphaltenes that justify the models used in this and similar work for describing asphaltene phase behavior.

5.1. Asphaltene Physical and Chemical Properties

Asphaltenes are a solubility class defined as the fraction of oil which is destabilized by n-heptane and solubilized by toluene. They are particularly heavy and appear as brown or black sticky semisolids in petroleum samples. Because they are defined as a range of solubilities, asphaltenes are polydisperse and the different subcomponents of the asphaltene fraction may exhibit markedly different phase and rheological behavior. These facts make asphaltenes particularly difficult to characterize, especially with the contention by some that the asphaltene fraction contains up to 20,000 constituents[70].

The asphaltene molecule is characterized by a poly-nuclear aromatic core with varying aliphatic chains, small amounts of heteroatoms like sulfur, nitrogen and oxygen and trace elements of metals like vanadium, nickel and iron. Hydrogen to carbon (H/C) ratios vary between 1.0 and 1.2 for asphaltene molecules and the heteroatom content is only a few percent[71]. The density of asphaltenes has been reported to be in the range of 1.13 – 1.20 g/cc [72].

The measurement of asphaltene molecular weight has been of great contention due to the inherent difficulties in using standard laboratory techniques to generate consistent measurements for a molecule as strongly self-associating as asphaltenes. Unfortunately, defining asphaltenes as a solubility class immediately precludes the determination of a precise molecular weight. Additionally, asphaltenes can vary greatly in their size and structure distributions from one crude oil to the next. Groenzin and Mullins[73] summarize some of the measurements that have been performed on molecular weight, showing that a range from about 400 to 10,000 g/mol have been found experimentally. Groenzin and Mullins argue that the asphaltene molecules found in petroleum reservoirs are on the smaller end of the spectrum, estimating the molecular weight to be 500 to 1000 g/mol. Their conclusion rests in large part on their finding that the fused ring system known to characterize asphaltenes possess on average about 7 rings, whereas the high molecular weight estimates offered would require systems with upwards of 20 fused rings. They attribute the large molecular weight measurements reported by other groups to the aggregation of several particles, which obfuscates the molecular weight measurement of a single asphaltene molecule.

Yarronton et al.[74] agreed with these findings, using vapor pressure osmometry (VPO) to estimate the average monomer molecular weight of their asphaltene sample at around 850 g/mol. They also found that approximately 90 wt% of their sample self-associated and the nanoaggregate molecular weight was more on the order of 10,000 to 20,000 g/mol. For the purposes of modeling asphaltenes in petroleum sources, the molecular weight of asphaltenes is more accurately represented by the self-associated molecular weight, given that self-association in hydrocarbon solutions is one of the most characteristic features of asphaltenes. In other words, asphaltenes exist in solution as self-associated clusters, so though the asphaltene monomer may have a molecular weight on the order of 500 g/mol, this is not representative of the asphaltenes that are extracted during the production of hydrocarbon fuels.

5.1.1. Modeling Methods for Asphaltene Phase Behavior

Asphaltene precipitation has been shown to be a strong function of temperature, pressure and composition, naturally leading to the use of thermodynamic models to describe asphaltene phase behavior. There are two distinct routes in modeling the thermodynamics of asphaltene particles in solution: the first is the colloidal approach, in which asphaltenes are assumed to be stabilized by forming colloids with resin particles, and the second is the solubility approach, in which asphaltenes are solubilized by the properties of the bulk solvent. These two viewpoints further highlight the contention that still exists regarding asphaltene structure and the functions that structure plays in determining behavior.

Sisco et al.[75] provided an overview of the colloidal models, but this class of models for asphaltene stability is not a focus of this thesis as the authors have shown the colloidal models to be difficult to apply to practical problems and it is unlikely that the colloidal description of asphaltene behavior matches reality. Thus, the solubility models – specifically, equations of state – are the preferred method in this work.

The solubility model postulates that asphaltenes are solubilized by chemically similar components in oil and that primarily dispersion forces drive asphaltene phase behavior. Given that resins are the most chemically similar components in the oil, resins act to solubilize asphaltenes and a reduction in the resin content would encourage asphaltene precipitation, a conclusion also inherent to the colloidal model. Treating asphaltene thermodynamics by the solubility approach has allowed the use of liquid-liquid and solid-liquid equilibria techniques. The success of these techniques in predicting asphaltene phase behavior helps toward validation of the solubility assumption.

The modeling approach in this work then follows from the observed phenomena and adheres to the solubility approach. Asphaltenes can be dissolved in a liquid phase or split into a separate phase as pressure, temperature and/or composition changes. This can be modeled by liquid-liquid equilibrium (LLE) between an asphaltene-rich organic phase and asphaltene-lean organic phase. Asphaltene precipitation is assumed to be thermodynamically reversible and phase behavior dominated by dispersion forces. Some solubility models – like Regular Solution Theory[76–79] or Flory Huggins Zuo[80–82] – treat the bulk solvent as a pseudo-pure component in equilibrium with an asphaltene component, while the EOS models do not make this simplifying assumption. For describing the full range of behaviors observed for crude oils – asphaltene instability, bubble points, critical points, etc. – the EOS models are the preferred approach, and the choice of EOS should be that model which captures best the physics of the asphaltic crude oil system. As discussed in Section 4.5, PC-SAFT is arguably the best model for these applications.

5.1.1.1. Monodisperse vs. Polydisperse Asphaltene

Asphaltenes are a solubility class containing all constituents of the stock tank oil that are insoluble in n-pentane and soluble in toluene, meaning that there is no precise chemical structure for asphaltenes and that the many different molecular structures that yield solubilities in the defined range qualify as asphaltenes. The physicochemical properties of the bulk asphaltene fraction of the oil are a function of the relative amounts of the various types of asphaltenes contained therein and their respective structures. Because asphaltenes are a broad solubility class, the various molecules included in the asphaltene fraction can have very different structure and properties and the relative amounts of the various molecules is dependent on the crude oil source from which the asphaltenes originate.

Vargas et al.[83] have shown that polydispersity can have a large effect on both phase behavior and the deposition profile. A plot of the cloud point pressure versus the mass fraction of a polydisperse polymer in a solvent is presented in **Figure 5.1**. The phase behavior is predicted quite accurately when the molecular weight distribution of the polydisperse polymer is well-characterized.

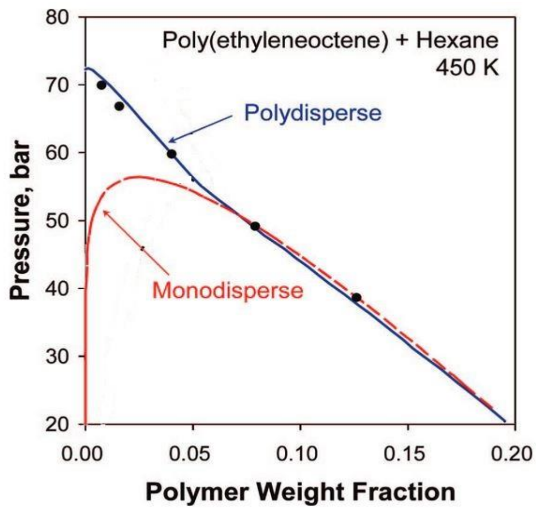


Figure 5.1. Cloud-point curve for poly(ethylene-octene) + hexane at 450 K. Experimental data from de Loos et al. SAFT simulation from Jog et al.[84]

Under the assumption of a monodisperse polymer, the accuracy of the predictions suffers markedly at low polymer concentrations. At high polymer concentrations, the precipitating phase is a light polymer-lean phase and the phase boundary is determined mostly by the average molecular weight of the polymer. Thus, the phase boundaries for monodisperse and polydisperse polymers are nearly identical at high concentrations. However, at low concentrations, the precipitating phase is a heavy polymer-rich phase and the heaviest polymers determine the phase boundary, and the monodisperse polymer does not capture this behavior.

Given that asphaltenes are a polydisperse class of components, the same limitations in predicting polydisperse polymer phase behavior should be expected. This greatly hinders the assumption of monodispersity when considering asphaltenes, as it has often been observed that the wells exhibiting the worst deposition behavior are often those with low asphaltene content. Thus, it is especially important that polydispersity considerations be made in wells with low asphaltene content as these are likely to cause deposition problems and the assumption of monodispersity has been shown to be especially poor in these types of systems.

Considering the additional complexity of a polydisperse asphaltene component means that more model tuning parameters are required, and more tuning parameters usually means that more experimental data for tuning will also be required. When obtaining experimental data is prohibitively expensive, this can be an undesirable consequence. More tuning parameters also can hint at a deficiency in the model where the tuning parameters act more as empirical corrections than descriptions of actual physical phenomena.

5.2. Crude Oil Characterization

Crude oils contain thousands of components with vastly different chemical structures. Obtaining a full and detailed compositional analysis for thousands of components is experimentally infeasible and applying an EOS to such a system would be computationally unmanageable. Instead, the crude oil is lumped into a reasonable number of pseudo-components with a characteristic set of EOS parameters. In the context of this thesis, “characterization” is defined as the process of defining a set of representative pseudo-components for crude oil, their compositions, and EOS parameters.

The compositional analyses performed by service laboratories list the compositions with approximately 30-50 components. The compositions of nonhydrocarbon gases like H_2S, CO_2 and N_2, and light hydrocarbon components (lighter than C_6) are reported in detail, while components including C_6 and heavier (C_6^+ fractions) are reported as boiling cuts corresponding to single carbon numbers (i.e. C_7, C_8, …, etc.). Components are then distributed into these single carbon number fractions according to their normal boiling temperatures as recommended by Katz and Firoozabadi[85]. For example, the C_9 pseudo-fraction contains all components with normal boiling temperatures between 126.1-151.3 °C, where the low end of the range corresponds to a boiling point that is 0.5 °C higher than that of n-C_8 and the high end corresponds to a normal boiling point that is 0.5 °C higher than that of n-C_9. Katz and Firoozabadi reported boiling ranges, average density, and average molecular weight for each carbon fraction up to C_{25}.

Service laboratories usually perform the compositional analysis up to C_{25} since the average properties for these pseudo-components can be determined from the work of Katz and Firoozabadi. The composition, density and molecular weight of the remaining portion of the crude oil (pseudo-fraction C_6^+) are calculated by subtracting off the contribution of the lighter components from the properties of whole sample, which are also measured experimentally.

Composition measurements are performed after flashing the reservoir fluid sample at standard pressure and temperature (1 bar and 15 °C), and the relative amounts of gas and liquid are measured and reported as gas-to-oil ratio (GOR). The compositions are reported for flashed gas and flashed liquid separately. Then, the reservoir fluid composition is determined by mathematically combining the measured gas and liquid compositions using the measured GOR.

Various crude oil characterization procedures using different equations of state have been proposed over the past few decades. A single carbon number characterization procedure proposed by Pedersen[86] is among the most popular of these procedures due to its simplicity. Pedersen's characterization scheme takes in a minimal amount of experimental data and can fully characterize an oil without requiring parameter tuning, though of course tuning can be done if more experimental data are available. The Pedersen characterization scheme was originally developed for the cubic EOS and then extended to handle PC-SAFT parameter estimation[87]. Some characterization methods make use of probability distribution functions to represent the composition of the C_7^+ fraction[88,89], while others are based on the SARA analysis of the stock tank oil[55–59].

The methodology for separating the crude oil into SARA fractions (Saturates, Aromatics, Resins, and Asphaltenes) is shown in **Figure 5.2**. The asphaltenes are first separated from the stock tank oil (STO, also called flashed liquid or dead oil) by addition of n-alkane precipitant, which induces asphaltene precipitation from the maltene phase. The maltenes are then separated into saturates, aromatics, and resins by chemical adsorption. This analysis provides the mass composition of the flashed liquid in terms of the total content of each SARA fraction of the crude oil. The SARA-based oil characterization techniques of Punnapala and Vargas[63] and Tavakkoli et al.[64] uses the compositions measured from the SARA analysis to define pseudo-components and compositions for the EOS model.

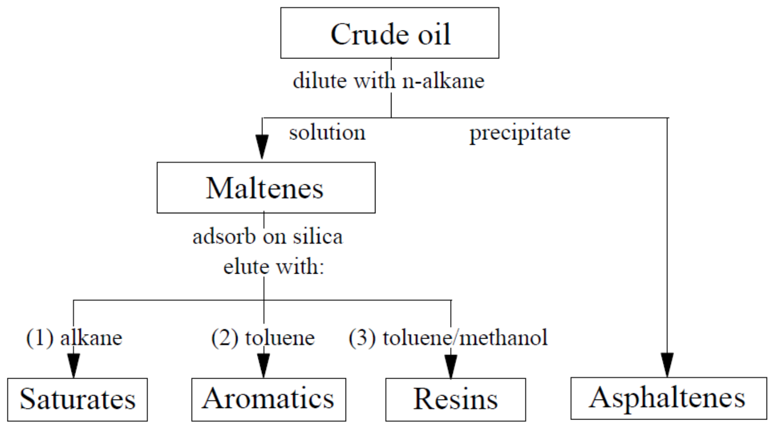


Figure 5.2. Separation scheme for SARA components.[91]

Regardless of methodology, the purpose of crude oil characterization is to describe the physics of a complex crude oil mixture with the minimum number of pseudo-components that are still able to produce accurate modeling results. However, a “reasonable” number of pseudo-components depends mainly on the objective of the modeling study and the type of crude oil under investigation. Generally, heavy crude oils (API gravity ≤ 30)[92] require more pseudo-components than light crude oils, especially oils that can liquid-liquid phase split. If the objective is modeling only vapor-liquid equilibrium for a reservoir fluid (e.g. bubble points, constant composition expansion, etc.), it is not necessary to characterize the heavy ends in detail since these properties are driven mainly by the light ends. Díaz et al.[93] showed that modeling Athabasca bitumen with solvent mixtures composed of 5 or 40 heavy-end pseudo-components yield similar results for bubble pressure using the Advanced PR EOS. Abutaqiya et al.[94] also validated this idea with light crude oils from the Middle East; the authors proposed a lumping scheme in which all flashed liquid components are combined into a single fraction and showed that bubble points of oil blends with different injection gases, as well as other volumetric properties for reservoir fluids (e.g. density, oil formation factor…etc.), are modeled accurately.

If the objective is modeling properties for which the heavy ends drive phase behavior (dew points, liquid drop out in gas condensates, or asphaltene precipitation), then more detailed characterization of the heavy ends is required to capture these phenomena. Pedersen et al.[92] showed that the choice of 6 or 22 pseudo-components to characterize the heavy ends significantly affects the predictions of liquid drop out from gas condensates using the SRK EOS. For asphaltene precipitation modeling, the number of pseudo-components depends on whether only asphaltene onsets are desired or if more extensive knowledge of the solubility of asphaltenes inside the unstable region is desired. If the purpose of the asphaltene precipitation study is to model only the onset of precipitation (AOP), usually 3-8 pseudo-fractions are used to represent the maltenes, while a single pseudo-fraction is used to represent the asphaltenes[61,63,95]. If modeling asphaltene solubility in the unstable region, then it is necessary to characterize the asphaltene pseudo-fraction as several cuts to account for asphaltene polydispersity. Tavakkoli et al.[64] used 4 asphaltene pseudo-fractions in modeling asphaltene precipitation from light crude oils.

5.3. PC-SAFT Characterization with SARA Analysis

The PC-SAFT + SARA analysis approach to crude oil characterization[60,61,63,64,83,90] has been in continuous development over the past years, principally at Rice University. In general, these developments followed the same lumping procedure but differed in the methodology for parameter estimation. The description of the SARA-based method in this section follows the latest development as implemented by Punnapala and Vargas[63] for monodisperse asphaltenes and Tavakkoli et al.[64] for polydisperse asphaltenes. The two approaches differ only in the characterization of the asphaltenes pseudo-fraction. Truncated versions of these characterization procedures are presented in this thesis. A schematic of the SARA-based characterization is shown in **Figure 5.3**. In this method, the flashed gas and flashed liquid from the PVT report are characterized separately and then mathematically combined using the experimental GOR to form the live reservoir fluid.

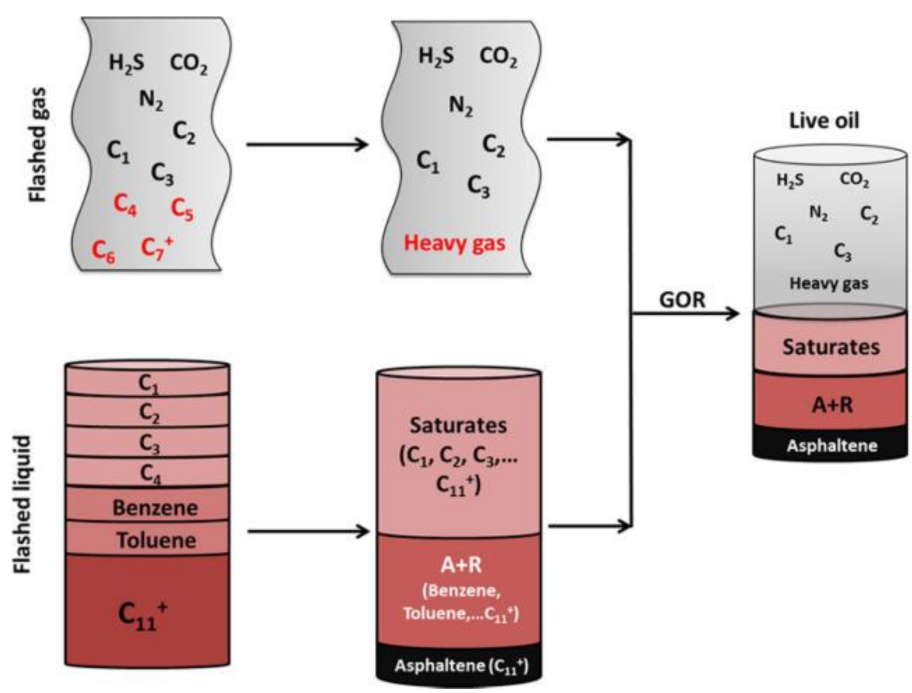


Figure 5.3. SARA-based crude oil characterization schematic.[94]

5.3.1. Characterization of Flashed Gas

The flashed gas is usually modeled as a mixture containing N_2, CO_2, H_2S, methane, ethane, propane and a heavy gas lumped fraction (C_{4+}). A different lumping scheme can be chosen, e.g. (C_{2+}), to reduce the number of components and thus increase computing speed, but the model may become less predictive. The gas phase is largely responsible for affecting the bubble pressure and considering each of the flashed gas components individually results in a more predictive model. All components in the flashed gas lumping scheme are considered saturates.

5.3.2. Characterization of Flashed Liquid

The liquid fraction is characterized based on the flashed liquid composition and the SARA analysis. In the monodisperse asphaltene approach of Punnapala and Vargas, the flashed liquid (also called STO or dead oil) is characterized as three pseudo-components: Saturates, Aromatics + Resins (A+R), and Asphaltenes. The normal alkanes, branched alkanes, and cycloalkanes are all lumped into the Saturates pseudo-fraction. The aromatics are lumped into the A+R pseudo-fraction. Boiling cuts lower than C_{10} are assumed paraffinic and are lumped into the Saturates. The C_{10}^+ pseudo-fraction is distributed into the three groups such that the weight fraction of each group matches the reported SARA analysis. The molecular weight value for C_{10}^+ in Asphaltenes is assumed to be 1700 g/mol as an initial guess only for calculating molecular weights and molar compositions of Saturates and A+R. This value can be tuned later to match experimental data (usually AOP experiments). The molecular weight values for C_{10}^+ in Saturates and A+R are set as an initial guess such that the calculated STO molecular weight matches the reported value. In the polydisperse asphaltene approach of Tavakkoli et al., the primary difference is that the Asphaltene fraction is de-lumped into four pseudo-components assumed to follow a gamma distribution in molecular weight.

After characterizing the flashed gas and flashed liquid, the phases are combined mathematically using the experimental GOR to determine the live oil composition. Once this task is completed, the composition and molecular weight of each component in the live oil is known. The next step is to determine the PC-SAFT parameters for each of these components.

5.3.3. PC-SAFT Parameter Estimation

The phase behavior of asphaltenes is largely determined by the polarizability of the molecules – not the polarity – meaning that association can often be safely neglected[56,96] and only three PC-SAFT parameters are required for each component: the number of segments per molecule (m_i), the temperature-independent diameter of each molecular segment (σ_i), and the dispersion energy (ε_i/k_B).

The PC-SAFT parameters for nonpolar hydrocarbons and light gases are taken from the work of Gross and Sadowski[36]; select components relevant to this work are shown in **Table 5.1**. The parameters for the pseudo-components are calculated using correlations originally published by Gonzalez et al.[60] and later modified by Punnapala and Vargas to provide consistency across the pseudo-component set. Punnapala and Vargas defined an aromaticity parameter (γ) that varies from 0 to 1, where 0 corresponds to n-alkanes and 1 to poly-nuclear aromatics (PNA). Their correlations for the PC-SAFT parameters as a function of molecular weight (M_W) and aromaticity (γ) are shown in **Table 5.2**.

Table 5.1. PC-SAFT parameters for pure components.

Component	**M_W g/mol**	**m**	**σ Å**	**ε/k_B K**
H_2S	34.08	1.6517	3.0737	227.34
N_2	28.01	1.2064	3.3130	90.96
CO_2	44.01	2.0729	2.7852	169.21
C_1	16.04	1.0000	3.7039	150.03
C_2	30.07	1.6069	3.5206	191.42
C_3	44.10	2.0020	3.6184	208.11

Table 5.2. Correlations for PC-SAFT parameters of Heavy Gas and SARA components.

Parameter	**Expression**
m	$(1-\boldsymbol{\gamma})(0.0257\boldsymbol{M_W} + 0.8444) + \boldsymbol{\gamma}(0.0101\boldsymbol{M_W} + 1.7296)$
σ (Å)	$(1-\boldsymbol{\gamma})\left(4.047 - \frac{4.8013\ln(\boldsymbol{M_W})}{\boldsymbol{M_W}}\right) + \boldsymbol{\gamma}\left(4.6169 - \frac{93.98}{\boldsymbol{M_W}}\right)$
ε/k_B (K)	$(1-\boldsymbol{\gamma})\exp\left(5.5769 - \frac{9.523}{\boldsymbol{M_W}}\right) + \boldsymbol{\gamma}\left(508 - \frac{234100}{\boldsymbol{M_W}^{1.5}}\right)$

5.3.3.1. Monodisperse Asphaltene

In the monodisperse asphaltene approach of Punnapala and Vargas, the Heavy Gas and Saturates pseudo-components are assumed to consist purely of n-alkanes ($\gamma_{HG} = \gamma_{Sat} = 0$), and the aromaticity of A+R is tuned to match the bubble pressure and saturated liquid density of the crude oil. The molecular weight and aromaticity of Asphaltenes are tuned to match experimental values for onset of precipitation (AOP) measured by HPHT NIR spectroscopy. The approach of Punnapala and Vargas requires tuning three parameters (γ^{A+R}, M_W^{Asph}, γ^{Asph}) to experimental data to fully specify the PC-SAFT parameters for all pseudo-components.

5.3.3.2. Polydisperse Asphaltene

To account for the effects of asphaltene polydispersity, Tavakkoli et al.[64] proposed a characterization scheme that mirrors that of Punnapala and Vargas[63] but treats the Asphaltene pseudo-component as multiple subfractions polydisperse in molecular weight. A slight modification to that approach is presented here, where instead of lumping the aromatics and resins, they are separated into distinct pseudo-components and then resins are included as part of the molecular weight distribution of asphaltenes. This molecular weight distribution is characterized by an average molecular weight ($\bar{M}$) and tunable distribution parameter (α):

$$f(r) = \frac{1}{M_m \Gamma(\alpha)} \left[\frac{\alpha}{(\bar{r}-1)}\right]^{\alpha} (r-1)^{\alpha-1} \exp\left[\frac{\alpha(1-r)}{(\bar{r}-1)}\right] \tag{5.1}$$

where M_m is the molecular weight of the monomer, which is assumed to be 600 g/mol, and r and $\bar{r}$ are calculated by:

$$r = M_W/M_m \qquad\qquad \bar{r} = \bar{M}/M_m \tag{5.2}$$

The aromaticity of the Aromatics, Resins and Asphaltenes pseudo-components are assumed to be equal ($\gamma_{ARA} = \gamma_{Arom} = \gamma_{Resin} = \gamma_{Asph}$) and this value is tuned to bubble pressure and saturated liquid density data. The molecular weight distribution parameters for the Resins and polydisperse Asphaltenes ($\bar{M}, \alpha$) are fit to onset data generated by the indirect method[66] instead of the customary high-pressure NIR spectroscopy method. In this scheme, asphaltenes are extracted from the stock tank oil and characterized based on the precipitating solvent. Asphaltenes that are soluble in n-C_6 but insoluble in n-C_5 are categorized as $C_{5\text{-}6}$ Asphaltenes, $C_{6\text{-}7}$ Asphaltenes include those that are soluble in n-C_7 but insoluble in n-C_6, and so on. Precipitation curves are generated with the indirect method by measuring light absorbance of the supernatant liquid as a function of n-alkane precipitant amount. The gamma distribution parameters are then tuned to the 90 vol% n-alkane precipitant measurements as these points have negligible kinetic effects on the precipitation measurement.

Tavakkoli et al.[64] showed the effect of sample aging time on asphaltene precipitation amounts for indirect method measurements. Only at high concentrations of precipitant (~90 vol%) did the experimental points for 1 hour, 1 day, and 1 week of aging time align. This indicates that sample equilibration occurs much faster for concentrations far from the onset concentration, thus making these experiments the preferred data for tuning an equilibrium model like PC-SAFT. If sufficient sample equilibration is not allowed to occur, then kinetic effects obfuscate the true equilibrium state.

5.4. Thermodynamic Modeling with PC-SAFT

5.4.1. Modeling Results of Crude Oil C2

For crude oil C2, the PVT properties and SARA analysis are shown in **Table 5.3** and asphaltene fractionation analysis is shown in **Table 5.4**. These data, along with the compositional analysis, are especially relevant for the oil characterization routine and tuning of the thermodynamic model.

Table 5.3. PVT properties and SARA analysis for oil C2.

Property	Value
P^{bub}(psia) at 259 °F	1905
ρ^{bub}(g/cc) at 259 °F	0.761
Zero-flash GOR (scf/stb)	381
ρ^{STO}(g/cc) at 60 °F, 1 atm	0.9042
M^{STO}(g/mol)	290.3
Saturates (wt%)	49.9
Aromatics (wt%)	19.8
Resins (wt%)	19.2
C_5 Asphaltenes (wt%)	11.1

Table 5.4. Asphaltene fractionation for oil C2.

Asphaltene Fraction	wt%
C_5 Asph	11.1
C_6 Asph	10.9
C_7 Asph	10.1
C_8 Asph	8.9

Additionally, the indirect method is used to obtain asphaltene precipitation amounts as a function of n-alkane volume. These data are used to tune the molecular weight distribution parameters for the gamma distribution function. Indirect method experiments were run at room temperature (~20 °C), and the gamma distribution parameters were fit to data generated at high n-alkane concentration. The experimental measurements for asphaltene precipitation at 90 vol% and the PC-SAFT predictions after tuning are shown in **Figure 5.4**. The fitting is performed at high n-alkane concentration to minimize the effect of kinetics on asphaltene precipitation. Tavakkoli et al.[64] showed that for precipitant concentrations in the vicinity of the onset, where the driving force for precipitation is low, establishing equilibrium takes much longer than for precipitant concentrations far from the onset where the driving force for precipitation is high. The parameter set for characterized crude oil C2 with tuned values for the pseudo-component parameters is given in **Table 5.5** and binary interaction parameter values in Appendix C.

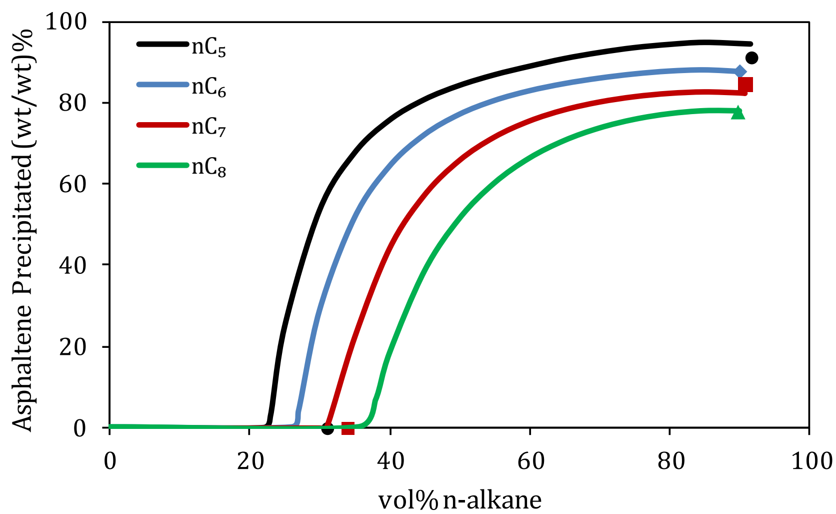


Figure 5.4. Indirect Method measurements at 90 vol% precipitant and corresponding PC-SAFT modeling results. Asphaltene onset concentrations (AOC) are shown but were not included in the parameter tuning.

Table 5.5. PC-SAFT parameters for oil C2.

Component	**z** **mol**	**M_W** **g/mol**	**m**	**σ** **Å**	**ε/k_B** **K**	**γ**
H_2S	0.000	34.08	1.6517	3.0737	227.34	
N_2	0.280	28.01	1.2064	3.3130	90.96	
CO_2	0.410	44.01	2.0729	2.7852	169.21	
C_1	26.079	16.04	1.0000	3.7039	150.03	
C_2	7.899	30.07	1.6069	3.5206	191.42	
C_3	6.805	44.10	2.0020	3.6184	208.11	
Heavy Gas	7.584	69.54	2.6315	3.7541	230.43	

Saturates	36.424	202.62	6.0518	3.9211	252.12	
Aromatics	9.069	322.93	6.8182	4.1654	374.78	0.56
Resins	4.371	649.61	12.3600	4.2641	391.14	0.56
Asph1	0.030	998.98	18.2869	4.2989	395.50	0.56
Asph2	0.115	1026.71	18.7573	4.3006	395.69	0.56
Asph3	0.163	1085.72	19.7583	4.3041	396.07	0.56
Asph4	0.771	1708.11	30.3165	4.3261	398.25	0.56

Traditional high-pressure asphaltene onset experiments were performed and used to validate the PC-SAFT parameters obtained from tuning to indirect method data. AOP was not detected for the live oil experiments performed at 120 °F and 259 °F, which agrees with the model predictions, and one of the AOP points detected for 55% gas injection was matched by the model. The point that is not matched is at low temperature, which is known to be a difficult condition for obtaining reliable equilibrium data. Additionally, the service laboratory reported that the onset occurred at a high pressure during sample cooling. Their intention was to cool the sample first and then perform the standard depressurization experiment, but asphaltene precipitation occurred before reaching the set-point temperature. Thus, the low temperature data point for the 55% gas injection case should only be viewed as a lower bound on the onset, and the model should predict an onset temperature higher than what was reported, as it does. **Figure 5.5** shows the asphaltene phase envelope for the live oil and 55% gas injection case, for which AOP data were collected. The composition of the injection gas is given in **Table 5.6**. The PT and Px projections of the asphaltene phase envelopes for crude oil C2 with gas injection are shown in **Figure 5.6**.

Table 5.6. Composition of injection gas for oil C2.

Component	wt%
N_2	0.32
CO_2	0.57
C_1	67.10
C_2	9.87
C_3	10.97
Heavy Gas	11.17

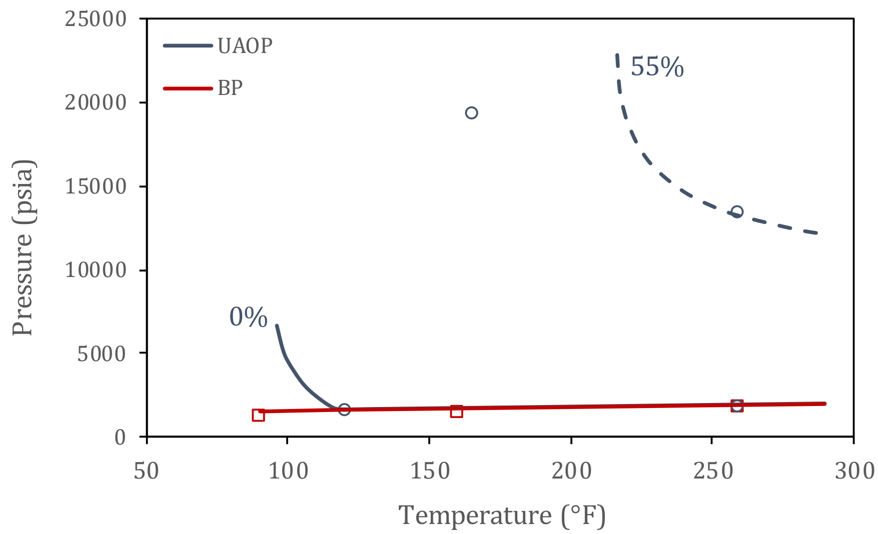


Figure 5.5. Asphaltene phase envelope for oil C2 at 0 mol% gas injection (solid lines) and 55 mol% gas injection (dashed line) with experimental data. Red squares: BP, Blue circles: AOP.

From the phase diagrams in **Figure 5.6**, it is clear that the asphaltene instability region grows with increasing gas injection, which is expected behavior. When the UAOP curve and LAOP curve are far apart, this means that asphaltenes are unstable across a wide range of pressure. Asphaltenes are stable at any pressure, temperature and composition for which the UAOP, BP and LAOP curves overlap. When these curves overlap, it means that no liquid-liquid separation will occur at the specified conditions (P-T or P-z). Instead, only vapor-liquid separation occurs.

However, these plots are not necessarily sufficient to convey information most relevant to deposition. Properties like density and composition of the bulk and onset phases have direct influence on the deposition tendency and, therefore, convey more relevant information.

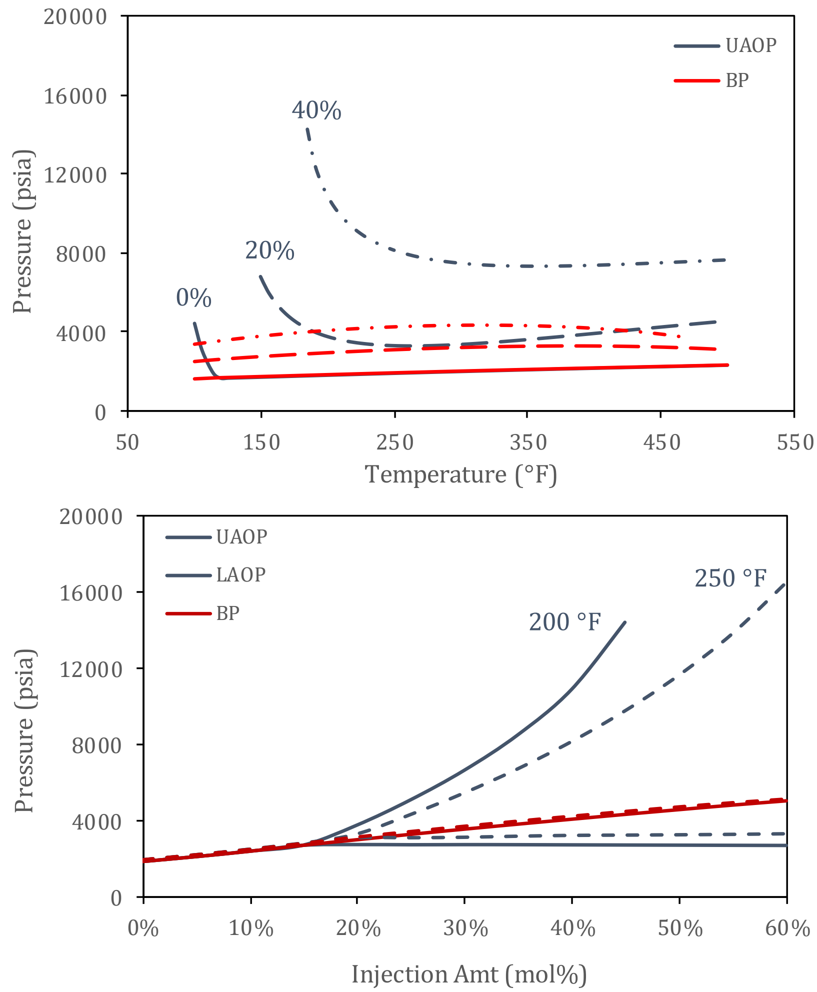


Figure 5.6. Asphaltene phase envelope for oil C2. (Top) PT-projection with three gas injection amounts (mol%), (Bottom) Px-projection with two isotherms.

To show the properties relevant to deposition, plots for phase amounts, densities, compositions, and component partitioning in the onset phase are shown a function of pressure along isotherms of interest. These properties are shown for 20% injection

(**Figure 5.7**) and 40% injection (**Figure 5.8**) for C2. The easiest way to interpret these plots, which are referred to collectively as "dP plots", is to view them as a production profile from reservoir to surface of a well operating isothermally. As we move from right-to-left on the dP plots, isothermal fluid depressurization is being simulated, and the plots show how the properties relevant to asphaltene deposition change as a function of pressure. The vertical dashed lines on the dP plots correspond to saturation pressures (UAOP, BP, LAOP, moving right-to-left). Plot A and plot C show the mass fractions and densities, respectively, of the bulk (L_1) and onset phases (L_2). Plot B shows the mass composition of the Asphaltene and Saturates pseudo-components in the bulk phase (dashed line) and onset phase (solid line), and plot D shows the ratio of the mass of each pseudo-component in the onset phase to the sum of the masses of that pseudo-component in the bulk and onset phases.

At high pressures, the fluid is a single liquid phase (L_1). As pressure decreases, the first vertical dashed line is crossed, which corresponds to the UAOP. At the UAOP, the single-phase fluid splits into two liquid phases, a bulk phase (L_1) and an onset phase (L_2). The onset phase is traditionally more dense and richer in asphaltenes than the bulk phase, but this does not have to be true, as will be shown for C2. Between the UAOP and the next vertical line, which corresponds to the BP, there are two liquid phases. At the BP, a gas phase forms, and three phases exist between this line and the last vertical dashed line, which corresponds to the LAOP. At the LAOP, enough gas has been liberated from the liquid phases that the bulk and onset phases are sufficiently similar that they combine, leaving one liquid and one gas phase. At all pressures below the LAOP, two phases exist.

The choice of which phase is called “bulk” and “onset” is usually straight-forward, but the case of C2 will show that this is sometimes not as clear. In this paper, the “bulk phase” is defined as that phase which is stable above the UAOP and the “onset phase” is that which forms at the UAOP. Usually, the mass of the onset phase in the asphaltene instability region remains small relative to the bulk phase. In these cases, the bulk phase completely solubilizes the onset phase above the UAOP and below the LAOP. However, for the 40% injection case shown in **Figure 5.8**, the bulk phase solubilizes the onset phase above the UAOP and the onset phase solubilizes the bulk phase below the LAOP.

In the 20% injection case, the onset phase is actually less dense and leaner in asphaltenes than the bulk phase, whereas the onset phase is denser and richer in asphaltenes for the 40% injection case. This switching from lighter onset phase to heavier onset phase occurs at around 30% injection for this case, and the phase-switching behavior is a function of temperature. The plots also reveal that though liquid-liquid phase-splitting occurs for both gas injection cases, neither the bulk nor onset phases have an asphaltene composition normally associated with a depositing phase. For the 20% case, partitioning of the heavy components strongly favors the bulk phase (shown by **Figure 5.7**.D) and does not show much variation between components, which is why the compositions are relatively constant across the asphaltene unstable region (**Figure 5.7**.B). However, for the 40% case, the asphaltene components partition strongly to the onset phase and all asphaltenes partition to the onset phase well before the BP (**Figure 5.8**.D). The other heavy components show preference to the onset phase but not as strongly as asphaltenes, thus the bulk and onset phase compositions show more variation in the 40 % case than in the 20% case.

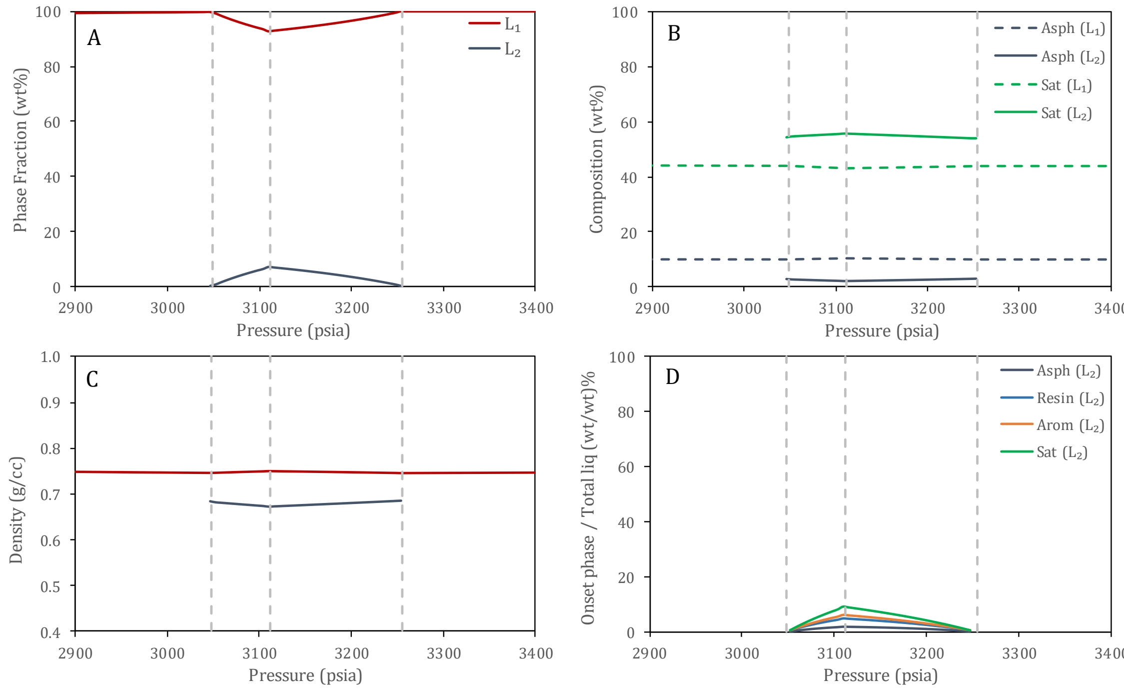


Figure 5.7. Bulk (L_1) and onset (L_2) phase properties as a function of pressure for oil C2 at 250 °F and 20 mol% gas injection. Dashed vertical lines denote saturation pressures (from right-to-left: UAOP, BP, LAOP).

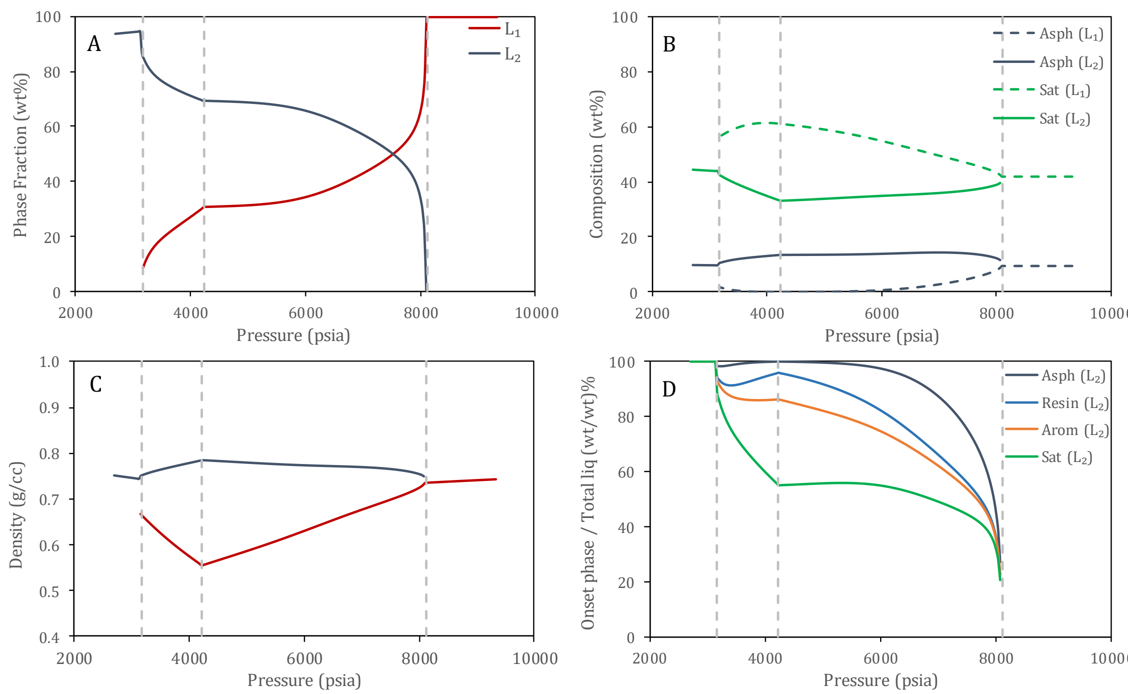


Figure 5.8. Bulk (L_1) and onset (L_2) phase properties as a function of pressure for oil C2 at 250 °F and 40 mol% gas injection. Dashed vertical lines denote saturation pressures (from right-to-left: UAOP, BP, LAOP).

The primary intent of the modeling study is to assess the likelihood and/or potential magnitude of the asphaltene deposition problem for crude oil C2. Precipitation is only a necessary condition of asphaltene deposition, but asphaltenes may precipitate in such a small amount that the deposition problem is negligible, or the precipitated asphaltenes may aggregate into large clusters such that they are carried with the bulk flow of oil instead of diffusing to the wall and forming solid deposits, or, as is observed in this case, the onset phase might be lean in asphaltenes such that any point in the region where two liquids exist is actually not a candidate for potential deposition problems. Even at high gas injection amounts, where the onset phase is denser than the bulk phase, the asphaltene composition is still relatively low and decreases as a function of pressure as other components co-precipitate with asphaltenes. A proper study of asphaltene precipitation and deposition must make clear these distinctions so that informed decisions can be made when choosing how to operate a well.

The observation of an asphaltene-lean onset phase yielded by the PC-SAFT model was partially validated qualitatively with images of the onset phase obtained from the service laboratory and shown in **Figure 5.9**. The image on the left shows the onset phase with asphaltenes as soft fluid-like particles while the image on the right shows more rigid solid-like asphaltene particles.

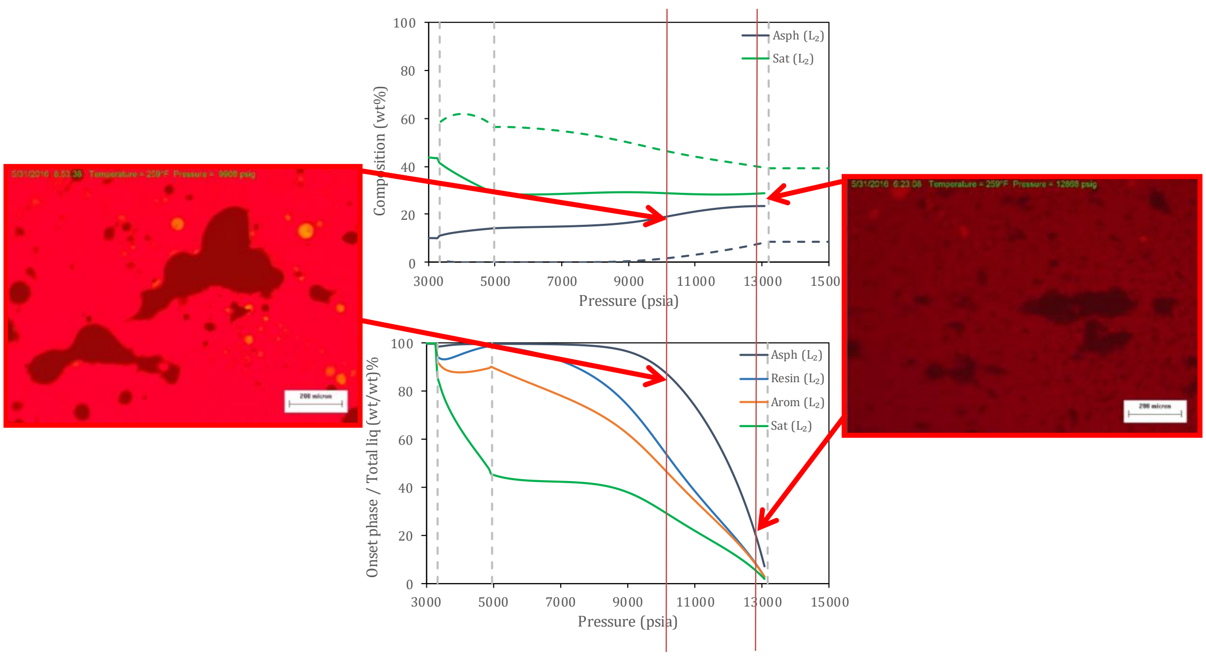


Figure 5.9. Images of onset phase of oil C2 with 55% gas injection taken during depressurization experiment. Asphaltenes appear as soft fluid-like particles in the left image and as hard solid-like particles in the right image. These differences are likely due to composition differences in the solvent.

For the pressure corresponding to the image taken near the UAOP, PC-SAFT predicts the onset phase to have an asphaltene composition of around 24 wt%, while in the other image, PC-SAFT predicts an asphaltene composition around 18 wt%. The onset phase is also becoming richer in good asphaltene solvents as pressure decreases, and the images shown agree with this conclusion. In the image taken near the UAOP, the asphaltene particles are more rigid and solid-like, while at the lower pressure, the asphaltene particles are softer and more liquid-like. The asphaltene particle near the UAOP is more rigid than the one at lower pressure likely because the solvent composition near the UAOP is a better precipitant than that at lower pressure. Additionally, the brightness of the continuous phase significantly increases at the lower pressure, indicating that a large fraction of the heavier components have precipitated out. This agrees with the PC-SAFT modeling results that show nearly 60 wt% Resins and 90 wt% Asphaltenes precipitating out of the bulk phase.

5.4.2. Modeling Results of Various Crude Oils

To highlight the anomalous behavior of crude oil C2, case studies from previous works were used to compare the bulk and onset phase properties. Simulation parameters for crude oil U8 were reported by Panuganti et al.[61], J1 by Jamaluddin[97], and S14 by Abutaqiya et al.[98] These parameters, along with the associated injection gas compositions, are provided in Appendix C. A comparison of relevant PVT properties and SARA analyses of C2, U8, J1, and S14 is provided in **Table 5.7**. Phase envelopes for crudes U8, J1, and S14 are shown in **Figure 5.10**, **Figure 5.11**, and **Figure 5.12**, respectively. The dP plots for U8 and J1 at 250 °F and 40 mol% gas injection are shown in **Figure 5.13** and **Figure 5.14**, respectively.

All studied crudes show wider asphaltene unstable regions as gas injection increases, meaning that the bulk oil becomes an increasingly worse solvent. This behavior is expected because the injection gases are rich in asphaltene precipitants and more gas corresponds to a lower solubility of asphaltenes in the bulk oil. Temperature increases, on the other hand, can either increase or decrease the solubility of asphaltenes in the bulk oil, depending on the location in the phase diagram.

Table 5.7. Comparison of PVT properties for C2, U8, J1, S14.

Property	C2	U8	J1	S14
GOR (scf/stb)	381	385	900	356
ρ^{STO}(g/cc)	0.9042	0.8239	0.8654	0.8290
M^{STO}(g/mol)	290	196	229	181
Saturates (wt%)	49.9	80.6	57.4	68.6
Aromatics (wt%)	19.8	17.4	30.8	22.3
Resins (wt%)	20.1	1.5	10.4	6.5
Asphaltenes (wt%)	10.2	0.5	1.4	0.5
Reference		75	97	98

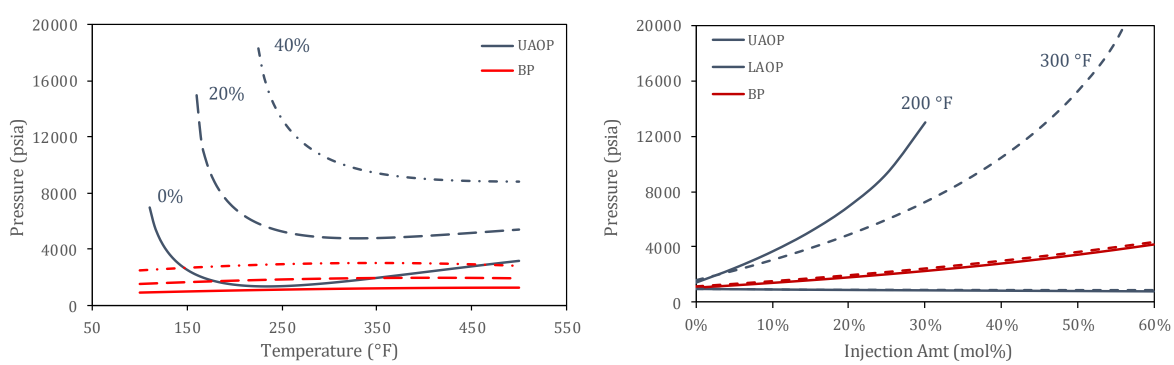


Figure 5.10. Asphaltene phase envelope for crude oil U8. (L) PT-projection with three gas injection amounts (mol%), (R) Px-projection with two isotherms.

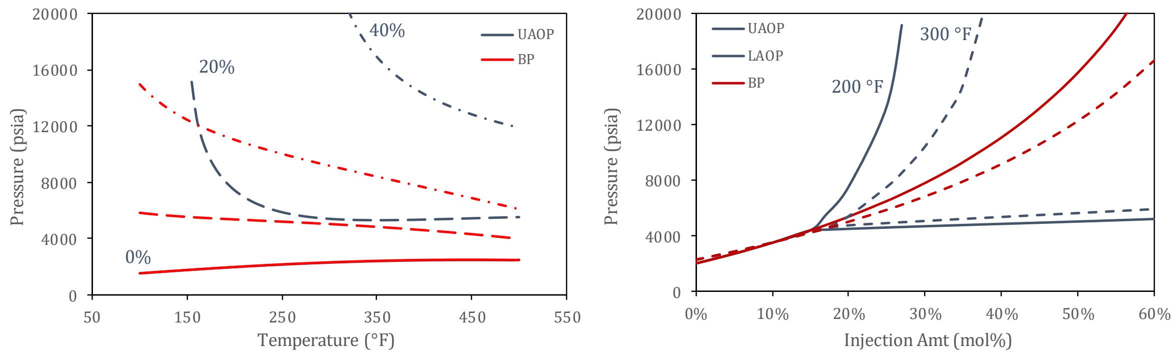


Figure 5.11. Asphaltene phase envelope for crude oil J1. (L) PT-projection with three gas injection amounts (mol%), (R) Px-projection with two isotherms.

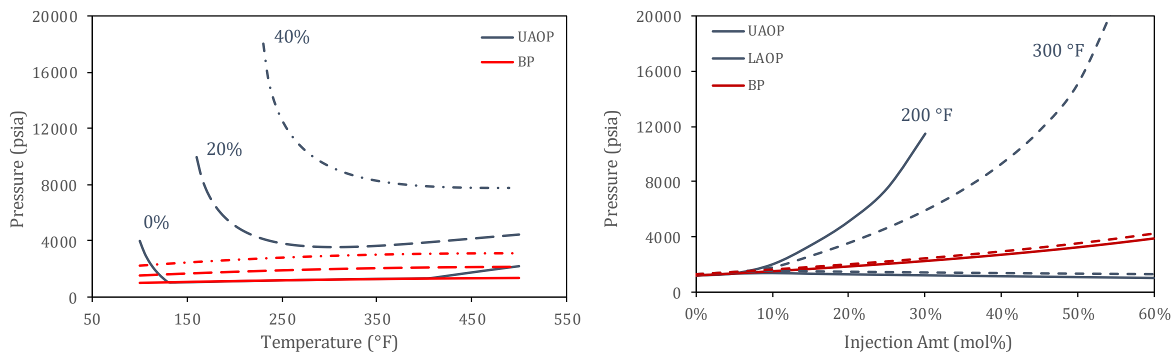


Figure 5.12. Asphaltene phase envelope for crude oil S14. (L) PT-projection with three gas injection amounts (mol%), (R) Px-projection with two isotherms.

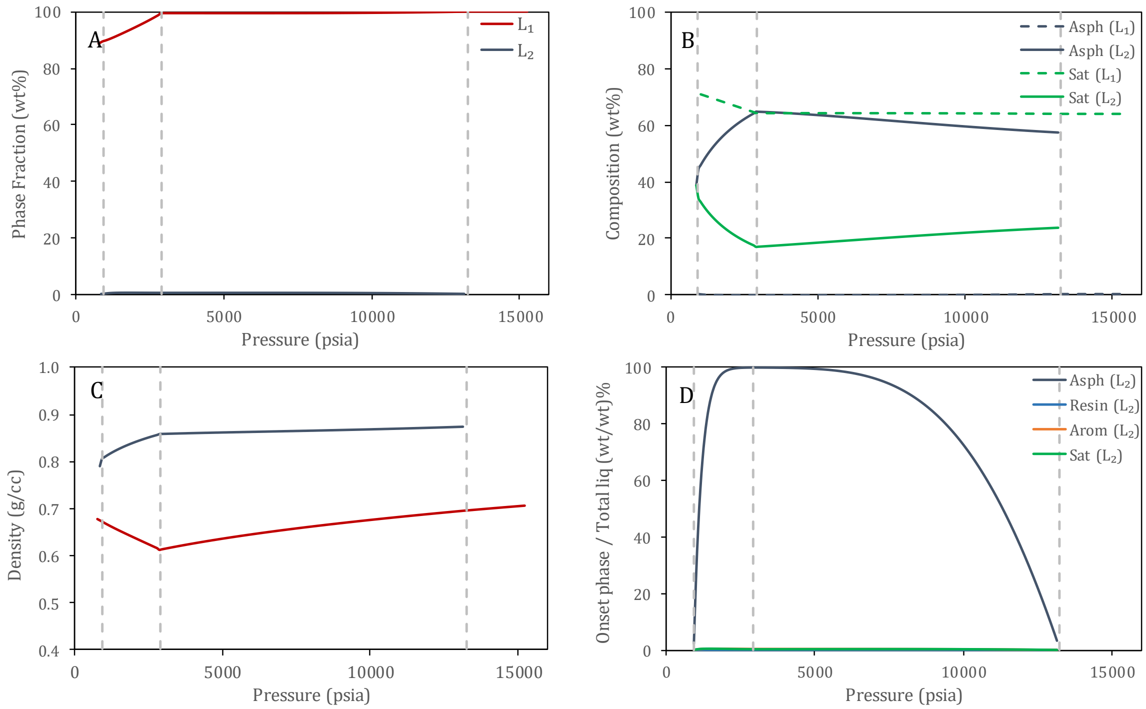


Figure 5.13. Bulk (L_1) and onset (L_2) phase properties as a function of pressure for U8 at 250 °F and 40 mol% gas injection. The onset phase is more dense and richer in asphaltenes than the bulk phase.

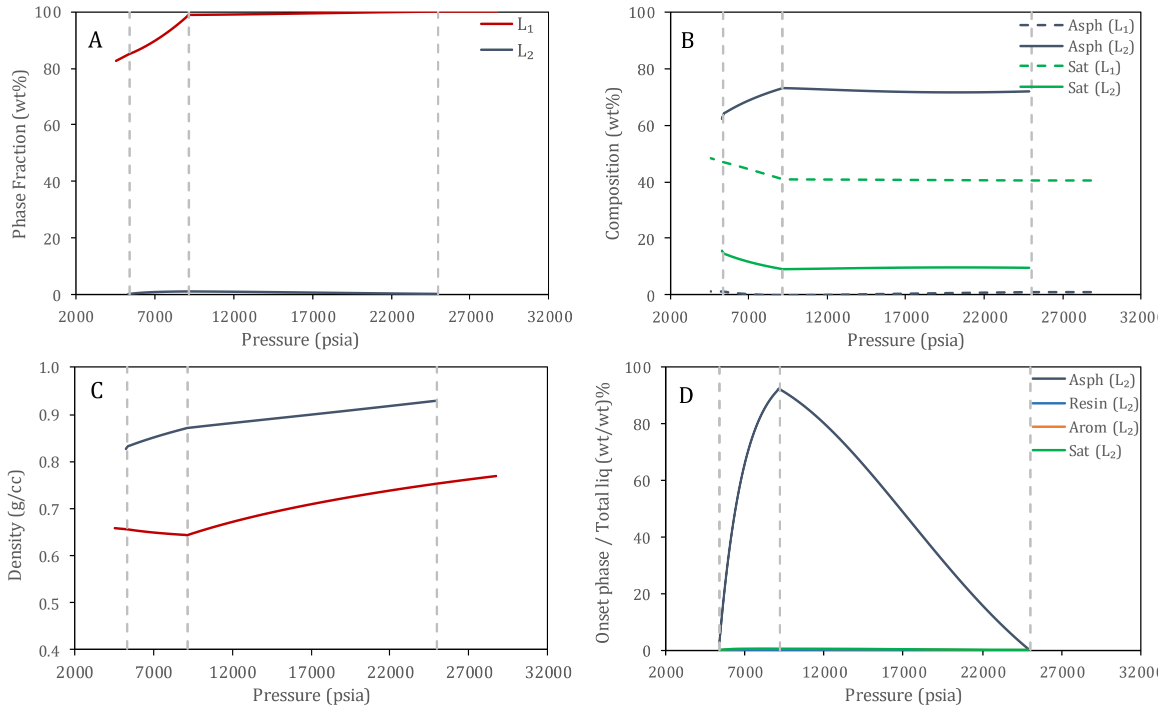


Figure 5.14. Bulk (L_1) and onset (L_2) phase properties as a function of pressure for J1 at 250 °F and 40 mol% gas injection. The onset phase is more dense and richer in asphaltenes than the bulk phase.

Comparing the density and composition of the bulk and onset phases predicted for crude oil C2 to U8 and J1 shows that C2 has significantly different onset phase properties which affect the rheological properties and deposition behavior of the oil. Crude oils U8, J1, and S14 show similar and consistent behavior for various gas injection amounts and various temperatures. Compared to C2, the weight fraction of the onset phase is always low, the density of the onset phase is high, and asphaltenes partition strongly to the onset phase while other heavy components partition strongly to the bulk phase. For C2, the non-asphaltene heavy components (Saturates, Aromatics, Resins) migrate much more strongly to the onset phase, which causes the phase fraction of the onset phase to be higher but also the density to be lower because the onset phase is composed by many non-asphaltene components.

5.4.3. Discussion and Conclusions

Based on the observations made in this modeling study and the partial validation provided by the images of the onset phase, we speculate that a key factor in explaining another anomalous phenomenon – that being the asphaltene deposition problem being worse in wells with lower asphaltene content – might be explained based on the compositions of the precipitating phase. Leontaritis and Mansoori[99] identified two oil fields in Venezuela – the Mata-Acema and Boscan fields – in which the Mata-Acema crudes with lower asphaltene content (0.4 to 9.8 wt%) showed catastrophic deposition problems while the Boscan crudes with high asphaltene content (17.2 wt%) showed no deposition problems. The explanation for this counter-intuitive result offered by Leontaritis and Mansoori was that the Boscan fields "contain something" that works against asphaltene flocculation. However, more recent evidence suggests that this mechanistic understanding of asphaltene deposition might not be correct.

First, it may be beneficial to promote asphaltene flocculation, a phenomena Leontaritis and Mansoori suggest might not be occurring in the Boscan fields that explain their lack of asphaltene deposition problems. Larger asphaltene particles are more likely to be carried with the bulk flow rather than diffusing toward the pipe wall and depositing. Chemical dispersants that seek to inhibit asphaltene particle growth have been shown to worsen the deposition problem[68,100,101], which is likely occurring because asphaltenes that would have otherwise aggregated and then carried off with the bulk flow of oil are now remaining as small particles prone to depositing. Second, asphaltene-rich reservoirs may be showing phase behavior like what is being shown for crude oil C2. A second liquid phase may be forming but the composition and density of that phase may be such that asphaltene particles remain soft in solution and are not candidates for deposition. Further experimental study should be able to reveal which, if any, properties of the onset phase change the deposition behavior of asphaltenes.

It is important that more experimental evidence is gathered to validate the modeling predictions generated in this study. If the model is to be believed, then many conditions that would be identified as problematic based only on the formation of a second liquid phase might not actually pose problems because of the compositions and densities of those phases. Whether or not this conclusion is justified needs to be validated more extensively. There is immense value in establishing reliable property correlations – either qualitatively or, preferably, quantitatively – between the behavior of the asphaltene particles and the composition and density of the fluid medium in which asphaltenes are dissolved. Because phase compositions play a key role in the rheological properties of a system, it is important to determine how accurately the compositional variations predicted by PC-SAFT for C2 are reflected in reality. Any deposition model relies on accurate compositional information, so the thermodynamic model must be able to model compositions correctly if crude oils like C2 are to be modeled accurately.

Chapter 6

A New Hybrid Equation of State for Nonpolar Chain Molecules

6.1. Introduction

The cubic-plus-chain (CPC) equation of state is a framework for modifying the standard cubic equations of state by adding a term to account for energy change due to chain formation. This idea is conceptually similar to the cubic-plus-association (CPA)[24] equation of state that hybridizes the standard cubic models with the SAFT association term proposed by Chapman et al.[32] Conversely, CPC adds the SAFT chain term to the cubic equation of state, making it a conceptually attractive model for describing the physics of nonpolar chain-like molecules, such as high molecular weight n-alkanes and polymer systems, which are known to be poorly described by standard cubic models.

The RK equation of state in pressure form is given by:

$$P = \frac{nRT}{V - B} - \frac{A}{V(V + B)} \tag{6.1}$$

where A and B are calculated by the following mixing rules:

$$A = \sum_i \sum_j n_i n_j \sqrt{\mathrm{a}_i \mathrm{a}_j}(1 - k_{ij}) \qquad B = \sum_i n_i b_i \tag{4.6}$$

and pure-component molecular attraction (a_i) and excluded volumes (b_i) are given by:

$$\mathrm{a}_i = \Omega_\mathrm{a} \frac{R^2 T_c^2}{P_c} \alpha \tag{4.10}$$

$$b_i = \Omega_b \frac{R T_c}{P_c}$$

where Ω_a and Ω_b are constants specific to the EOS tuned to match the critical point. The critical temperature (T_c) and critical pressure (P_c) are specific to the components and typically obtained experimentally. For RK EOS, $\Omega_\mathrm{a} = 0.42748$ and $\Omega_b = 0.08664$.

The alpha-function for RK EOS has a different functional form than for SRK and PR.

$$\alpha = (T/T_c)^{-\frac{1}{2}} \tag{6.2}$$

In the CPA equation of state proposed by Kontogeorgis et al.[24], the pressure of the fluid is calculated by combining Eq. (6.1) with the association term from SAFT.

$$P = \frac{nRT}{V - B} - \frac{A}{V(V + B)} + \frac{nRT}{V} \sum_S \left[\frac{1}{X^S} - \frac{1}{2}\right] \rho \frac{\partial X^S}{\partial \rho} \tag{6.3}$$

where X^S is the unbonded monomer fraction of site S. The parameters a_c and α (used for calculating A) and B are retuned to match saturation pressure and density. T_c and P_c are not used as model parameters in CPA as they are in RK, and CPA sacrifices the ability to match the critical point. Additionally, CPA requires an optimization routine inside the volume solver to converge X^S, which makes it much more computationally expensive than the standard cubic models and more on par with models like SAFT.

Conversely, CPC retains the expressions in Eqs. (4.6) and (4.10) and the parameters T_c and P_c as in RK. The constants Ω_a and Ω_b are reformulated as a function of the chain length (m) such that the critical point is matched for CPC for any positive value of m, which is tuned for each pure component to match saturation pressure and density reported in the literature[102]. The pressure of the fluid is calculated by combining Eq. (6.1) with the chain term from SAFT.

$$P = \frac{nRT}{V}\left(1 - \frac{mB}{V - B}\right) - \frac{mA}{V(V + B)} - nRT\left[\frac{(m-1)}{g(\rho)}\frac{\partial g(\rho)}{\partial \rho}\right] \tag{6.4}$$

where m is the chain length or segment number and $g(\rho)$ is the radial distribution function.

6.2. Theory

The model molecule of the cubic EOS is a sphere with attractive potential. The magnitude of the attraction force is captured by "A" and the excluded volume of the sphere is given by "B", which are both calculated from values of T_c and P_c. The assumption of a spherical molecule works well for low molecular weight n-alkanes that could be reasonably represented by this model, but long-chain molecules like high molecular weight n-alkanes and polymers certainly violate the spherical molecule assumption by any reasonable standard. Some versions of the cubic EOS offer a modification to the attraction force to account for temperature-dependence (as in Eq. (6.2)) that is often interpreted as a shape factor that relaxes the hard-sphere assumption. Even with this modification, the cubic EOS model molecule is, at best, a sphere of variable attraction energy, volume, and rigidity.

To address this shortcoming, CPC employs the cubic EOS to describe the physics of the monomer and then uses the SAFT chain term to bond the monomer beads to form linear chains. The model molecule in CPC is a smaller, homogeneous monomer bonded tangentially to form chains, which is much more consistent with actual polymer physics than the representation given by the cubic EOS. A graphical representation comparing the model molecule in the cubic EOS to that in CPC is given in **Figure 6.1**.

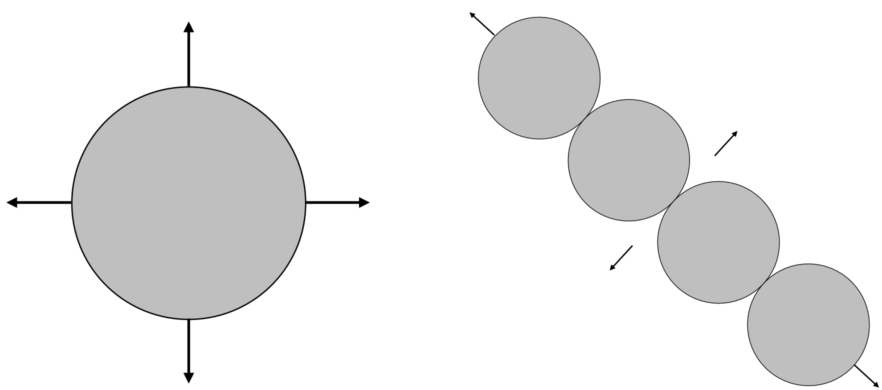


Figure 6.1. (L) cubic EOS model molecule. (R) CPC model molecule.

In terms of the reduced residual Helmholtz, the cubic-plus-chain (CPC) equation of state is written:

$$F = \frac{A^R(T, V, \boldsymbol{n})}{RT} = ma^{\text{mon}} + a^{\text{chain}} \tag{6.5}$$

where the monomer term (mon) accounts for the repulsive (rep) and attractive (att) contributions of the cubic EOS and the chain term accounts for the monomer bonding contribution from SAFT.

The monomer and chain terms of CPC are given by:

$$a^{\mathrm{mon}} = a^{\mathrm{rep}} + a^{\mathrm{att}} \tag{6.6}$$

$$a^{\mathrm{rep}} = -n \ln(1-\beta) \tag{6.7}$$

$$a^{\mathrm{att}} = -\frac{A}{BRT} \ln(1+\beta) \tag{6.8}$$

$$a^{\mathrm{chain}} = -n(m-1) \ln[\mathrm{g}(\beta)] \tag{6.9}$$

where RK has been chosen as the monomer equation of state. CPC simplifies to only the monomer term when $m = 1$. Additionally, the reduced volume (β) is defined as a ratio of the excluded volume of the chain to the total volume of the system: $\beta = mB/V$.

In terms of the compressibility factor, CPC can be written:

$$Z = 1 + m\beta \left[\frac{n}{1-\beta} - \frac{A}{BRT(1+\beta)}\right] - n(m-1)\beta \frac{\mathrm{g}'(\beta)}{\mathrm{g}(\beta)} \tag{6.10}$$

Writing the equation of state in terms of β instead of V is useful for applying numerical methods to the volume root-finding procedure as β is bound between 0 (min density) and 1 (max density), instead of between the excluded volume of the chain (mB) and ideal gas volume (V^{ig}).

In this work, RK is used for the monomer term and the radial distribution function (g) is that proposed by Elliott et al.[103], which is the same used for more recent versions of CPA[104].

$$\mathrm{g}(\beta) = \frac{1}{1 - 0.475\beta} \tag{6.11}$$

Because of the addition of the chain term, CPC is not the same equation of state as RK and will not match the critical point if all constants are left unmodified. To force CPC to match the critical point, Ω_a and Ω_b are re-derived as functions of chain length (m) and critical compressibility factor (Z_c). The coefficients presented in this work correspond to RK with the Elliott radial distribution function.

$$\Omega_\mathrm{a} = \frac{1}{m^2}\left[\frac{\beta_c Z_c^2}{\lambda^{\mathrm{mon}}} + (m-1)\beta_c Z_c \frac{\lambda^{\mathrm{chain}}}{\lambda^{\mathrm{mon}}}\right] \qquad \Omega_b = \frac{1}{m}\beta_c Z_c \tag{6.12}$$

$$Z_c = mZ_c^{\mathrm{mon}} - (m-1)Z_c^{\mathrm{chain}} \tag{6.13}$$

$$Z_c^{\mathrm{mon}} = \frac{1}{1-\beta_c}\left[1 + \frac{\beta_c}{\lambda^{\mathrm{mon}}(1+\beta_c)}\right]^{-1} \tag{6.14}$$

$$Z_c^{\mathrm{chain}} = \frac{\beta_c}{1+\beta_c}\left[\frac{\lambda^{\mathrm{chain}}}{\lambda^{\mathrm{mon}}} + \frac{40}{(40-19\beta_c)}\left[1 + \frac{\beta_c}{\lambda^{\mathrm{mon}}(1+\beta_c)}\right]^{-1}\right] \tag{6.15}$$

$$\lambda^{\mathrm{mon}} = -\beta_c \frac{\beta_c^2 + 2\beta_c - 1}{(1+\beta_c)^2} \qquad \lambda^{\mathrm{chain}} = \frac{840\beta_c}{(40-19\beta_c)^2} \tag{6.16}$$

All left-hand-side variables in Eqs. (6.12) - (6.16) are functions of β_c, which is calculated from a Newton-Raphson solver for a sixth-order polynomial in β_c, with coefficients for the polynomial given in **Table 6.1**.

Table 6.1. 6th order polynomial coefficients for solving β_c.

Coefficient	Expression
c_0	$-128{,}000 - 128{,}000\, m_0$
c_1	$566{,}400 + 566{,}400\, m_0$
c_2	$-249{,}840 - 748{,}800\, m_0$
c_3	$-145{,}562 + 188{,}800\, m_0$
c_4	$36{,}366 + 182{,}400\, m_0$
c_5	$45{,}486$
c_6	$-13{,}718 - 60{,}800\, m_0$

Polynomial of the form: $0 = c_0 + \cdots + c_6 \beta_c^6$

$m_0 = (1 - m)/m$

For the standard cubic equations of state, Ω_a and Ω_b and critical compressibility (Z_c) are constants that are specific to each model. For CPC, these variables are functions of the chain length (m), as shown in **Figure 6.2** and **Figure 6.3**. Note that for $m = 1$, Ω_a and Ω_b reduce to 0.42748 and 0.08664, respectively, which are the default values for RK EOS. The trends shown in **Figure 6.2** indicate that as the chain length increases, the attraction energy (A) and excluded volume (B) decrease to satisfy the criticality condition. Although Z_c is a function of m for CPC, the dependence is weak and the predicted Z_c is approximately $1/3$ regardless of chain length, which is the value predicted by RK EOS. The choice of the radial distribution function determines the magnitude of the effect that m has on Z_c.

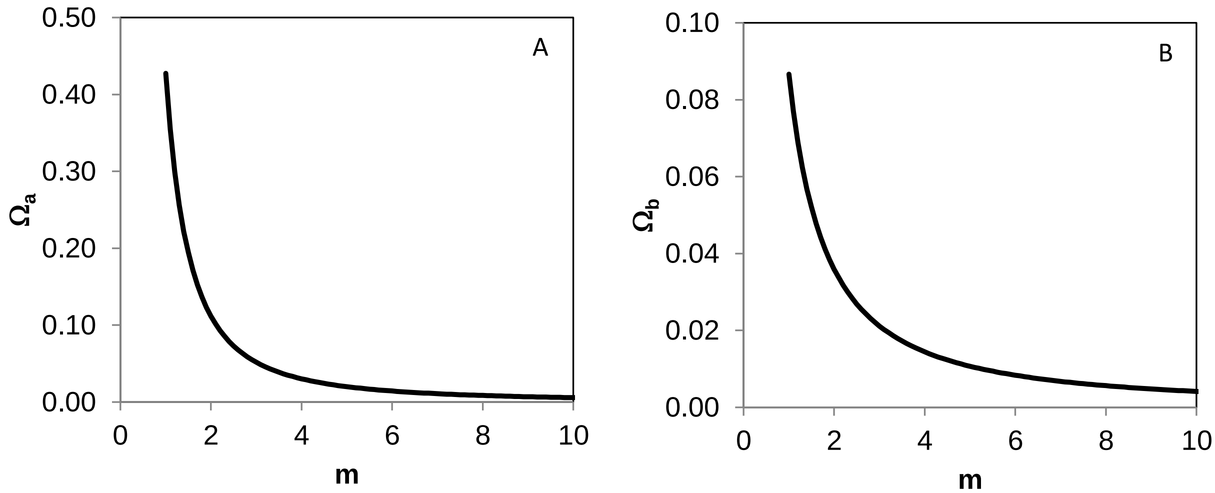


Figure 6.2. Dependence of the coefficients (A) Ω_a and (B) Ω_b on the chain length (m) in CPC EOS.

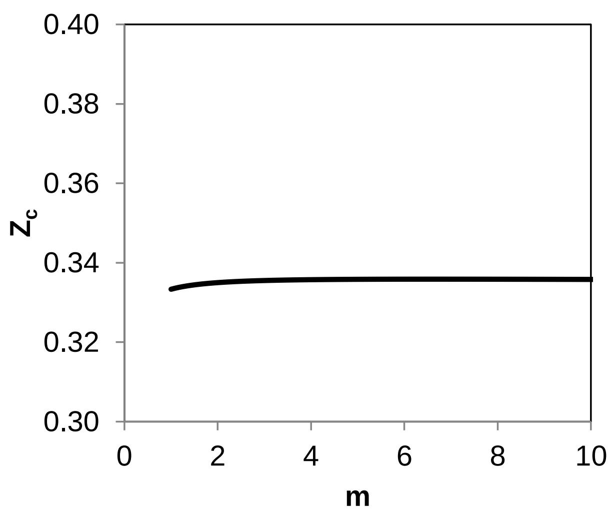


Figure 6.3. Effect of chain length (m) on the critical compressibility predicted by CPC.

6.3. CPC Parameter Optimization

6.3.1. Methodology

The input parameters for CPC are T_c, P_c, and m. The critical temperature (T_c) and critical pressure (P_c) are often obtained experimentally and available in the literature for most pure compounds[105], while the chain length (m) is tuned to match reference data for vapor pressure and density[102] in the reduced temperature range $T_r = 0.5 - 0.9$.

Because of short-comings in the RK and Soave alpha-functions[106], alternative formulations have been proposed by various researchers – including Mathias and Copeman[107], Melhem[108], and Twu[109] – to correct physically inconsistent results and to improve vapor pressure predictions. Any of these or similar alpha-functions can be substituted into CPC for the standard alpha-function, which would not alter the expressions for Ω_a, Ω_b and their auxiliary variables in Eqs. (6.12) - (6.16) because Ω_a and Ω_b are fit strictly to match the critical point while the alpha-function describes deviations from the vapor pressure curve away from the critical point.

The objective function to be minimized for parameter tuning is given by:

$$OBJ = \alpha_P \sum_{k=1}^{N_{pt}} \left| \frac{P_{CPC}^{\mathrm{sat}} - P^{\mathrm{sat}}}{P^{\mathrm{sat}}} \right| + \alpha_\rho \sum_{k=1}^{N_{pt}} \left| \frac{\rho_{CPC}^{\mathrm{sat}} - \rho^{\mathrm{sat}}}{\rho^{\mathrm{sat}}} \right| \tag{6.17}$$

where α_P and α_ρ are the weighting factors of saturation pressure and density, respectively. The sampling points are equally spaced between $T_r = 0.5 - 0.9$ and reference data for saturation pressure and density are generated from correlations provided by DIPPR[102].

The m parameter for each component was tuned independently from other components to ensure that the expected trends – such as m increasing as a function of n-alkane molecular weight – were observed without being forced. For each pure component, m is allowed any positive value and the tuning procedure searches the solution space to find m that minimizes the objective function.

6.3.2. Results and Discussion

The optimized values of chain length (m) for components C_1 to n-C_{25} along with critical properties are provided in **Table 6.2.** The chain length m in CPC increases with increasing carbon number for the n-alkanes, as expected. This suggests that m in CPC can be correlated to M_W for n-alkanes in a similar manner as m is correlated to M_W in PC-SAFT[36]. **Figure 6.4** shows that the relation between m and M_W is virtually linear for n-alkanes. The regressed line in **Figure 6.4** is forced to an intercept of zero. The resulting straight line indicates that m/M_W has, to a good approximation, a constant value of 0.04155 for n-alkanes.

Table 6.2. Critical properties and chain length (m) for select n-alkanes.

Component	M_W **g/mol**	T_c **K**	P_c **bar**	m **-**
C_1	16.04	190.6	45.99	0.8973
C_2	30.07	305.3	48.72	1.3301
C_3	44.10	369.8	42.48	1.7047
C_4	58.12	425.1	37.96	2.0854
C_5	72.15	469.7	33.70	2.6278
C_6	86.18	507.6	30.25	3.1708
C_7	100.20	540.2	27.40	3.7948
C_8	114.23	568.7	24.90	4.4125
C_9	128.26	594.6	22.90	4.9937
C_{10}	142.28	617.7	21.10	5.6575
C_{11}	156.31	639.0	19.50	6.2173
C_{12}	170.33	658.0	18.20	6.8653
C_{13}	184.36	675.0	16.80	7.5387
C_{14}	198.39	693.0	15.70	8.0164
C_{15}	212.41	708.0	14.80	8.7008
C_{16}	226.44	723.0	14.00	9.1951

C_{17}	240.47	736.0	13.40	10.0490
C_{18}	254.49	747.0	12.70	10.6229
C_{19}	268.52	758.0	12.10	11.1571
C_{20}	282.55	768.0	11.60	11.7654

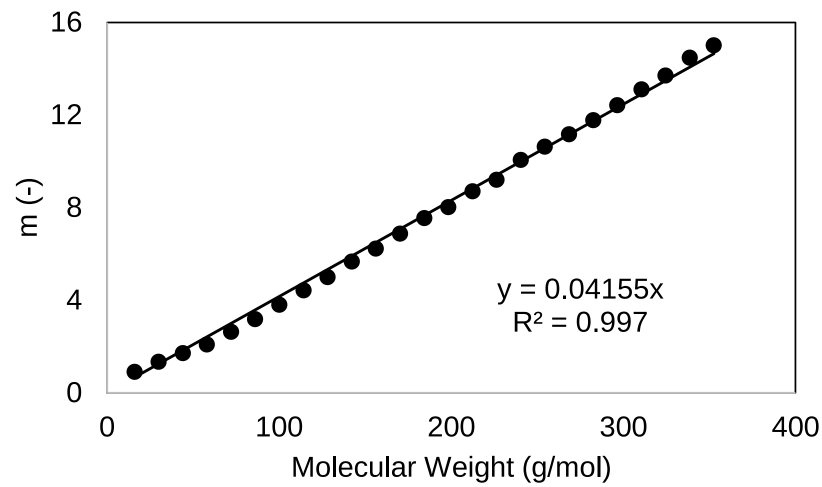


Figure 6.4. Chain length (m) in CPC as a function of M_W for n-alkanes.

Because CPC is forced to match the critical point and tuned to minimize differences between simulation and reference data for saturation pressure and density, it is expected to outperform the standard RK EOS even for short-chain alkanes like propane, but the conceptual framework of CPC is designed specifically for long-chain nonpolar molecules like polymers. Saturation pressure and density plots comparing RK, CPC, and PC-SAFT[36] to reference data[102] are given in Figure 6.5 for propane, Figure 6.6 for n-decane, and Figure 6.7 for pentadecane. Subcooled liquid density plots comparing RK, CPC, and PC-SAFT to reference data are shown in Figure 6.8 and Figure 6.9. Time comparisons for these three equations of state are shown in **Figure 6.10.** To solve the volume loop and fugacity coefficient equations, CPC is approximately 1.5x slower than RK whereas PC-SAFT is approximately 5.6x slower.

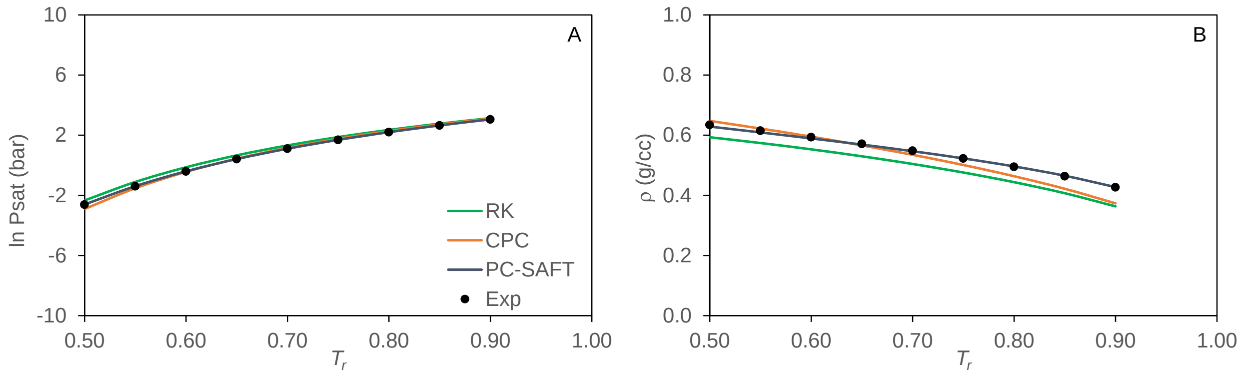


Figure 6.5. Saturation pressure (A) and liquid density (B) for C_3 from $T_r = 0.5 - 0.9$.

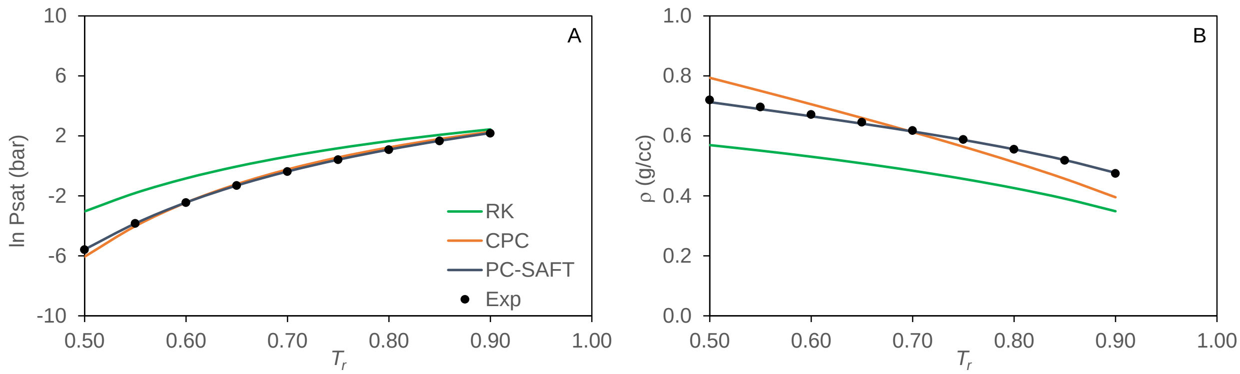


Figure 6.6. Saturation pressure (A) and liquid density (B) for n-C_{10} from $T_r = 0.5 - 0.9$.

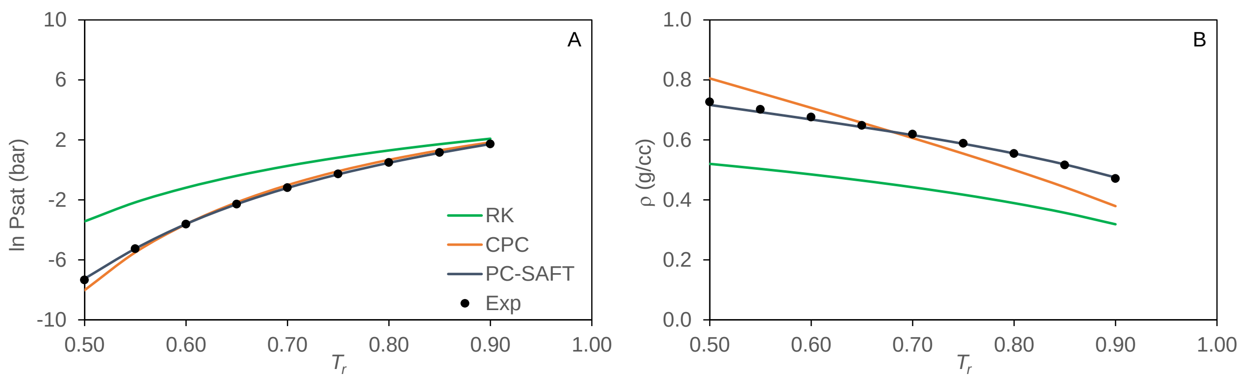


Figure 6.7. Saturation pressure (A) and liquid density (B) for n-C_{15} from $T_r = 0.5 - 0.9$.

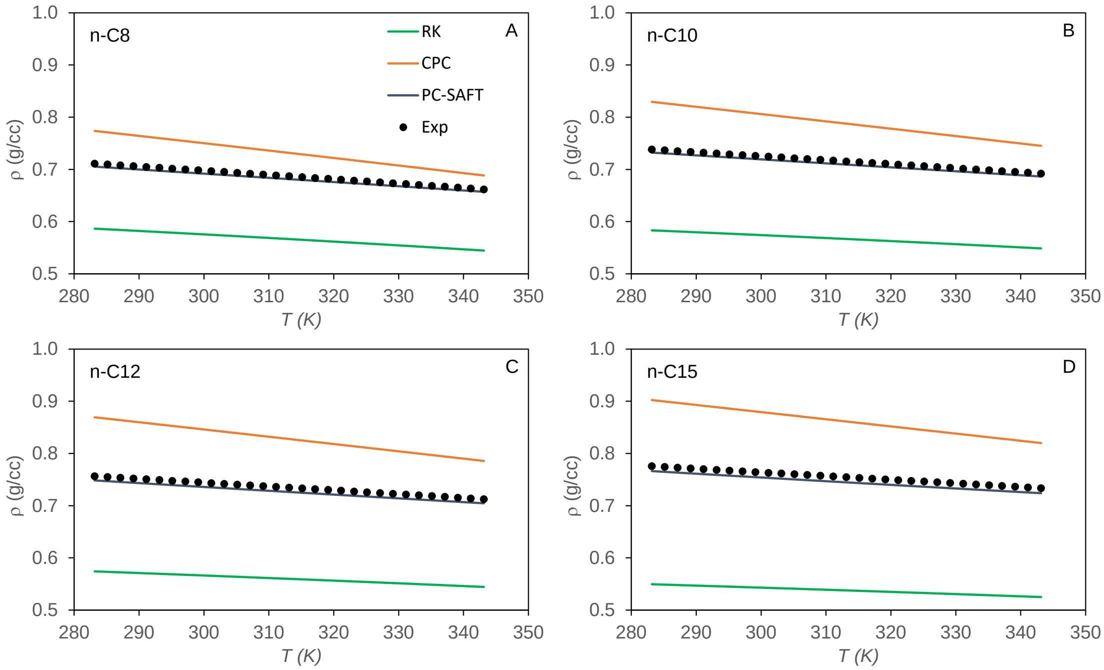


Figure 6.8. Density as a function of temperature at $P = 1$ bar for (A) n-C_8 (B) n-C_{10} (C) n-C_{12} (D) n-C_{15}.

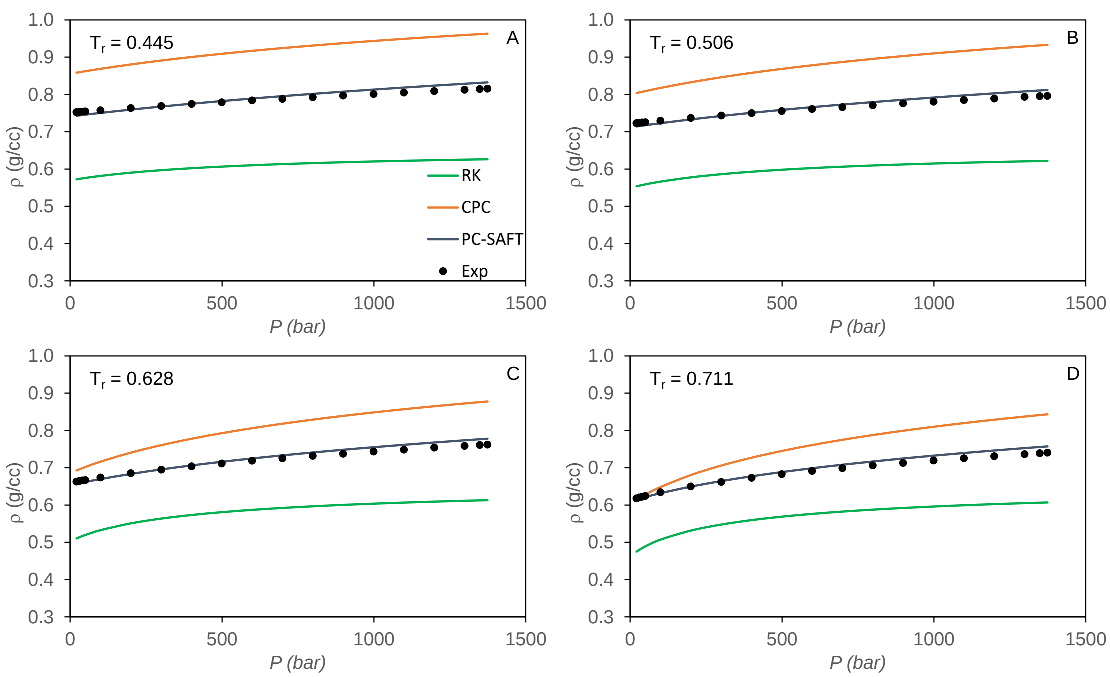


Figure 6.9. Density as a function of pressure for n-C_{12} at (A) T_r = 0.445, (B) T_r = 0.506, (C) T_r = 0.628, and (D) T_r = 0.711.

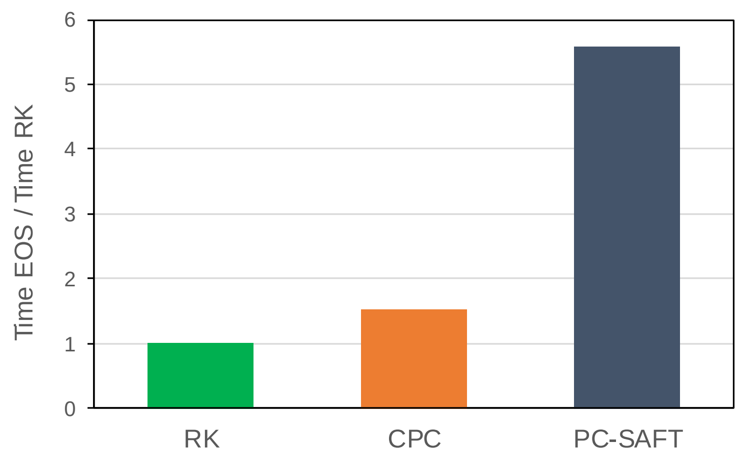


Figure 6.10. Time comparison for RK, CPC and PC-SAFT to solve volume loop and fugacity coefficient equations. Simulations were executed for n-C_1 to n-C_{10} between $T_r = 0.5 - 0.9$ and $P_r = 0.5 - 0.9$. Comparison presented as ratio per elapsed time for RK.

6.4. CPC Function Behavior

6.4.1. Solving the Volume Roots for CPC

Newton-Raphson algorithms are often the most efficient techniques for converging to the volume roots that satisfy the equation of state. Newton-Raphson (NR) algorithms for volume root-finding require, at minimum, differentiating the pressure and/or compressibility factor expressions with respect to volume or density. For higher-order Newton-Raphson algorithms, higher-order derivatives are required. In this section, an approach is outlined for solving the volume roots for CPC with a second-order NR procedure with analytic derivatives of the compressibility factor (Z) with respect to reduced volume (β).

To avoid mistakes in differentiation and in programming, it is recommended that all properties of interest be recast in terms of derivatives of the residual Helmholtz function. Each term of the residual Helmholtz function can be differentiated individually and then combined to give the final expression. The compressibility factor equation is given in Eq. (6.10) and the derivatives of the compressibility factor, along with other important properties for CPC, are provided in Appendix B.

The root-finding procedure preferred by the authors is nearly identical to that proposed by Michelsen and Mollerup[1] for the standard cubic EOS models. In their formulation, each successive approximation for β is obtained from either a first-order or second-order Newton-Raphson formula:

$$\beta^{j+1} - \beta^{j} = \Delta \qquad \text{1st order NR} \tag{6.18}$$

$$\beta^{j+1} - \beta^{j} = \Delta\left(1 + \frac{1}{2}\frac{\Delta}{\Delta_1}\right) \qquad \text{2nd order NR} \tag{6.19}$$

where Δ and Δ_1 are calculated by:

$$\Delta = -\frac{h(\beta^{j})}{h_{\beta}(\beta^{j})} \qquad \Delta_1 = -\frac{h_{\beta}(\beta^{j})}{h_{\beta\beta}(\beta^{j})} \tag{6.20}$$

When $|\Delta/\Delta_1| \leq 1$, the second-order NR approximation is used. If $(\Delta/\Delta_1) < -1$, then (Δ/Δ_1) is set equal to -1 before execution of Eq. (6.19). In the case that $\Delta > \Delta_1$, the calculation is unlikely to reach a solution due to either divergence or the presence of a stationary point. In this case, a new value of β is estimated from:

$$\beta^{j+1} - \beta^{j} = \Delta_1 \tag{6.21}$$

If more than one root is found, then the stable root is that corresponding to a lower Gibbs energy.

Michelsen and Mollerup defined $h(\beta)$ as the difference between the product βZ calculated from the EOS and βZ evaluated at $V = B$. Because the excluded volume of the molecule for CPC is given by mB, the function $h(\beta)$ and its derivatives are given as:

$$h(\beta) = \beta Z - \frac{PmB}{nRT} \tag{6.22}$$

$$h_\beta = Z + \beta \frac{\partial Z}{\partial \beta} \tag{6.23}$$

$$h_{\beta\beta} = 2\frac{\partial Z}{\partial \beta} + \beta \frac{\partial^2 Z}{\partial \beta^2} \tag{6.24}$$

6.4.2. Isotherms for CPC

The pressure form of CPC in terms of the reduced volume β is given by:

$$P = \frac{nRT}{B}\left[\frac{\beta}{1-\beta} - \frac{A}{BRT}\left(\frac{\beta^2}{1+\beta}\right) + \frac{(1-m)}{m}\left(\frac{\beta}{1-0.475\beta}\right)\right] \tag{6.25}$$

which can be rearranged to yield a polynomial in β of the form:

$$\beta^4 + C_3\beta^3 + C_2\beta^2 + C_1\beta + C_0 = 0 \tag{6.26}$$

where the coefficients of the polynomial are given by:

$$C_3 = -\frac{59}{19} + \frac{B^2P}{A} - \frac{21}{19}\frac{BRT}{A} + \frac{40}{19m}\frac{BRT}{A} \tag{6.27}$$

$$C_2 = \frac{40}{19} - \frac{40}{19}\frac{B^2P}{A} - \frac{21}{19}\frac{BRT}{A}$$

$$C_1 = \frac{-B^2P}{A} - \frac{40}{19m}\frac{BRT}{A}$$

$$C_0 = \frac{40}{19}\frac{B^2P}{A}$$

Eq. (6.26) indicates that CPC is 4th order in β, one order higher than RK EOS. The order of β in CPC depends on the RDF and can become higher than four if a different RDF is selected.

At a specified temperature and pressure, CPC yields four solutions for β, whereas the standard cubic EOS models yield three. The additional root predicted by CPC is found to always correspond to a value of $\beta > 1$, which is outside the physical range ($0 \leq \beta \leq 1$). Within the physical range, CPC predicts a maximum of three real roots, similar to the standard cubic models. The highest of these real roots corresponds to a liquid reduced volume and the lowest corresponds to a vapor reduced volume, while the intermediate root does not have a physical meaning. An important implication of this behavior is that the predicted isotherms using CPC are similar in shape to those predicted by the standard cubic EOS within the physical range. **Figure 6.11** shows P-V isotherms from CPC and RK for n-pentane at different temperatures.

The similar behavior of CPC as compared to standard cubic EOS within the physical range implies that they have similar computational characteristics. When using the volume solver algorithm described in Section 6.4.1, only the physical region is searched for the solution, meaning that the convergence behavior for CPC is similar to that of RK since in the physical region they behave similarly.

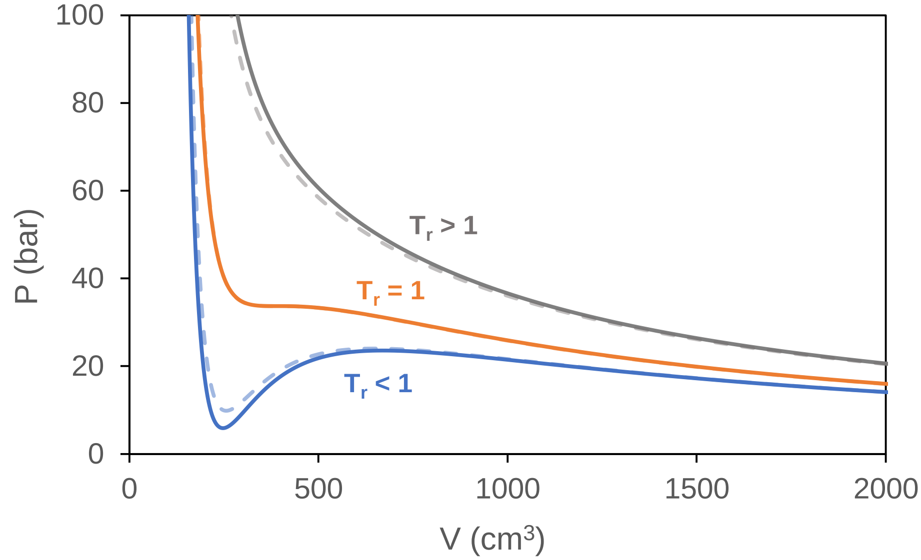


Figure 6.11. P-V isotherms predicted by CPC (solid lines) and RK (dashed lines) for n-pentane.

6.5. Applying CPC to Polymer Phase Behavior

The SAFT family of EOS models have shown vast improvement over the standard cubic EOS models in describing polymer phase behavior, mainly due to the addition of the chain term. Whereas the standard cubic EOS models treat molecules as spherical segments of variable volume and interaction energy, the SAFT models treat molecules as chains of spherical segments. Because of the problems inherent to the cubic EOS model molecule, polymers represent a particularly challenging system, and various researchers have tried modifying the mixing rules or the form of the A and B parameter calculations[110–113]. However, modifications that do not affect the underlying equation of state – and the molecular representation that causes long-chain molecules to be poorly described – will sacrifice property predictions in some areas.

Because the cubic-plus-chain EOS is specifically designed to handle long-chain nonpolar molecules by affecting the underlying physics of the molecule, including its structure and interaction energy, it is a particularly attractive model for describing polymer phase behavior. The approach for tuning the CPC parameters for polymers is also intuitive. Because polymers do not have critical points, it is meaningless to tune polymer simulation parameters to fit some phenomena that cannot be observed in practice, so parameter tuning is different than that presented in Section 6.3. Instead, polymers are treated as chains of monomer building blocks in which the monomers themselves are tuned via the standard procedure. Thus, polypropylene, whose structure is shown in **Figure 6.12**, can be viewed as a chain of repeating propane monomers. The CPC model molecule representation of polypropylene, shown also in **Figure 6.12**, is then the CPC propane monomer repeated m^{PP} times.

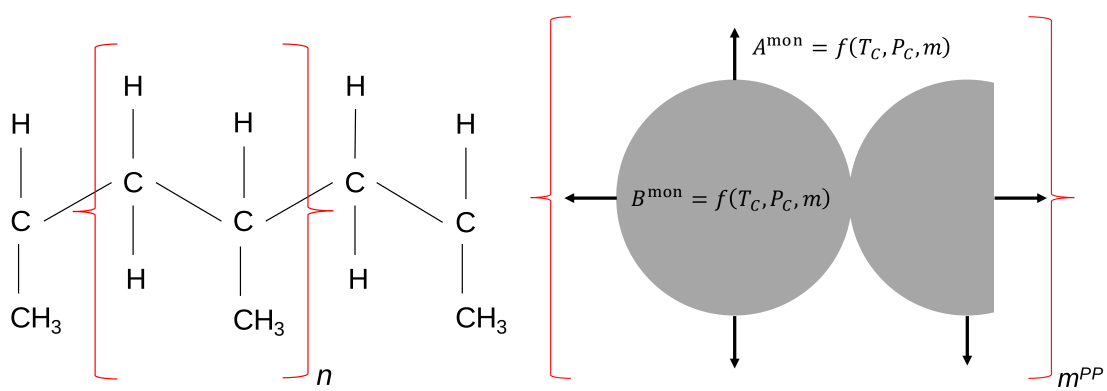


Figure 6.12. Polypropylene (L) actual molecular structure (R) CPC model molecule structure. A^{mon} and B^{mon} are calculated from the pure-component parameters of propane. The number of repeating propane monomer units in the CPC representation of polypropylene (m^{PP}) is tuned to match liquid density.

Density predictions for polypropylene with CPC and PC-SAFT are shown in **Figure 6.13** and **Figure 6.14** and compared to experimental data[114]. For both EOS models, only the chain length (m) was treated as an adjustable parameter. CPC outperforms PC-SAFT in these predictions across the broad range of temperatures and pressures studied. The total error in density and the EOS parameters for polypropylene are listed in **Table 6.3**.

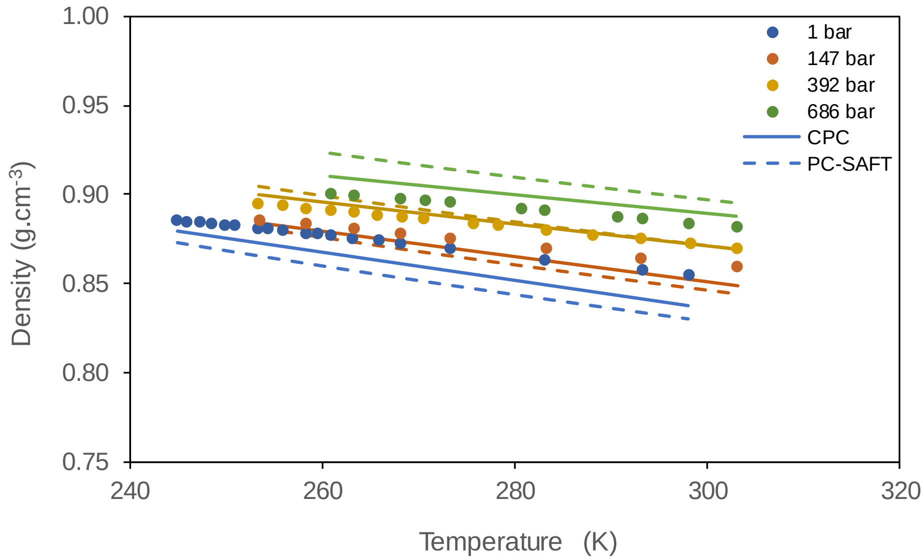


Figure 6.13. Density predictions for polypropylene as a function of temperature for CPC (solid) and PC-SAFT (dashed).

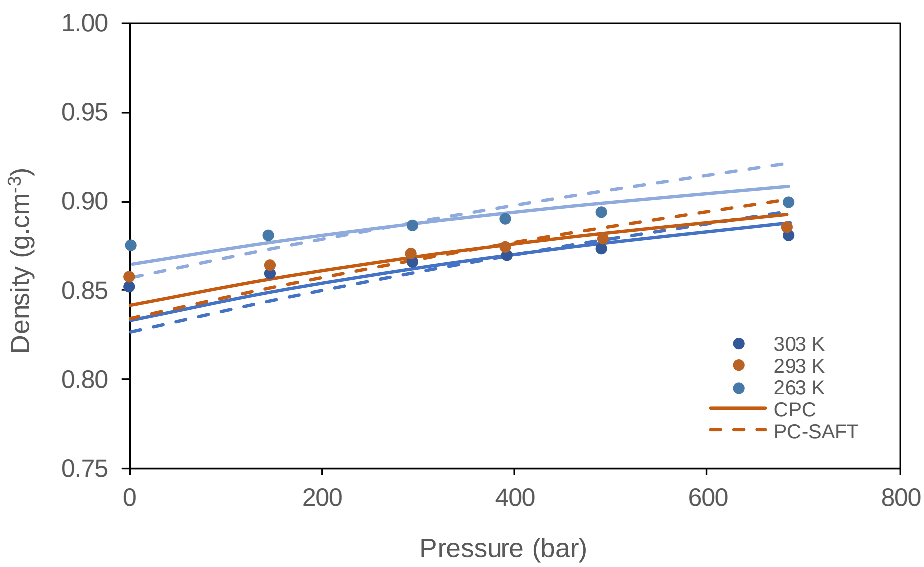


Figure 6.14. Density predictions for polypropylene as a function of pressure for CPC (solid) and PC-SAFT (dashed).

Table 6.3. Polypropylene simulation parameters for CPC and PC-SAFT with % error in density for predictions shown in **Figure 6.13** and **Figure 6.14**.

		CPC				PC-SAFT				
	M_W kg/mol	m/M_W -	T_c^{mon} K	P_c^{mon} bar	m^{mon} -	m/M_W -	σ Å	ε/kB K	ρ^{CPC} % APD	ρ^{PCS}
PP	15.70	0.0177	369.8	42.48	1.7047	0.0246	4.10	217.0	0.63	1.12

For CPC, the monomer parameters are used to calculate Ω_a and Ω_b as outlined in Eqs. (6.12) - (6.16), which will then yield the same values for A and B used for the monomer. The chain length of the polymer (m^{poly}) affects the equation of state through the pressure equation and radial distribution function but not in the values of the monomer interaction energy (A^{mon}) and monomer excluded volume (B^{mon}).

$$P = \frac{nRT}{V}\left(1 - \frac{m^{\text{poly}}B^{\text{mon}}}{V - B^{\text{mon}}}\right) - \frac{m^{\text{poly}}A^{\text{mon}}}{V(V + B^{\text{mon}})} - nRT\left[\frac{\left(m^{\text{poly}} - 1\right)}{\text{g}(\rho)}\ \frac{\partial \text{g}(\rho)}{\partial \rho}\right] \qquad (6.28)$$

6.6. Conclusions and Future Work

CPC outperforms RK for prediction of saturation pressure and density across the family of n-alkanes. For low molecular weight n-alkanes which are represented reasonably well by the spherical molecule assumption implicit in the cubic equation of state, the chain term is relatively unimportant and the predictions with CPC do not show significant improvement. However, for high molecular weight n-alkanes, the spherical molecule assumption in the cubic equation of state is especially poor and predictions suffer severely. The physics implicit in the CPC model – primarily in the representation of the model molecule as spherical homogenous beads covalently bonded to form chains

– matches much more closely to how long-chain n-alkanes exist in reality, so CPC shows dramatically better predictions for these types of systems than RK shows.

The new equation of state proposed in this work is primarily a proof-of-concept of the CPC framework, showing that the addition of the chain term to the cubic equation of state can offer substantial improvements to the model without adding significant computational time. The framework is not restricted to a specific equation of state for the monomer or radial distribution for the chain term. For some systems, RK, SRK or PR paired with another radial distribution function – such as those proposed by Carnahan and Starling[41] or Gow[115] – could show better results than the version of CPC (RK + Elliott chain term) proposed here. There are numerous variations of CPC that can be constructed from hybridizing any cubic equation of state with a suitable radial distribution function, and future work should focus on identifying which combinations are best for which systems.

Additionally, CPC is an attractive model for heavy hydrocarbon systems like crude oils. Standard cubic equations of state such as SRK and PR are still being used in the oil and gas industry for crude oil modeling because they are much less computationally expensive than the more advanced equations of state – such as PC-SAFT and CPA. CPC is only marginally slower than RK and yet provides significant improvement in density predictions. CPC does not require an internal minimization procedure nested in the volume-solving loop that the association models require. Therefore, CPC offers a reasonable balance between computational time and accuracy in density predictions which makes it an attractive model for engineering applications, especially in the oil and polymer industries.

Chapter 7

Predicting Phase Behavior of Nonpolar Mixtures from Refractive Index at Ambient Conditions

7.1. Introduction

A generalized cubic equation of state in pressure-explicit form was introduced previously in Section 4.4:

$$P = \frac{RT}{V - B} - \frac{A}{(V + \delta_1 B)(V + \delta_2 B)} \tag{4.8}$$

with the variables A and B calculated from mixing rules for the pure component values of a_i and b_i, respectively. The standard van der Waals mixing rules are:

$$A = \sum_i \sum_j n_i n_j \sqrt{\mathrm{a}_i \mathrm{a}_j}\left(1 - k_{ij}\right) \qquad B = \sum_i n_i b_i \tag{4.6}$$

where n_i is the mole number of component i and k_{ij} is the binary interaction parameter between components i and j. The binary interaction parameters are often tuned to match saturation pressure data. The variables A and B represent mixture averages of the molecular interaction energies and excluded volumes, respectively. The pure-component values of a_i and b_i are calculated by:

$$\mathrm{a}_i = \Omega_\mathrm{a} \frac{R^2 T_c^2}{P_c} \alpha \tag{4.10}$$

$$\alpha = \left[1 + \kappa(\omega)\left[1 - \sqrt{T/T_c}\right]\right]^2$$

$$b_i = \Omega_b \frac{R T_c}{P_c}$$

The critical temperature (T_c), critical pressure (P_c) and acentric factor (ω) are component-specific parameters while δ_1, δ_2, Ω_a, Ω_b are constants specific to the EOS. The values of these constants, as well as $\kappa(\omega)$, are shown in **Table 4.1** for two cubic EOS: Soave-Redlich-Kwong (SRK), and Peng-Robinson (PR) equations of state. The SRK[22] and PR[23] equations of state are the most commonly used cubic EOS in the oil and gas industry.

With the critical properties (T_c, P_c, ω) and binary interaction parameters (k_{ij}) defined for all components present in the mixture, modeling with cubic EOS can be performed across a wide range of temperatures and pressures, and phase behavior predictions – including saturation pressure, density, thermal expansion, solubility parameter, etc. – can be generated.

Evangelista and Vargas[116] proposed correlations relating critical properties for nonpolar hydrocarbons to simple laboratory measurements performed at ambient conditions. These correlations are referred hereafter as the EV correlations:

$$T_C(K) = 373.49\, F_{RI}^{2/3} M_W^{2/7} \tag{7.1}$$

$$P_C(\text{bar}) = 2.70 \times 10^4\, F_{RI}^{8/5} M_W^{-1} \tag{7.2}$$

$$\hat{V}_C(\text{cm}^3/\text{mol}) = 0.47\, F_{RI}^{-8/9} M_W^{6/5} \tag{7.3}$$

$$\omega = 8.62 \times 10^{-4} F_{RI}^{-1} M_W \tag{7.4}$$

where M_W has units of g/mol and F_{RI} is calculated from refractive index or density measurements performed at 20 °C and 1 atm. The expression relating F_{RI} to refractive index is given as:

$$F_{RI} = \frac{n_D^2 - 1}{n_D^2 + 2} \tag{7.5}$$

where n_D is the refractive index measured at the yellow sodium-D line.

If refractive index is not reported, then F_{RI} can be calculated from a mass density correlation proposed by Vargas and Chapman[28]:

$$F_{RI} = 0.5054\rho - 0.3951\rho^2 + 0.2314\rho^3 \approx \frac{\rho}{3} \tag{7.6}$$

where ρ is mass density in units of g/cm^3.

Applying the EV correlations within the cubic EOS framework, critical properties for multicomponent nonpolar solvent mixtures can be calculated from simple volumetric or optical measurements without knowledge of the composition of the mixture.

7.2. Methodology

7.2.1. Experimental Methodology

Experimental data for density and refractive index of nonpolar hydrocarbon mixtures were reported by Wang et al.[117] Density measurements were performed with the Anton Paar DMA 4500 digital vibrating U-tube densitometer and refractive index measurements were performed with the automatic Anton Paar WR refractometer under the sodium-D wavelength with a measuring range of 1.30 to 1.72. Table 7.1 provides a list of suppliers and chemical purity for the studied components. Wang et al.[117] also performed refractive index and density measurements on the mixtures reported in **Table 7.2**. These mixtures constitute the pseudo-pure fractions used in this modeling study.

Table 7.1. Supplier, purity, and experimental data measured at 25 °C and 1 bar for studied hydrocarbons.[117]

Component	Supplier	Purity mol%	n_D -	ρ g/cm³
n-C_6	Alfa Aesar	99	1.3724	0.6554
n-C_7	Sigma-Aldrich	≥ 99	1.3852	0.6797
n-C_{10}	Sigma-Aldrich	≥ 99	1.4097	0.7266
n-C_{12}	Sigma-Aldrich	≥ 99	1.4195	0.7452
Benzene	OmniSolv	≥ 99.7	1.4975	0.8736
Toluene	Sigma-Aldrich	≥ 99.9	1.4938	0.8622
p-Xylene	Alfa Aesar	99	1.4929	0.8567

Refractive Index measurement uncertainty: u(T) = 0.06 °C, u(n_D) = 0.0002 with a 95% confidence interval
Density measurement uncertainty: u(T) = 0.02 °C, u(ρ) = 0.0004 g/cm3 with a 95% confidence interval

Table 7.2. Properties of selected mixtures measured at 20 °C and 1 bar to be used as pseudo-components.[117]

	Mixtures			Amount (mol%)			M_W	n_D	ρ
	Comp 1	**Comp 2**	**Comp 3**	z_1	z_2	z_3	**g/mol**	**-**	**g/cm³**
B1	n-C_7	n-C_{12}	-	63.0	37.0	-	126.18	1.4047	0.716
B2	n-C_7	Toluene	-	47.9	52.1	-	96.00	1.4345	0.764
B3	Toluene	p-Xylene	-	53.6	46.4	-	98.64	1.4957	0.864
T1	n-C_6	n-C_{10}	n-C_{12}	33.5	33.3	33.2	132.83	1.4080	0.722
T2	n-C_6	n-C_{12}	Toluene	33.3	33.4	33.3	116.24	1.4254	0.751
T3	Toluene	Benzene	p-Xylene	33.3	33.7	33.0	92.04	1.4968	0.867

7.2.2. Modeling Methodology

The classical example for applying equations of state to pure components or mixtures is to use defined components with known compositions and critical properties reported in the literature[103]. However, many applications of EOS models are for complex systems – such as crude oils – for which a full, detailed compositional analysis is infeasible to obtain experimentally and the critical properties of some components are either not reported or might not be readily measurable. For these cases, researchers have developed characterization techniques with correlations to calculate critical properties for the otherwise undefined fractions of the system. One such example is the characterization technique of Pedersen et al.[87] that uses density and molecular weight correlations to specify the critical properties for the heavy carbon number fractions of a crude oil[118].

A similar modeling approach is proposed in this section, with critical properties for pseudo-pure fractions calculated from refractive index or density measurements. The critical properties for each pseudo-component are calculated using the EV correlations with the refractive index and molecular weight measurements that are reported in **Table**

7.2. Critical properties for pure components used in this modeling study are shown in **Table 7.3**. To compare the approach proposed in this section (called the "lumped" approach because the multicomponent solvent is treated as a single pseudo-component) to the standard modeling approach, saturation curves are generated with PR EOS for both approaches. The systems modeled in this section are formed by adding a light key component (propane, hydrogen sulfide or carbon dioxide) in various proportions to the solvent fractions reported in **Table 7.2** with 1-methylnaphthalene (1-MN) as the heavy key component. In the lumped approach, critical properties of the pseudo-pure solvents are calculated from the EV correlations, while the standard approach uses defined components with critical properties reported in the literature[103].

Table 7.3. Simulation parameters for studied pure components.

Component	M_W **g/mol**	T_c **K**	P_c **bar**	ω **-**
C_3	44.096	369.85	42.48	0.1520
H_2S	34.081	373.10	90.00	0.1005
CO_2	44.010	304.13	73.77	0.2230
n-C_6	86.175	507.82	30.18	0.2979
n-C_7	100.202	540.13	27.27	0.3498
n-C_{10}	142.285	617.70	21.20	0.4890
n-C_{12}	170.339	658.20	18.24	0.5750
Benzene	78.114	562.16	48.98	0.2080
Toluene	92.141	591.79	41.04	0.2630
p-Xylene	106.168	616.20	35.11	0.3180
1-MN	142.200	772.00	36.00	0.3416

7.3. Modeling Results

Phase diagrams for the studied systems are shown in Figure 7.1 with critical points represented by open squares. The molar composition of these systems is given by the ratios associated with each curve. For example, 80:20:10 means that 80 parts by mole of the mixture consists of the light key, 20 parts is the mixed solvent and 10 parts is the heavy key. For the lumped solvent approach, the amount of the solvent fraction is precisely the second number in the composition vector. In the standard de-lumped approach, that solvent amount is subdivided into defined components and amounts according to the molar compositions reported in **Table 7.2**. In addition to saturation pressure, various properties of the bulk solvent phase are of interest, and it is important that the proposed modeling approach does not introduce significant error to these predictions. **Figure 7.2**, **Figure 7.3**, and **Figure 7.4** show parity plots comparing thermal expansion coefficient, isothermal compressibility, and density, respectively, of the saturated liquid phase calculated by the proposed approach (EV) to the standard modeling approach (Std). Points lying below the 45° line signify under-predictions by the EV approach.

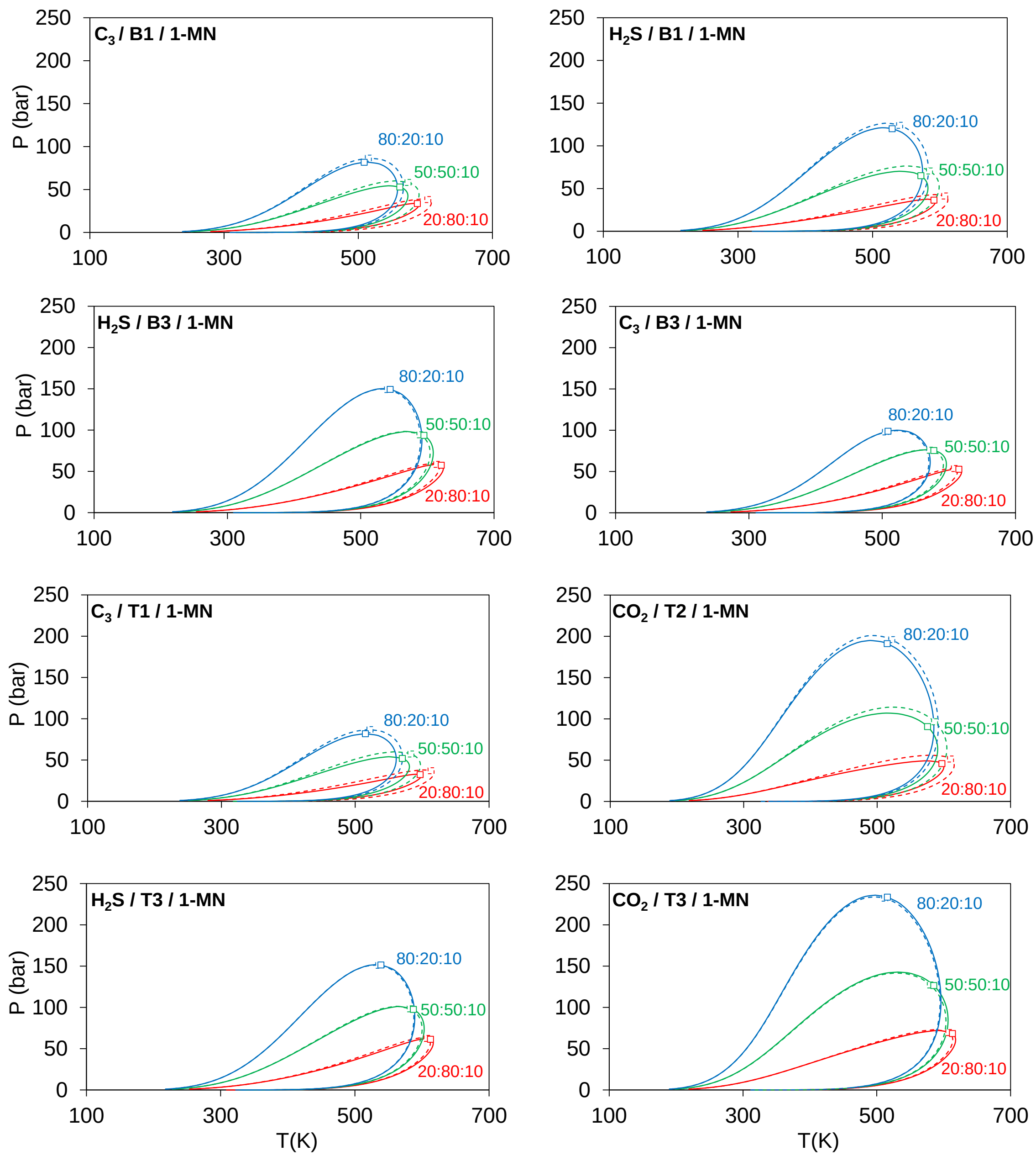


Figure 7.1. Phase diagrams generated with PR EOS for the solvents listed in **Table 7.2** with various light keys (LK) and 1-MN as the heavy key (HK). Solid lines: lumped solvent approach (EV). Dashed lines: standard de-lumped approach (Std). Composition vector given in molar amount by [LK:Solvent:HK].

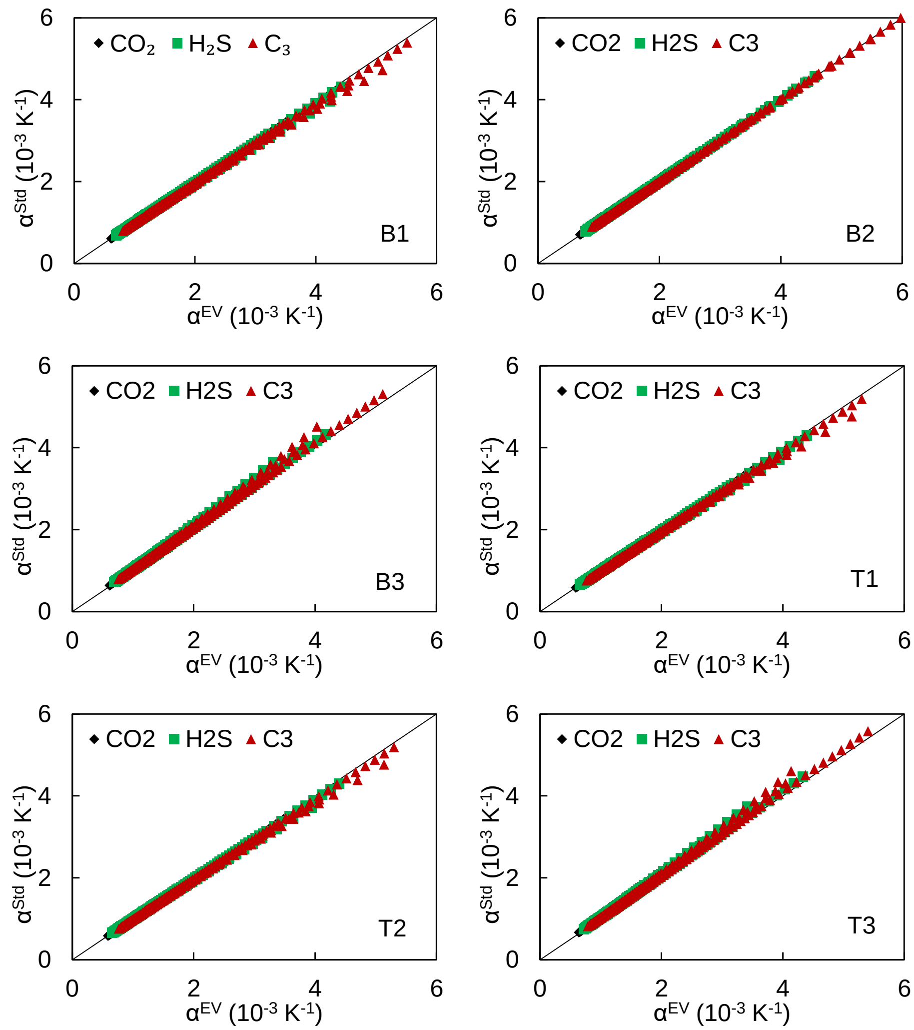


Figure 7.2. Thermal expansion coefficient (α) parity plots comparing the lumped solvent approach (EV) to the standard (Std) de-lumped modeling approach with PR EOS.

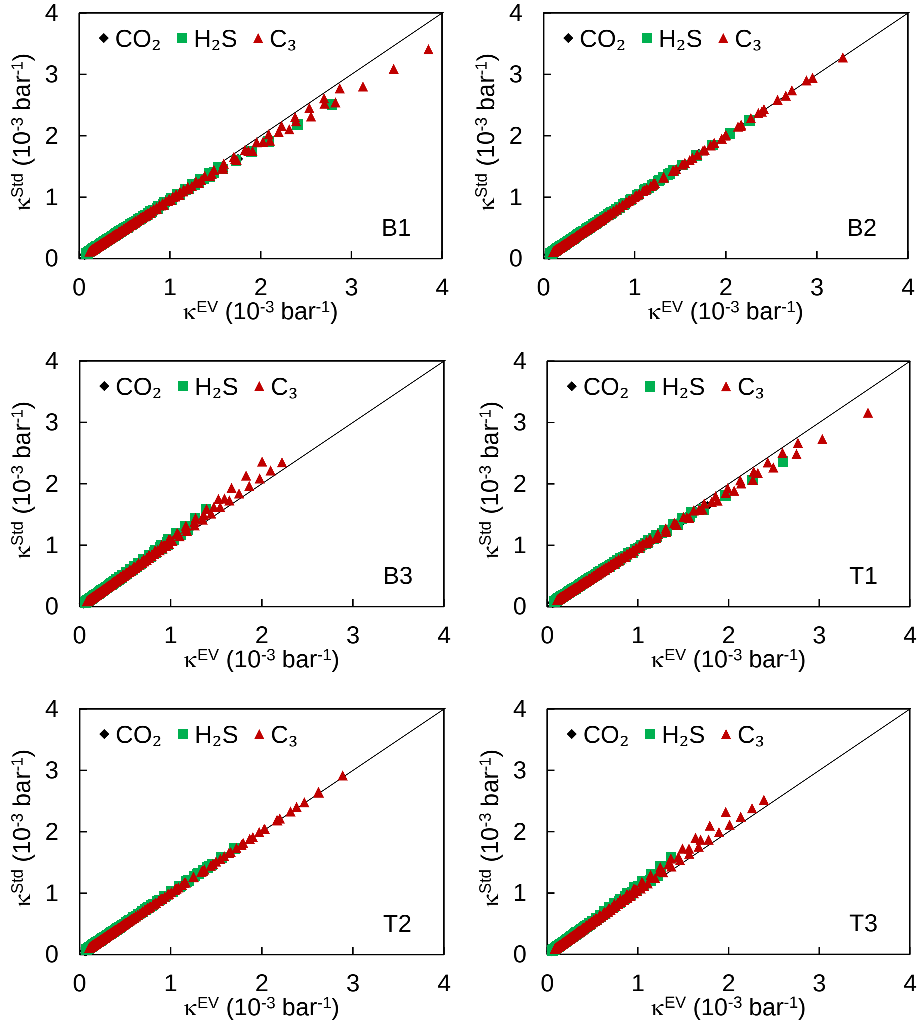


Figure 7.3. Isothermal compressibility (κ) parity plots comparing the lumped solvent approach (EV) to the standard (Std) de-lumped modeling approach with PR EOS.

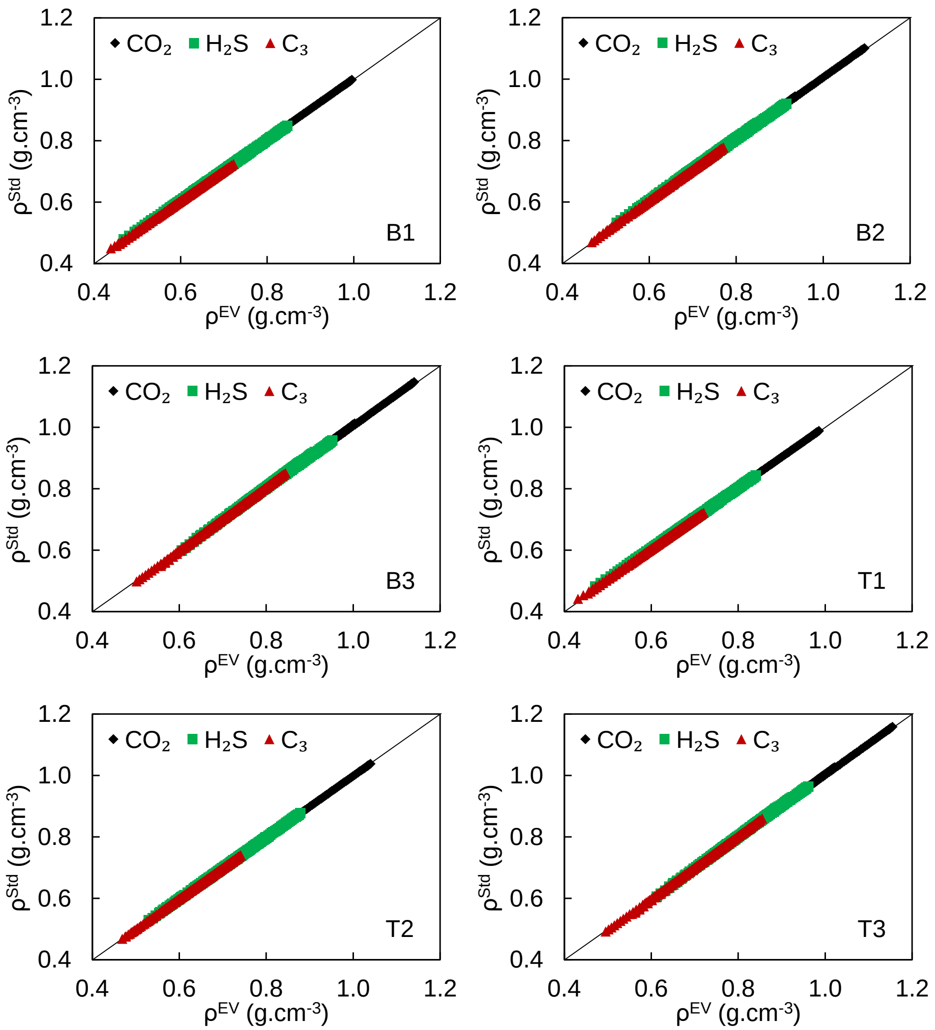


Figure 7.4. Density (ρ) parity plots comparing the lumped solvent approach (EV) to the standard (Std) de-lumped modeling approach with PR EOS.

7.4. Discussion

Bar plots for cohesive energy and solubility parameter for the pure components studied in this work are shown in Figure 7.5 and Figure 7.6, respectively, showing that PR EOS reproduces the trend in cohesive energy much better than it does for solubility parameter. The reference data are obtained from a correlation for solubility parameter proposed by Wang[92].

$$\delta = 52.042\ F_{RI} + 2.904 \tag{7.7}$$

With the definition of solubility parameter as the square root of the residual internal energy multiplied by density, Wang's solubility parameter correlation can be rewritten in terms of residual internal energy.

$$-U^R = \frac{MW}{\rho}(52.042\ F_{RI} + 2.904)^2 \tag{7.8}$$

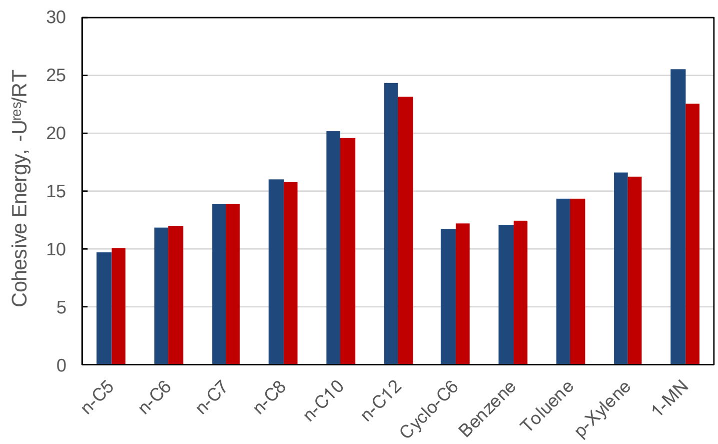


Figure 7.5. Cohesive energy for selected pure components at 20 °C and 1 bar from Eq. (7.8) [blue] and PR EOS [red].

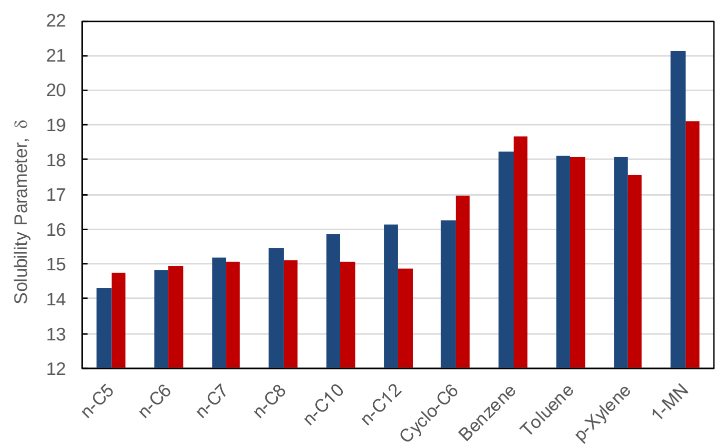


Figure 7.6. Solubility parameter for selected pure components at 20 °C and 1 bar from Eq. (7.7) [blue] and PR EOS [red].

Based on solubility parameter alone, one might expect that 1- MN would be always the appropriate choice as heavy key for the selected mixtures and that the solvent components are sufficiently different from 1-MN that the latter will be largely driving the phase-split at the dew point. However, the poor volumetric property predictions that are a well-known drawback of the cubic EOS cause the solubility parameter of 1-MN to be under-predicted, as compared to Wang's correlation, and the cohesive energy of n-C_{12} is greater than that of 1-MN at some conditions.

Because the form of the equation of state and the mixing rules remain consistent, deviations between the lumped and de-lumped approaches can be explained by only two factors: the accuracy of the EV correlations to describe the solvent to which they are applied and the implicit assumption that all components in the lumped solvent partition equally. **Figure 7.7**, which shows plots for cohesive energy and solubility parameter of the solvent mixtures, suggests that the EV correlations describe nonpolar solvent mixtures very well in comparison to the standard de-lumped approach.

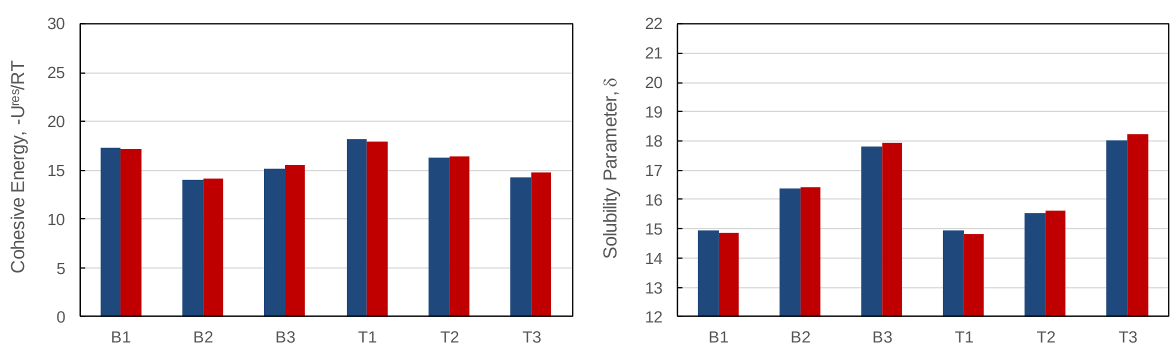


Figure 7.7. Cohesive energy and solubility parameter plots for solvent mixtures at 20 °C and 1 bar from PR EOS with standard (Std) approach [blue] and lumped solvent (EV) approach [red].

However, regardless of the accuracy of the correlations, the lumped approach will always deviate from the de-lumped approach to some degree because of the limitation of the partitioning assumption. The magnitude of the deviations, though, might be negligible depending on the system and how the solvent is lumped.

In general, saturation pressure predictions for the lumped and de-lumped approaches agree closely when the component(s) most responsible for driving the phase-split are well characterized and their partitioning with respect to the solvent are only negligibly affected by lumping. For example, the calculated bubble pressure of a mixture containing propane with toluene, p-xylene and 1-MN will be essentially the same with the lumped and de-lumped approaches because the solubility of propane in the solvent is only negligibly affected by lumping the heavy components and only a small amount of the heavy components will partition into the vapor phase at the bubble point. In these cases, the equi-partitioning assumption causes very little deviation to property predictions. However, if n-butane were added to the solvent, the bubble pressure calculation for the lumped and de-lumped approaches would not agree as closely. Propane and n-butane have similar volatilities, and the solubility of n-butane in the solvent has a non-negligible effect on bubble pressure that would be obfuscated by lumping n-butane with the heavy components. The same argument applies for the dew pressure calculation. The more dissimilar the heavy key is from the components that constitute the solvent, the lesser impact lumping the solvent will have on dew pressure predictions.

Though all systems studied show very good saturation pressure predictions – due mostly to the accuracy of the EV correlations – some solvents (B1, T1, T2) showed much more deviation between the lumped and standard approaches, particularly in the dew region, than the other solvents. The primary reason this deviation occurs is because the heavy key (1-MN) and the heaviest component in the solvent for these cases (n-C_{12}) have similar cohesive energy. To describe the phase-splitting process accurately in the dew region, it is important that there is sufficient resolution in the heavy components that are driving the phase-split. When one of the solvent components that is important to this process is lumped in with other solvent components, then its effect is diluted and the prediction suffers. When the difference in the cohesive energy between the heavy key and solvent components is sufficiently large, which is the case for the solvents that do not contain n-C_{12} (B2, B3, T3), then the deviations between the lumped and standard approaches are negligible.

The modeling study performed in this work is a proof-of-concept for the EV correlations and the lumped solvent approach. With the accuracy of the approach established, it is important to understand how and when the proposed approach can be applied to practical systems. The approach is primarily of interest for defining the simulation parameters of unknown fractions of a complex mixture. If all components and their compositions in a mixture are known in advance, then the standard approach would be always preferable unless the mixture exceeds some limitation on the number of components allowed in a simulation. In these cases, the EV correlations offer a method to lump similar components into a solvent fraction and calculate the critical properties of the pseudo-component.

However, the primary reason these correlations were developed was to describe fractions of complex crude oils for which a detailed compositional analysis is not readily available. The crude oil can be lumped into as many or few fractions as desired and then the critical properties of each fraction can be obtained from simple refractive index or density measurements performed on each fraction at ambient conditions. The number of component fractions and whether the light end and/or heavy end requires more resolution depends on which properties are being predicted and the required accuracy of the predictions. If only bubble pressures are desired, then a simple lumping approach with all components heavier than n-hexane lumped into a single fraction is often sufficient, as shown by Abutaqiya et al.[95] If dew pressures or asphaltene onsets are desired, then the heavy end must be characterized in more detail, as shown in the more customary crude oil characterization methodologies[54,55,62,65,87].

It was expected that solubility parameter would offer an indication as to why certain solvent mixtures showed consistently worse predictions than others, but there was no apparent trend in solubility parameter that suggested when the lumped solvent approach predictions might suffer. Cohesive energy, on the other hand, did show such a trend. When the difference in cohesive energy between the light and/or heavy key and all solvent components is large, then the lumped solvent approach will introduce only negligible error to saturation pressure predictions. The Wang correlation for residual internal energy in Eq. (7.8) (which is equivalent to cohesive energy) offers a means to check on whether de-lumping the crude oil fractions might be appropriate.

Eq. (7.8) requires only refractive index (or density) and molecular weight measurements, and PR EOS with EV correlations was shown to match the residual internal energy correlation quite well, suggesting that cohesive energy can be used as a heuristic for determining the number of fractions into which the crude oil should be separated in order to generate simulation parameters that are representative of the actual mixture.

7.5. Conclusions

Critical property correlations proposed by Evangelista and Vargas[116] were evaluated for their accuracy in describing nonpolar solvent mixtures. The correlations, along with the proposed lumped solvent approach, show very good predictive capability for saturation pressures and bulk liquid properties like density, thermal expansion, and isothermal compressibility. This suggests that the approach is suitable to be applied to nonpolar crude oil fractions for which the composition is unknown. The EV correlations offer a simple expression for relating easily measured volumetric or optical properties to critical properties, and the lumped solvent approach shows small deviations with respect to the standard modeling approach. Additionally, it was shown that cohesive energy calculated by PR EOS with EV correlations matches well with the Wang correlation for cohesive energy (Eq. (7.8)), and differences in cohesive energy between the heavy key and the components comprising the solvent was shown to correlate with accuracy of the lumped solvent approach in matching dew point predictions. When the heavy key and solvent are sufficiently different, the lumped solvent modeling approach produces very good predictions for saturation pressure in the dew region. When the heavy key and solvent components have similar cohesive energy, dew pressure predictions suffer if the solvent is lumped.

Chapter 8

Conclusions and Future Work

8.1. Cubic-Plus-Chain (CPC) Equation of State

A new equation of state for nonpolar chain molecules was proposed in this work with testing performed for short-chain and long-chain n-alkanes. The current version of CPC combines the RK equation of state with the Elliott radial distribution function (RDF), but the cubic-plus-chain (CPC) hybrid model is more of a framework than an equation of state itself. The CPC framework can handle any cubic EOS with a suitable RDF.

The next iteration of CPC should modify either or both contributions to test which combination of EOS and RDF generates the most accurate model. The most immediate task for improving CPC is identifying an RDF that shifts the critical volume (or critical compressibility factor) more strongly as a function of the chain length. As shown in **Figure 6.3**, the Elliott RDF has negligible effect on critical compressibility and thus the poor volume predictions in the critical region that are a well-known drawback of the cubic EOS are not rectified by the addition of the chain term. The constraint of fitting to the critical point can also be relaxed for CPC, which would likely result in a better match for volume predictions but suffer for saturation pressure predictions near the critical point. Removing the critical point constraint should be done with care, as leaving the three CPC parameters (T_c, P_c, m) free to match saturation pressure and density across a short range of T_r could produce parameter sets that do not predict other thermodynamic properties well. If the critical point matching criteria is relaxed, additional tuning criteria should be included to ensure physical consistency.

8.2. Critical Property Correlations for Mixed Solvents

Utilizing the EV correlations (Eqs. (7.1) - (7.4)) within the cubic EOS framework, critical properties for multicomponent nonpolar solvent mixtures can be calculated from simple volumetric or optical measurements without knowledge of the composition of the mixture, and then phase behavior predictions can be performed. **Figure 7.1** - **Figure 7.4** show that the lumped solvent approach with critical properties calculated from the EV correlations match very well the standard de-lumped solvent approach with critical properties obtained from the literature.

The most attractive feature of these correlations is that the input properties are easily measured from experiments performed at ambient conditions, so only a small sample of solvent is required to generate critical properties that can then be used for phase behavior modeling. Additionally, the lumped solvent modeling framework presented in this work is flexible, so a very complex mixed solvent – such as crude oil –

can be fractionated into multiple cuts and the molecular weight and refractive index for these cuts can be easily measured.

8.3. Asphaltene Phase Behavior for Deepwater Reservoirs

The asphaltic crude oil modeling study discussed in Section 5.4.1 showed anomalous thermodynamic property predictions for the onset phase. The heavy non-asphaltene components partitioned to the onset phase much more strongly than in past modeling studies, which translates to densities and compositions of the bulk and onset phases that are different than what is normally observed. Because many non-asphaltene components co-precipitate into the onset phase, this phase becomes lighter and leaner in asphaltenes than phases that have traditionally been identified as problematic for asphaltene deposition. Partial validation of the thermodynamic modeling observations was shown in **Figure 5.9**.

Future work in this area should focus on validating the anomalous trends observed, mainly with respect to the density and composition of the onset phase, and also establishing relationships between phase density and composition and the tendency of that phase to deposit. Though two liquid phases might exist at a given pressure and temperature, it is possible that neither phase has properties conducive to deposition. Operating a well in the asphaltene unstable region has traditionally been a concern for producers, but asphaltene deposition might not be a threat for occurring even when precipitation does occur.

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Appendix A

Thermodynamic properties calculated from the reduced residual Helmholtz function (F) and its derivatives.

$$F = \frac{A^R(T, V, \boldsymbol{n})}{RT} \doteq [\mathrm{mol}]$$

Pressure

$$P = -RT\left(\frac{\partial F}{\partial V}\right)_{T,\boldsymbol{n}} + \frac{nRT}{V}$$

$$\left(\frac{\partial P}{\partial V}\right)_{T,\boldsymbol{n}} = -RT\left(\frac{\partial^2 F}{\partial V^2}\right)_{T,\boldsymbol{n}} - \frac{nRT}{V^2}$$

$$\left(\frac{\partial P}{\partial T}\right)_{V,\boldsymbol{n}} = -RT\left(\frac{\partial^2 F}{\partial T\,\partial V}\right)_{T,\boldsymbol{n}} + \frac{P}{T}$$

$$\left(\frac{\partial P}{\partial n_i}\right)_{T,V} = -RT\left(\frac{\partial^2 F}{\partial V\,\partial n_i}\right)_{T} + \frac{RT}{V}$$

Volume

$$\left(\frac{\partial V}{\partial P}\right)_{T,\boldsymbol{n}} = \left(\frac{\partial P}{\partial V}\right)_{T,\boldsymbol{n}}^{-1}$$

$$\left(\frac{\partial V}{\partial T}\right)_{T,\boldsymbol{n}} = -\left(\frac{\partial P}{\partial T}\right)_{V,\boldsymbol{n}} \Big/ \left(\frac{\partial P}{\partial V}\right)_{T,\boldsymbol{n}}$$

$$\left(\frac{\partial V}{\partial n_i}\right)_{T,P} = \bar{V}_i = -\left(\frac{\partial P}{\partial n_i}\right)_{T,V} \Big/ \left(\frac{\partial P}{\partial V}\right)_{T,\boldsymbol{n}}$$

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_{P,\boldsymbol{n}}$$

$$\kappa = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_{T,\boldsymbol{n}}$$

Compressibility Factor

$$Z = 1 - \frac{V}{n}\left(\frac{\partial F}{\partial V}\right)_{T,n} = 1 + \beta\left(\frac{\partial F}{\partial \beta}\right)_{T,n}$$

$$\frac{\partial Z}{\partial \beta} = \beta\left(\frac{\partial^2 F}{\partial \beta^2}\right)_{T,n} + \left(\frac{\partial F}{\partial \beta}\right)_{T,n}$$

$$\frac{\partial^2 Z}{\partial \beta^2} = \beta\left(\frac{\partial^3 F}{\partial \beta^3}\right)_{T,n} + 2\left(\frac{\partial^2 F}{\partial \beta^2}\right)_{T,n}$$

$$\frac{\partial Z}{\partial V} = \frac{\partial Z}{\partial \beta}\frac{\partial \beta}{\partial V}$$

$$\frac{\partial^2 Z}{\partial V^2} = \frac{\partial \beta}{\partial V}\left[\frac{\partial^2 Z}{\partial \beta^2}\frac{\partial \beta}{\partial V} + \frac{\partial Z}{\partial \beta}\frac{\partial^2 \beta}{\partial V^2}\right] + \frac{\partial Z}{\partial \beta}\frac{\partial^2 \beta}{\partial V^2}$$

Fugacity Coefficient

$$\ln \phi_i = \left(\frac{\partial F}{\partial n_i}\right)_{T,V} - \ln Z$$

$$\left(\frac{\partial \ln \phi_i}{\partial T}\right)_{P,\boldsymbol{n}} = \left(\frac{\partial^2 F}{\partial T\,\partial n_i}\right)_V + \frac{1}{T} - \frac{\bar{V}_i}{RT}\left(\frac{\partial P}{\partial T}\right)_{V,\boldsymbol{n}}$$

$$\left(\frac{\partial \ln \phi_i}{\partial P}\right)_{P,\boldsymbol{n}} = \frac{\bar{V}_i}{RT} - \frac{1}{P}$$

$$n\left(\frac{\partial \ln \phi_i}{\partial n_j}\right)_{T,P} = n\left(\frac{\partial^2 F}{\partial n_i\,\partial n_j}\right)_{T,V} + 1 - \frac{n}{RT}\bar{V}_i\left(\frac{\partial P}{\partial n_j}\right)_{T,V}$$

Residual Bulk Properties

$$\frac{S^R(T,V,\boldsymbol{n})}{R} = -T\left(\frac{\partial F}{\partial T}\right)_{V,\boldsymbol{n}} - F$$

$$\frac{C_V^R(T,V,\boldsymbol{n})}{R} = -T^2\left(\frac{\partial^2 F}{\partial T^2}\right)_{V,\boldsymbol{n}} - 2T\left(\frac{\partial F}{\partial T}\right)_{V,\boldsymbol{n}}$$

$$\frac{C_P^R(T,V,\boldsymbol{n})}{R} = \frac{C_V^R(T,V,\boldsymbol{n})}{R} - \frac{T}{R}\left(\frac{\partial P}{\partial T}\right)_{V,\boldsymbol{n}}^2\left(\frac{\partial P}{\partial V}\right)_{T,\boldsymbol{n}}^{-1} - n$$

$$U^R(T,V,\boldsymbol{n}) = A^R(T,V,\boldsymbol{n}) + S^R(T,V,\boldsymbol{n})$$

$$\delta^2 = -\frac{U^R(T,V,\boldsymbol{n})}{V}$$

Appendix B

Derivatives of the reduced residual Helmholtz function for the cubic-plus-chain (CPC) equation of state. The monomer terms (repulsive and attractive) match the generic cubic equation of state given in Eqs. (4.8), (4.9) and (4.11). The monomer contribution for CPC is given by the RK equation of state, for which $\delta_1 = 1, \delta_2 = 0$.

$$F = \frac{A^R(T, V, \boldsymbol{n})}{RT} = \bar{m}(a^{\text{rep}} + a^{\text{att}}) + a^{\text{chain}} \tag{B.1}$$

$$a^{\text{rep}} = -n \ln(1 - \beta) \tag{B.2}$$

$$a^{\text{att}} = -\frac{A}{BRT(\delta_1 - \delta_2)} \ln\left[\frac{1 + \delta_1\beta}{1 + \delta_2\beta}\right] \tag{B.3}$$

$$a^{\text{chain}} = -\sum_i n_i(m_i - 1) \ln \mathrm{g}(\beta) \tag{B.4}$$

Compressibility Factor

The compressibility factor and its derivatives can be calculated from volume or density derivatives of the Helmholtz energy function:

$$Z = 1 - \frac{V}{n}\left(\frac{\partial F}{\partial V}\right)_{T,n} = 1 + \beta\left(\frac{\partial F}{\partial \beta}\right)_{T,n}$$

$$\frac{\partial Z}{\partial \beta} = \beta\left(\frac{\partial^2 F}{\partial \beta^2}\right)_{T,n} + \left(\frac{\partial F}{\partial \beta}\right)_{T,n}$$

$$\frac{\partial^2 Z}{\partial \beta^2} = \beta\left(\frac{\partial^3 F}{\partial \beta^3}\right)_{T,n} + 2\left(\frac{\partial^2 F}{\partial \beta^2}\right)_{T,n}$$

Differentiation of Eq. (B.1) with respect to β yields:

$$\frac{\partial F}{\partial \beta} = m\left(\frac{\partial a^{\text{rep}}}{\partial \beta} + \frac{\partial a^{\text{att}}}{\partial \beta}\right) + \frac{\partial a^{\text{chain}}}{\partial \beta} \tag{B.5}$$

$$\frac{\partial^2 F}{\partial \beta^2} = m\left(\frac{\partial^2 a^{\mathrm{rep}}}{\partial \beta^2} + \frac{\partial^2 a^{\mathrm{att}}}{\partial \beta^2}\right) + \frac{\partial^2 a^{\mathrm{chain}}}{\partial \beta^2} \tag{B.6}$$

$$\frac{\partial^3 F}{\partial \beta^3} = m\left(\frac{\partial^3 a^{\mathrm{rep}}}{\partial \beta^3} + \frac{\partial^3 a^{\mathrm{att}}}{\partial \beta^3}\right) + \frac{\partial^3 a^{\mathrm{chain}}}{\partial \beta^3} \tag{B.7}$$

Differentiation of Eq. (B.2) with respect to β yields:

$$\frac{\partial a^{\mathrm{rep}}}{\partial \beta} = \frac{n}{1-\beta} \tag{B.8}$$

$$\frac{\partial^2 a^{\mathrm{rep}}}{\partial \beta^2} = \frac{n}{(1-\beta)^2} \tag{B.9}$$

$$\frac{\partial^3 a^{\mathrm{rep}}}{\partial \beta^3} = \frac{2n}{(1-\beta)^3} \tag{B.10}$$

Differentiation of Eq. (B.3) with respect to β yields:

$$\frac{\partial a^{\mathrm{att}}}{\partial \beta} = -\frac{A}{BRT}\left[(1+\delta_1\beta)(1+\delta_2\beta)\right]^{-1} \tag{B.11}$$

$$\frac{\partial^2 a^{\mathrm{att}}}{\partial \beta^2} = \frac{A}{BRT}\frac{\delta_1+\delta_2+2\delta_1\delta_2\beta}{(1+\delta_1\beta)^2(1+\delta_2\beta)^2} \tag{B.12}$$

$$\frac{\partial^3 a^{\mathrm{att}}}{\partial \beta^3} = -\frac{2A}{BRT}\frac{\delta_1^2(1+3\delta_2\beta+3\delta_2^2\beta^2)+\delta_2^2+\delta_1\delta_2(1+3\delta_2\beta)}{(1+\delta_1\beta)^3(1+\delta_2\beta)^3} \tag{B.13}$$

Differentiation of Eq. (B.4) with respect to β yields:

$$\frac{\partial a^{\mathrm{chain}}}{\partial \beta} = -\sum_i n_i(m_i-1)\frac{\mathrm{g}'(\beta)}{\mathrm{g}} \tag{B.14}$$

$$\frac{\partial^2 a^{\mathrm{chain}}}{\partial \beta^2} = -\sum_i n_i(m_i-1)\left[\frac{\mathrm{g}''(\beta)}{\mathrm{g}} - \frac{\mathrm{g}'(\beta)^2}{\mathrm{g}^2}\right] \tag{B.15}$$

$$\frac{\partial^3 a^{\mathrm{chain}}}{\partial \beta^3} = -\sum_i n_i(m_i-1)\left[\frac{\mathrm{g}'''(\beta)}{\mathrm{g}} - 3\frac{\mathrm{g}''(\beta)\mathrm{g}'(\beta)}{\mathrm{g}^2} + 2\frac{\mathrm{g}'(\beta)^3}{\mathrm{g}^3}\right] \tag{B.16}$$

Differentiation of the radial distribution function (g) with respect to β yields:

$$\mathrm{g} = \frac{1}{1 - 0.475\beta} \tag{B.17}$$

$$\mathrm{g}'(\beta) = \frac{0.475}{(1 - 0.475\beta)^2} \tag{B.18}$$

$$\mathrm{g}''(\beta) = \frac{0.45125}{(1 - 0.475\beta)^3} \tag{B.19}$$

$$\mathrm{g}'''(\beta) = \frac{0.643031}{(1 - 0.475\beta)^4} \tag{B.20}$$

Combining terms and plugging into the compressibility factor equation yields the compressibility form of CPC.

Fugacity Coefficient

The fugacity coefficient for component i in the mixture can be calculated from a mole number derivative of the residual Helmholtz energy function:

$$\ln \hat{\phi}_i = \left(\frac{\partial F}{\partial n_i}\right)_{T,V} - \ln Z \tag{B.21}$$

Differentiating the residual Helmholtz for CPC with respect to n_i yields:

$$\left(\frac{\partial F}{\partial n_i}\right)_{T,V} = \bar{m}_{n_i}(a^{\mathrm{rep}} + a^{\mathrm{att}}) + m\left(\frac{\partial a^{\mathrm{rep}}}{\partial n_i} + \frac{\partial a^{\mathrm{att}}}{\partial n_i}\right) + \frac{\partial a^{\mathrm{chain}}}{\partial n_i} \tag{B.22}$$

$$\frac{\partial a^{\mathrm{rep}}}{\partial n_i} = -\ln(1-\beta) - \frac{\beta_{n_i} - n}{1-\beta} \tag{B.23}$$

$$\frac{\partial a^{\mathrm{att}}}{\partial n_i} = -\frac{A\,\beta_{n_i}}{BRT(1+\delta_1\beta)(1+\delta_2\beta)} + \frac{A\,B_{n_i} - A_{n_i}\,B}{B^2RT}\ln\left[\frac{1+\delta_1\beta}{1+\delta_2\beta}\right] \tag{B.24}$$

$$\frac{\partial a^{\mathrm{chain}}}{\partial n_i} = -(m_i - 1)\ln \mathrm{g} - \sum_j n_j\left(m_j - 1\right)\frac{\mathrm{g}_{n_j}}{\mathrm{g}} \tag{B.25}$$

The mean segment diameter, reduced volume, radial distribution function, attraction energy, excluded volume and their derivatives with respect to mole numbers are calculated by:

$$\bar{m} = \sum_i x_i m_i \qquad \bar{m}_{n_i} = \frac{m_i}{n} - \frac{1}{n^2}\sum_j n_j m_j \tag{B.26}$$

$$\beta = \frac{\bar{m}B}{V} \qquad \beta_{n_i} = \frac{\bar{m}_{n_i}B + \bar{m}B_{n_i}}{V} \tag{B.27}$$

$$\mathrm{g} = \frac{1}{1 - 0.475\beta} \qquad \mathrm{g}_{n_i} = \frac{0.475\,\beta_{n_i}}{(1 - 0.475\beta)^2} \tag{B.28}$$

$$A = \sum_i \sum_j n_i n_j a_{ij} \qquad A_{n_i} = \sum_j 2\, n_j a_{ij} \tag{B.29}$$

$$B = \sum_i n_i b_i \qquad B_{n_i} = b_i \tag{B.30}$$

To apply these derivatives to the standard cubic EOS, set all $m_i = 1$. The chain term will vanish and properties like reduced volume – which is defined as $\bar{m}B/V$ for CPC and B/V for standard cubics – will adjust naturally.

Appendix C

Live oil composition, PC-SAFT simulation parameters, binary interaction parameters, and injection gas composition for crude oils studied in this work.

Table C1. PC-SAFT simulation parameters for U8.

Component	**z** **mol**	**M_W** **g/mol**	**m**	**σ** **Å**	**ε/k_B** **K**
H_2S	0.690	34.08	1.6517	3.0737	227.34
N_2	0.402	28.01	1.2064	3.3130	90.96
CO_2	1.087	44.01	2.0729	2.7852	169.21
C_1	19.060	16.04	1.0000	3.7039	150.03
C_2	4.285	30.07	1.6069	3.5206	191.42
C_3	4.742	44.10	2.0020	3.6184	208.11
Heavy Gas	10.320	71.78	2.6891	3.7611	231.42
Saturates	49.743	188.35	5.6850	3.9135	251.22
Aromatics	9.678	227.01	5.3559	4.0670	346.10
Asph	0.0083	6300	125.7444	4.2538	356.45

Table C2. PC-SAFT simulation parameters for J1.

Component	**z** **mol**	**M_W** **g/mol**	**m**	**σ** **Å**	**ε/k_B** **K**
H_2S	0.000	34.08	1.6517	3.0737	227.34
N_2	0.495	28.01	1.2064	3.3130	90.96
CO_2	14.583	44.01	2.0729	2.7852	169.21
C_1	27.334	16.04	1.0000	3.7039	150.03
C_2	0.000	30.07	1.6069	3.5206	191.42
C_3	0.000	44.10	2.0020	3.6184	208.11
Heavy Gas	21.917	44.60	1.9906	3.6382	213.45
Saturates	23.853	207.60	6.1797	3.9236	252.40
Aromatics	11.750	270.50	7.5295	3.9680	271.13
Asph	0.0673	1700	35.3571	4.2322	349.37

Table C3. PC-SAFT simulation parameters for S14.

Component	z mol	M_W g/mol	m	σ Å	ε/k_B K
H_2S	0.081	34.08	1.6517	3.0737	227.34
N_2	0.911	28.01	1.2064	3.3130	90.96
CO_2	2.335	44.01	2.0729	2.7852	169.21
C_1	16.839	16.04	1.0000	3.7039	150.03
C_2	4.987	30.07	1.6069	3.5206	191.42
C_3	5.138	44.10	2.0020	3.6184	208.11
Heavy Gas	6.596	66.29	2.5480	3.7432	228.89
Saturates	49.448	158.18	4.9095	3.8933	248.81
Aromatics	13.653	257.68	5.6706	4.1204	367.40
Asph	0.0121	5000	100.8119	4.2458	353.88

Table C4. PC-SAFT simulation parameters for C2.

Component	z mol	M_W g/mol	m	σ Å	ε/k_B K
H_2S	0.000	34.08	1.6517	3.0737	227.34
N_2	0.280	28.01	1.2064	3.3130	90.96
CO_2	0.410	44.01	2.0729	2.7852	169.21
C_1	26.079	16.04	1.0000	3.7039	150.03
C_2	7.899	30.07	1.6069	3.5206	191.42
C_3	6.805	44.10	2.0020	3.6184	208.11
Heavy Gas	7.584	69.54	2.6315	3.7541	230.43
Saturates	36.424	202.62	6.0518	3.9211	252.12
Aromatics	9.069	322.93	6.8182	4.1654	374.78
Resins	4.371	649.61	12.3600	4.2641	391.14
Asph1	0.030	998.98	18.2869	4.2989	395.50
Asph2	0.115	1026.71	18.7573	4.3006	395.69
Asph3	0.163	1085.72	19.7583	4.3041	396.07
Asph4	0.771	1708.11	30.3165	4.3261	398.25

Table C5. Binary interaction parameters for U8.

U8										
kij	H_2S	N_2	CO_2	C_1	C_2	C_3	HG	Sat	Arom	Asph
H_2S										
N_2	0.09									
CO_2	0.0678	0								
C_1	0.062	0.03	0.05							
C_2	0.058	0.04	0.097	0						
C_3	0.05	0.06	0.10	0	0					
HG	0.07	0.075	0.12	0.03	0.02	0.015				
Sat	0.09	0.14	0.13	0.03	0.012	0.01	0.005			
Arom	0.015	0.158	0.10	0.029	0.025	0.01	0.012	0.007		
Asph	0.015	0.16	0.18	0.07	0.06	0.01	0.01	-0.004	0	

Table C6. Binary interaction parameters for J1.

J1										
kij	H_2S	N_2	CO_2	C_1	C_2	C_3	HG	Sat	Arom	Asph
H_2S										
N_2	0.09									
CO_2	0.0678	0								
C_1	0.062	0.03	0.05							
C_2	0.058	0.04	0.097	0						
C_3	0.05	0.06	0.10	0	0					
HG	0.07	0.06	0.01	0.03	0.02	0.015				
Sat	0.09	0.12	0.12	0.029	0.012	0.01	0.01			
Arom	0.015	0.12	0.11	0.029	0.025	0.01	0.01	0.007		
Asph	0.015	0.25	0.11	0.029	0.06	0.01	0.01	0.007	0	

Table C7. Binary interaction parameters for S14.

S14										
kij	H_2S	N_2	CO_2	C_1	C_2	C_3	HG	Sat	Arom	Asph
H_2S										
N_2	0.09									
CO_2	0.0678	0								
C_1	0.062	0.03	0.05							
C_2	0.058	0.04	0.097	0						
C_3	0.05	0.06	0.10	0	0					
HG	0.07	0.075	0.12	0.01	0.02	0.015				
Sat	0.09	0.14	0.13	0.03	0.012	0.01	0.005			
Arom	0.015	0.158	0.10	0.029	0.025	0.01	0.012	0.007		
Asph	0.015	0.16	0.18	0.11	0.06	0.01	0.01	-0.004	0	

Table C8. Binary interaction parameters for C2.

C2								
kij	H_2S	N_2	CO_2	C_1	C_2	C_3	HG	Sat
H_2S								
N_2	0.09							
CO_2	0.0678	0						
C_1	0.062	0.03	0.05					
C_2	0.058	0.04	0.097	0				
C_3	0.05	0.06	0.10	0	0			
HG	0.07	0.075	0.12	0.01	0.02	0.015		
Sat	0.09	0.14	0.13	0.005	0.005	0.005	0.005	
Arom	0.015	0.158	0.18	0.012	0.01	0.01	0.012	0.007
Resins	0.015	0.158	0.18	0.013	0.01	0.007	0.016	-0.025
Asph	0.015	0.158	0.18	0.017	0.0115	0.01	0.01	-0.004

Table C9. Composition of injection gas for studied crudes.

Comp	**Injection Gas (Mol%)**			
	U8	**J1**	**S14**	**C2**
H_2S	0.49		0.031	0.00
N_2	0.83	100	0.65	0.23
CO_2	4.96		6.73	0.26
C_1	73.54		60.57	84.59
C_2	11.34		13.70	6.64
C_3	5.19		10.19	5.03
HG	3.65		8.13	3.25